Momentum Regime Shifts: A Bayesian Hidden Markov Model Approach to Cross-Sectional Stock Selection

By

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Abstract

This thesis investigates how a real-time regime signal interacts with cross-sectional momentum and finds that momentum is not a single stable anomaly but a regime-dependent family of horizon-specific effects: in calm markets, intermediate-horizon continuation dominates the cross-section; in the most acute panic months, beaten-down stocks across all horizons become more likely to outperform, while in milder panic months the cross-section retains continuation structure. A Bayesian Hidden Markov Model produces a filtered probability of market panic, which is combined with momentum at 12 lookback horizons in machine learning models to rank stocks. The regime signal functions as a context variable – it carries no direct linear return-predictive content but conditions how the model jointly uses the full momentum term structure in a given market state. Out-of-sample on 2011–2025 U.S. equities, only a nonlinear model (XGBoost) exploits this structure (Sharpe 1.11, six-factor alpha 24.1%, $t = 4.81$); both a classification-based and a regression-based linear model collapse, establishing that the interaction is inherently nonlinear. Cross-sectional z-score analysis confirms this regime-dependent structure in the raw data. The regime signal is essential (removing it reduces the Sharpe by more than 60%) and most valuable during stress, with the long-short spread widening from 1.3% to 3.1% per month. These findings are consistent with the interpretation that momentum's well-documented fragility arises partly from applying unconditional models to what is fundamentally a regime-dependent phenomenon.

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Management Summary

Cross-sectional momentum – buying recent winners and selling recent losers – is one of the most profitable and most fragile anomalies in finance. This thesis finds that momentum's fragility may arise because it is not a single stable pattern but a regime-dependent family of effects: in calm markets, stocks that rose over intermediate horizons continue to outperform; in panic, stocks that have fallen across all lookback horizons become more likely to outperform. The regime signal, derived from a Bayesian Hidden Markov Model estimated on real-time stress indicators, functions as a context variable that conditions how the model jointly uses the full momentum term structure rather than predicting returns directly. Only a nonlinear model (XGBoost) can exploit this structure; linear models with identical features fail entirely, including both a classification-based and a regression-based specification.

The thesis develops a two-stage framework (Section 3). In the first stage, a Bayesian two-state Hidden Markov Model is estimated via Gibbs sampling on four macrofinancial stress indicators (drawdown, dispersion, relative market participation, and credit spread), producing a filtered probability of market panic. In the second stage, this regime signal is combined with momentum at 12 lookback horizons in models of increasing complexity: a deterministic formula, logistic regression, Ridge regression, and XGBoost.

Out-of-sample on 167 months of U.S. equity data (2011–2025), only XGBoost survives long-short construction (Sharpe 1.11, six-factor alpha 24.1%). The regime signal is essential: removing it reduces the Sharpe by more than 60%. Cross-sectional z-score analysis confirms that momentum's predictive structure shifts with market conditions in the raw data, independent of any model. Stress tests show the strategy can absorb a 12-month recession but is vulnerable to prolonged fundamental deterioration where beaten-down stocks never recover.

Limitations include the single test period (2011–2025), US-only sample, and untested resilience to depression-style downturns. The thesis complements the exposure-scaling approach of Daniel and Moskowitz (2016) with a stock-selection channel: rather than adjusting how much to hold, the regime signal adjusts which stocks to hold.

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1 Introduction

Cross-sectional momentum, the tendency for recent stock market winners to continue outperforming and losers to continue underperforming, is one of the most robust anomalies in empirical finance (Jegadeesh and Titman, 1993). Yet it is also one of the most fragile: momentum portfolios suffered losses exceeding 50% during the 2009 market rebound (Daniel and Moskowitz, 2016). This tension between long-run profitability and vulnerability to regime shifts motivates the central question of this thesis: how does a real-time regime signal interact with the cross-sectional momentum anomaly, and what does this interaction reveal about momentum's underlying structure?

Existing responses to momentum's fragility adjust either exposure or horizon weights based on backward-looking indicators (Section 2). This thesis proposes an alternative: a Bayesian Hidden Markov Model estimated on real-time stress indicators produces a filtered probability of market panic, which enters a machine learning model alongside 12 momentum horizons so that cross-sectional stock selection adapts to the current regime.

The central claim is that this regime signal is not a direct return predictor but a *context variable*: it conditions how the model jointly uses the full momentum term structure in a given market state. No prior study has documented how the full term structure of cross-sectional momentum predictability reorganises across regimes, or that exploiting this reorganisation requires nonlinear modelling.

Out-of-sample evaluation on 167 months of U.S. equity data (2011–2025) yields three main findings. First, only XGBoost produces meaningful long-short returns (Sharpe 1.11, six-factor alpha 24.1%, $t = 4.81$); both linear baselines on identical features collapse – classification-based logistic regression to Sharpe -0.03 and regression-based Ridge to -0.57 (Appendix G.8) – establishing that the interaction is inherently nonlinear rather than an artefact of the training target. Second, the regime signal is essential – removing it reduces the Sharpe by more than 60% – and most valuable during stress, with the long-short spread widening from 1.3% to 3.1% per month in panic. Third, cross-sectional z-score analysis confirms this shift: in calm, the model's long leg selects stocks with a humped momentum profile peaking at intermediate horizons; in panic, the same leg selects stocks uniformly below average at every horizon from 1 to 12 months.

These findings answer three research questions: (i) does the regime signal's interaction with momentum require nonlinear modelling, and if so, why does a linear model fail? (ii) how does the cross-sectional structure of momentum change between calm and panic regimes? (iii) how robust is the regime-momentum interaction to alternative signal specifications, risk preferences, and prolonged market stress?

The thesis makes three contributions. First, it documents how the full term structure of cross-sectional momentum predictability reorganises between calm and panic regimes, with short-horizon signals flipping sign during stress. Second, it proposes and validates a framework in which a macrofinancial regime signal functions as a context variable for stock selection, conditioning how the model jointly weighs the full momentum term structure rather than predicting returns directly. Third, it establishes that this interaction is inherently nonlinear: both classification-based and regression-based linear models fail on identical features, while a tree-based model succeeds.

The remainder of the thesis is organised as follows. Section 2 reviews the momentum anomaly and its horizon structure, momentum crashes and exposure-based risk management, regime models and regime-conditional asset pricing, and machine learning in cross-sectional asset pricing. Section 3 develops the two-stage framework: the Bayesian HMM for regime classification and the three cross-sectional stock selection methods. Section 4 describes the data, HMM feature selection procedure, and cross-sectional features. Section 5 presents the out-of-sample results, including regime-conditional performance, stress tests, SHAP analysis, ablation studies, and the risk-aversion analysis. Section 6 summarises the findings, discusses limitations, and suggests directions for future research.

2 Literature Review

This review covers four strands of literature: the cross-sectional momentum anomaly and its horizon structure, momentum crashes and exposure-based risk management, regime models and their use in regime-conditional asset pricing, and the application of machine learning to cross-sectional return prediction. Each provides a piece of the framework this thesis assembles. None has, on its own, posed the question of how the full term structure of cross-sectional momentum reorganises across regimes.

2.1 The Momentum Anomaly and Its Horizon Structure

Jegadeesh and Titman (1993) established cross-sectional momentum as a pervasive feature of equity markets: portfolios that buy past winners and sell past losers earn significant returns over 3-to-12-month holding periods. The original strategy ranks stocks by cumulative returns over the past twelve months and has been replicated across international markets, asset classes, and sample periods (Jegadeesh and Titman, 2001; Asness et al., 2013).

The economic mechanism behind momentum remains debated. Behavioural explanations include gradual information diffusion (Hong and Stein, 1999) and the joint conservatism-representativeness mechanism of Barberis et al. (1998), while risk-based explanations link momentum to time-varying systematic risk. This thesis does not take a position on the underlying mechanism but exploits the empirical regularity that momentum's cross-sectional structure varies with market conditions.

A defining feature of momentum is its sensitivity to the lookback horizon. Short lookbacks of one to three months capture short-term reversal effects (Jegadeesh, 1990; Lehmann, 1990), intermediate horizons of six to twelve months capture continuation (Novy-Marx, 2012), and lookbacks beyond twelve months reflect long-term mean reversion (De Bondt and Thaler, 1985). Lee and Swaminathan (2000) document a momentum lifecycle in which the same signal evolves from continuation to reversal as information ages. Together these findings suggest that momentum is a family of horizon-specific patterns whose relative strength varies with market conditions, not a single signal. The framework here tests this directly by exposing all twelve monthly horizons jointly to the cross-sectional model rather than committing to a single lookback ex ante.

2.2 Momentum Crashes and Exposure-Based Solutions

Despite its long-run profitability, momentum is prone to severe crashes. Daniel and Moskowitz (2016) document that momentum portfolios suffered catastrophic losses during the 1932 and 2009 market rebounds, and show that these crashes are predictable: when the market has recently declined, past losers tend to be high-beta stocks that rally disproportionately during recoveries, while past winners are low-beta defensive stocks that lag. Barroso and Santa-Clara (2015) propose a simple risk-management response: scaling momentum exposure inversely with the strategy's recent realised volatility, a constant-volatility overlay that substantially raises the Sharpe ratio. Bianchi et al. (2022) pursue a related approach by constructing a crash indicator from the joint time variation of conditional volatility and skewness, again operating on the strategy's exposure rather than its composition. This thesis explores a complementary composition-side route: rather than scaling exposure during stress, the model reorganises which stocks populate each leg conditional on the regime, with the consequence that the leg-beta inversion underlying the crash mechanism does not occur (Appendix G.6).

The liquidity literature provides a theoretical basis for why momentum crashes reverse. Margin spirals (Brunnermeier and Pedersen, 2009) and institutional flow pressure (Vayanos and Woolley, 2013) create temporary price dislocations during stress that revert when selling pressure subsides, the mechanism central to the regime-momentum interpretation developed below.

What unites this strand is the form of remedy: each approach manages momentum's crash risk by scaling *how much* momentum to hold using backward-looking risk indicators. The framework here addresses the same risk through a different channel, adapting *which stocks* are held using a forward-looking regime signal derived from contemporaneous stress indicators. The two responses are complementary, not substitutes: a stock-selection channel sits alongside the exposure-scaling literature rather than replacing it.

2.3 Regime Models and Regime-Conditional Asset Pricing

Hidden Markov Models have a long history in financial econometrics, originating with Hamilton (1989) for business-cycle analysis and extended to Bayesian estimation via Gibbs sampling by Albert and Chib (1993), with the multivariate-emission generalisation now standard (Frühwirth-Schnatter, 2006; Hoff, 2009). A common finding in the regime-switching literature for equity markets is that returns are well described by a two-state model with a high-mean, low-volatility state and a low-mean, high-volatility state, where the bear state is less persistent but more volatile (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007). Hamilton's original application fitted the regime model to GNP growth rather than returns themselves, a design choice followed here by using macrofinancial stress indicators as observable inputs.

A separate strand asks whether asset-pricing anomalies are themselves regime-conditional. Cooper et al. (2004) establish that momentum profits depend on the prior market state, with strong continuation following up-markets and weaker or reversed performance following down-markets. More recent work integrates regime estimation into portfolio construction: Shu and Mulvey (2024) use a sparse jump model to identify factor-specific bull and bear regimes and feed the resulting classifications into a Black-Litterman framework for dynamic factor allocation.

The most directly related work is Goulding et al. (2023), who classify each month into one of four market cycles (Bull, Correction, Bear, Rebound) from the trailing one-month and rolling twelve-month market returns. In each cycle a different blending parameter a determines the weight on one-month versus twelve-month momentum: the composite signal is $a \cdot mom_1 + (1 - a) \cdot mom_{12}$, where $a = 0$ corresponds to pure twelve-month momentum and $a = 1$ to pure one-month momentum. Their dynamic strategy selects the optimal blending parameters for each non-Bull state via grid search on the training sample. The framework here generalises that approach along three axes: from a deterministic four-state classification based on trailing returns to a continuous panic probability from a Bayesian HMM on real-time stress indicators; from two horizons to the full twelve-horizon term structure; and from a parametric linear blend to a learned nonlinear combination. Each of these moves is necessary, individually, to pose a question GHM cannot: whether the cross-sectional structure of momentum reorganises across regimes. Since GHM design their strategies for long-only construction, a direct long-short performance comparison is not appropriate; Section 5 instead uses the GHM market-cycle variable as an alternative regime input within the XGBoost framework, providing a fair test of whether trailing-return-based regime definitions capture the same information as the HMM's stress indicators.

The unifying feature of this strand is that the regime variable enters portfolio decisions through a hand-coded rule, threshold, or parametric weight, never as a continuous learned feature. This thesis treats π_t^{filter} instead as an input passed to a learner that decides how to use it.

2.4 Machine Learning in Cross-Sectional Asset Pricing

The use of machine learning methods for cross-sectional stock selection has grown rapidly. Gu et al. (2020) compare a wide range of methods on a kitchen-sink characteristic set including multiple momentum horizons, finding that tree-based models and neural networks substantially outperform linear models by capturing nonlinear interactions among firm characteristics. Gradient-boosted trees (Chen and Guestrin, 2016) have proven particularly effective for tabular financial data because they handle missing values natively, capture nonlinear relationships and interactions without explicit specification, and are robust to irrelevant features. A central challenge in applying these methods to asset pricing is interpretability, addressed here through SHAP values (Lundberg and Lee, 2017; Lundberg et al., 2020), which decompose each prediction into feature-level contributions and admit an exact polynomial-time algorithm for tree ensembles.

The closest methodological precedent is Beckmeyer and Wiedemann (2025), who use machine learning to learn data-driven weights across past daily returns for a single momentum signal, constructing a dynamically reweighted strategy that outperforms the conventional twelve-minus-one specification. Their finding establishes that data-driven weighting of past returns outperforms fixed schemes. The framework here differs in two respects. First, the weighting is learned not over the daily history of one signal but over a vector of twelve monthly horizons. Second, and more importantly, the weighting is learned *conditional on a regime context variable*, isolating the regime-conditioning role as the primary contribution. The unconditional weighting in Beckmeyer and Wiedemann (2025) averages across calm and panic regimes; this thesis tests whether disaggregating that average reveals a structural reorganisation of the term structure between states.

Despite these advances, no published study examines how the full term structure of cross-sectional momentum predictability reorganises across regimes, or whether exploiting this reorganisation requires nonlinear modelling.

Position of This Thesis

Each strand contributes one ingredient that the others cannot supply on their own. The momentum literature provides the object of study, a family of horizon-specific signals whose relative strength is known to vary with conditions but has been treated unconditionally in standard practice. The crash literature documents the cost of this fragility, shows that exposure-scaling overlays can partially manage it, and motivates a complementary fix at the stock-selection level rather than at the exposure level alone. The regime-conditional asset-pricing literature provides the conditioning variable, although it has consistently been used as a switch or parametric weight rather than as a continuous learned feature. The machine-learning literature provides the function class capable of representing high-dimensional interactions, although its existing applications have addressed kitchen-sink characteristic sets or unconditional momentum weighting rather than regime-conditional momentum structure.

The methodology developed in this thesis combines these four ingredients into a two-stage fix. First, a Bayesian Hidden Markov Model identifies the prevailing regime in real time from macrofinancial stress indicators, producing a continuous filtered panic probability. Second, a tree-based learner consumes this probability alongside the full twelve-horizon momentum vector and learns to select stocks differently in each regime. Both stages are essential: an unreliable regime estimate misinforms the cross-sectional model, and a model that cannot represent regime-conditional shape interactions cannot exploit even a perfect regime estimate. The regime variable is treated as a feature rather than a switch, the momentum signal as a vector rather than a scalar, and the combination as a learned interaction rather than a parametric blend. Each individual move appears somewhere in the prior literature; the targeted combination, applied to the question of how the cross-section reorganises across regimes, does not. Section 3 develops this framework in detail, beginning with the Bayesian HMM and proceeding through the cross-sectional models that consume its output. As anticipated by the regime-as-feature framing, the resulting panic probability is best read as a non-priced state variable, conditioning the joint use of momentum horizons rather than carrying its own risk premium, an interpretation supported by the factor-alpha tests reported in Section 5.

3 Methodology

The empirical framework is two stages. The first stage estimates market regimes using a Bayesian Hidden Markov Model (Section 3.1): a two-state HMM is fit to four macro-financial stress indicators via Gibbs sampling, producing a monthly filtered probability π_t^{filter} that the market is in a panic state. Because this probability conditions only on information available at time t, it is a real-time signal with no look-ahead bias.

The second stage uses this regime signal to guide cross-sectional stock selection (Section 3.2). Three portfolio construction methods of increasing complexity serve distinct analytical roles. The deterministic formula (Method 0) tests whether simply varying the momentum lookback horizon with market conditions adds value. The logistic regression (Method 1) tests whether a linear model can exploit the regime signal alongside momentum. XGBoost (Method 2) tests whether nonlinear interactions between the regime signal and other features matter. Comparing M1 and M2 is the key diagnostic: if the regime signal functions as a context variable rather than a direct return predictor, a linear model should be unable to use it, while a tree-based model should find it important.

3.1 Stage 1: Regime Classification via Hidden Markov Model

3.1.1 Model Specification

This thesis employs a two-state ($K = 2$) Bayesian Hidden Markov Model to classify the market into latent regimes. The choice of $K = 2$ is validated empirically: Appendix F.4 shows that increasing to $K = 3, 4$, or 5 states does not improve downstream portfolio performance and introduces estimation instability. The hidden state $s_t \in \{0, 1\}$ represents the unobserved regime at month t, where $s_t = 0$ denotes the *calm* regime and $s_t = 1$ denotes the *panic* regime. The two states could equally be labelled bull/bear, expansion/recession, or low/high volatility; calm/panic is used throughout this thesis because it reflects the economic nature of the chosen stress features, which are most directly diagnostic of market stress rather than of trend direction or business-cycle phase. Crucially, as emphasized by Hamilton (1989), market regimes are not directly observable: an investor cannot know with certainty whether the market is currently in a calm or panic state, and classifications that look obvious in retrospect are genuinely difficult to identify in real time. The HMM formalises this uncertainty by treating regimes as hidden states that must be inferred probabilistically from observable market data (see Figure 1).

Observation equation Conditional on the regime, the vector of standardized features $\mathbf{z}_t \in \mathbb{R}^D$ (where $D = 4$) is drawn from a regime-specific multivariate Gaussian. The shorthand $\mathbf{z}_{1:t} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_t)$ denotes the full history of observations up to and including month t:

$$(\mathbf{z}_t \mid s_t = 0) \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \qquad (\mathbf{z}_t \mid s_t = 1) \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1). \tag{1}$$

Each regime is characterized by its own mean vector $\boldsymbol{\mu}_k \in \mathbb{R}^D$ and covariance matrix $\boldsymbol{\Sigma}_k \in \mathbb{R}^{D\times D}$, where $k \in \{0 \text{ (calm)}, 1 \text{ (panic)}\}$. The multivariate specification allows the model to capture not only level differences in individual features across regimes, but also changes in the correlation structure; for instance, stress features may become more correlated during panic periods. The Gaussian assumption is standard in the HMM literature. Appendix G.9 verifies its adequacy by re-estimating the model with multivariate Student-t emissions: regime classifications agree on at least 97.8% of months, filtered panic probabilities correlate at ≥ 0.98, and BIC does not improve, confirming that the Gaussian specification is robust for this application.

Transition equation The evolution of the hidden state follows a first-order Markov chain with a 2×2 transition matrix:

$$\mathbf{P} = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix}, \tag{2}$$

where $P_{ij} = P(s_t = j \mid s_{t-1} = i)$ and each row sums to one. The diagonal elements P_{00} (calm persistence) and P_{11} (panic persistence) govern how likely each regime is to continue into the next month. The expected duration of the calm regime is $1/(1 - P_{00})$ months, and likewise $1/(1 - P_{11})$ for the panic regime.

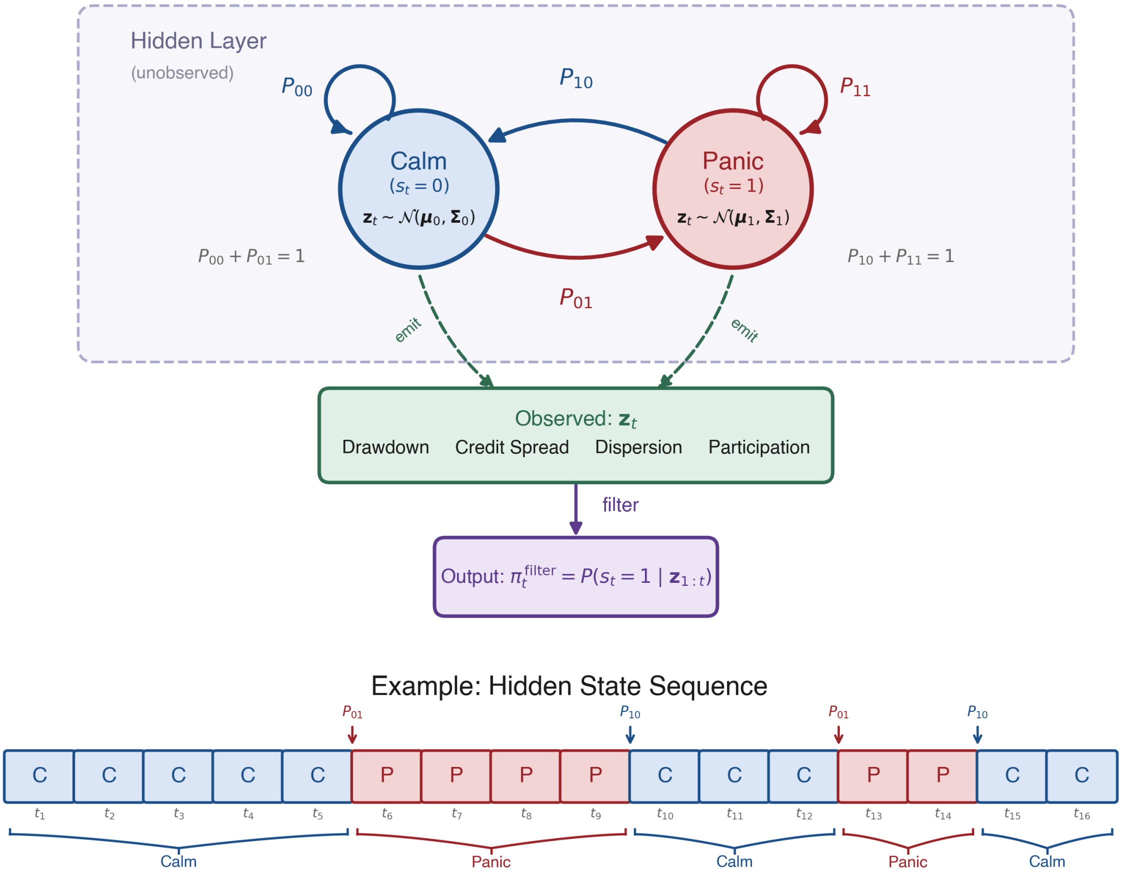


Figure 1: Architecture of the two-state Hidden Markov Model. Hidden states (Calm/Panic) evolve via transition probabilities P_{ij} and emit the four-dimensional observation vector $\mathbf{z}_t$. The forward filter inverts this process to produce $\pi_t^{\text{filter}} = P(s_t = 1 \mid \mathbf{z}_{1:t})$. The lower panel shows an example state sequence illustrating regime persistence and abrupt transitions.

3.1.2 Bayesian Estimation via Gibbs Sampling

The standard frequentist alternative is the Baum–Welch algorithm (Baum et al., 1970), a special case of Expectation-Maximisation (Dempster et al., 1977), which finds maximum likelihood point estimates.

Maximum likelihood estimation of regime-switching HMMs has several known shortcomings for the present application. The likelihood surface is multimodal, with at minimum two equivalent global optima arising from the invariance of the likelihood under state relabelling, and EM can converge to degenerate solutions in which one regime collapses around a single observation (Frühwirth-Schnatter, 2006). Both problems are exacerbated for rare regimes: the panic state in this sample contains only 91 monthly observations in training, where the asymptotic standard errors that frequentist inference relies on are unreliable and the information matrix can become near-singular at boundary transition probabilities (Hamilton, 1989, footnote 12). The Bayesian approach addresses these issues directly. Posterior sampling integrates over parameter uncertainty rather than relying on a point estimate; conjugate priors regularise rare-regime estimates with explicit, defensible weights (Section 3.1.3); and convergence diagnostics across multiple chains (Section 3.1.5) diagnose multimodality that would otherwise go unnoticed under a single EM run.

The Bayesian approach instead draws from the joint posterior:

$$p\big(\{s_t\}_{t=1}^{T},\ \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=0}^{1},\ \mathbf{P} \,\big|\, \mathbf{z}_{1:T}\big)\,.$$

This posterior is not available in closed form because the hidden states and model parameters depend on each other. Therefore we employ Gibbs sampling (Albert and Chib, 1993; Hoff, 2009), a Markov chain Monte Carlo (MCMC) method, which resolves this by alternating between three conditional updates: (i) the full hidden state path $\{s_t\}_{t=1}^{T}$, (ii) the emission parameters $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ for each regime, and (iii) the transition matrix $\mathbf{P}$. Conjugate priors ensure that each block can be sampled exactly from its conditional posterior.

3.1.3 Prior Specification

The priors encode two economic beliefs. First, before seeing the data, neither regime should be forced to have extreme values in any feature; the model should learn the characteristics of calm and panic states from the data. Second, regimes should be persistent rather than switching every few months.

Emission parameters For each regime k, the mean vector and covariance matrix are given a conjugate Normal-Inverse-Wishart (NIW) prior:[1]

$$\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_0, \boldsymbol{\Psi}_0), \qquad \boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}\left(\mathbf{m}_0, \frac{\boldsymbol{\Sigma}_k}{\kappa_0}\right). \tag{3}$$

The hyperparameters are chosen as

$$\mathbf{m}_0 = \mathbf{0}, \qquad \kappa_0 = 0.01, \qquad \nu_0 = D + 2, \qquad \boldsymbol{\Psi}_0 = \mathbf{I}_D(\nu_0 - D - 1), \tag{4}$$

where $D = 4$ is the number of features. After observing data, these prior hyperparameters update to posterior counterparts κ_n, $\mathbf{m}_n$, ν_n, and $\boldsymbol{\Psi}_n$ (Equation 9). Each hyperparameter controls a different aspect of the prior belief:

[1] The Inverse-Wishart is a probability distribution over positive-definite matrices, making it the natural prior for a covariance matrix: every sample is guaranteed to be symmetric and positive-definite. The Normal-Inverse-Wishart combines this with a conditional Normal prior on the mean. The prior is called *conjugate* because when the data are Gaussian, the posterior takes the same NIW form with updated hyperparameters, making each Gibbs update an exact draw rather than an approximation.

- $\mathbf{m}_0$ is the prior guess for each regime's mean vector. Setting $\mathbf{m}_0 = \mathbf{0}$ reflects that, since all features are standardized, there is no reason to expect either regime to be centered away from zero before seeing the data.

- κ_0 controls how strongly the prior mean $\mathbf{m}_0$ is trusted relative to the data. The very small value $\kappa_0 = 0.01$ makes this prior nearly uninformative, so the posterior mean is driven overwhelmingly by the observed data.

- ν_0 is the degrees of freedom of the Inverse-Wishart prior, interpretable as the number of pseudo-observations the prior is worth. The value $\nu_0 = D + 2 = 6$ is the smallest integer for which the prior expected value exists, yielding the most diffuse valid prior; a larger choice (e.g., $\nu_0 = 9$) would impose stronger confidence that the covariance is close to $\mathbf{I}_D$. With roughly 150 calm and 91 panic months in training, the posterior degrees of freedom ($\nu_n = \nu_0 + n_k$) far exceed the prior, so the data dominate the covariance estimates.

- $\mathbf{\Psi}_0$ is the scale matrix of the Inverse-Wishart prior. It is calibrated so that the prior expected covariance equals the identity matrix, $\mathbb{E}[\mathbf{\Sigma}_k] = \mathbf{\Psi}_0/(\nu_0 - D - 1) = \mathbf{I}_D$, implying unit marginal variance and no cross-feature correlation before observing the data.

Transition matrix Each row of the transition matrix is assigned an independent Dirichlet prior:

$$\mathbf{P}_{0,\cdot} \sim \mathrm{Dir}(9, 1), \qquad \mathbf{P}_{1,\cdot} \sim \mathrm{Dir}(1, 9). \tag{5}$$

Equivalently, in matrix form,

$$\boldsymbol{\alpha}_{\mathrm{dir}} = \begin{pmatrix} 9 & 1 \\ 1 & 9 \end{pmatrix}.$$

This prior favors persistence in both regimes: absent data, the prior mean transition matrix is

$$\mathbb{E}[\mathbf{P}] = \begin{pmatrix} 0.90 & 0.10 \\ 0.10 & 0.90 \end{pmatrix}. \tag{6}$$

In other words, the prior expects each regime to continue with probability 0.9, corresponding to an expected duration of roughly 10 months. This prior is informative enough to discourage pathological solutions but weak enough to be overridden by the data.

3.1.4 Gibbs Sampling Algorithm

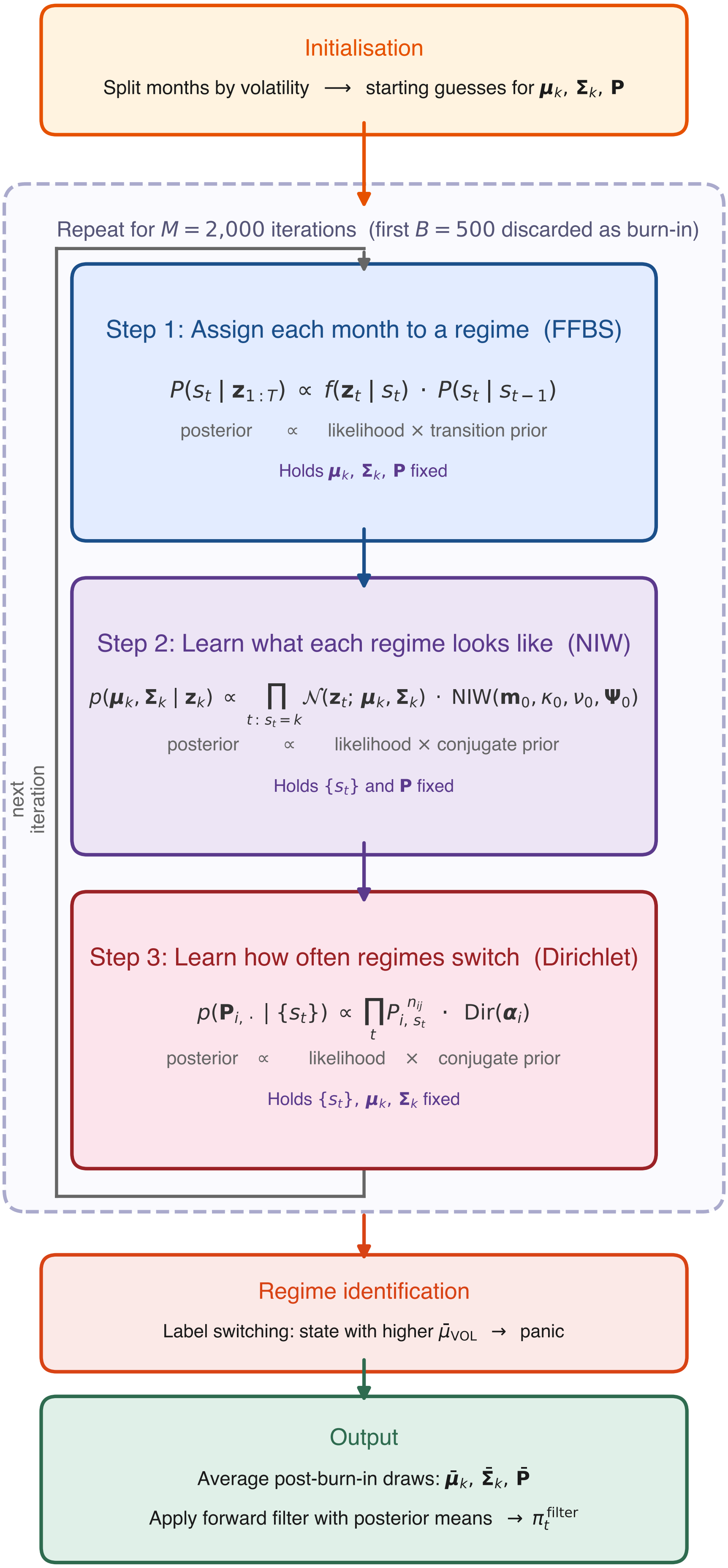
Initialization The Gibbs sampler requires starting values for the emission parameters and the transition matrix. A crude drawdown-based partition is used:[2] months with below-median drawdown DD_t^z (i.e., deeper drawdowns) are initially assigned to the panic state ($s_t = 1$), and the remaining months to the calm state ($s_t = 0$). The initial emission parameters are then computed from each group:

$$\boldsymbol{\mu}_k^{(0)} = \frac{1}{n_k} \sum_{t:\, s_t = k} \mathbf{z}_t, \qquad \boldsymbol{\Sigma}_k^{(0)} = \mathrm{Cov}(\{\mathbf{z}_t : s_t = k\}),$$

where n_k is the number of months assigned to regime k. The initial transition matrix is set to the prior mean, $\mathbf{P}^{(0)} = \mathbb{E}[\mathbf{P}]$ (Equation 6). These initial parameters give the first forward pass (described below) emission densities that can distinguish the two states.

Iterative updates Each iteration cycles through the three conditional updates described in Section 3.1.2. Figure 2 summarizes the cycle graphically.

[2]The choice of initialization has negligible impact on the posterior: the sampler is run for 2,000 iterations with 500 discarded as burn-in, and convergence diagnostics (Section 3.1.5) confirm that the chain mixes well regardless of the starting partition.


Initialisation
Split months by volatility $\longrightarrow$ starting guesses for $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$
Repeat for $M = 2{,}000$ iterations (first $B = 500$ discarded as burn-in)
Step 1: Assign each month to a regime (FFBS)
$P(s_t \mid \mathbf{z}_{1:T}) \propto f(\mathbf{z}_t \mid s_t) \cdot P(s_t \mid s_{t-1})$
posterior $\propto$ likelihood × transition prior
Holds $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$ fixed
Step 2: Learn what each regime looks like (NIW)
$p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k \mid \mathbf{z}_k) \propto \prod_{t:\, s_t = k} \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \cdot \mathrm{NIW}(\mathbf{m}_0, \kappa_0, \nu_0, \boldsymbol{\Psi}_0)$
posterior $\propto$ likelihood × conjugate prior
Holds $\{s_t\}$ and $\mathbf{P}$ fixed
Step 3: Learn how often regimes switch (Dirichlet)
$p(\mathbf{P}_{i,\cdot} \mid \{s_t\}) \propto \prod_t P_{i,\, s_t}^{n_{ij}} \cdot \mathrm{Dir}(\boldsymbol{\alpha}_i)$
posterior $\propto$ likelihood × conjugate prior
Holds $\{s_t\}$, $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$ fixed
next iteration
Regime identification
Label switching: state with higher $\bar{\mu}_{\mathrm{VOL}}$ $\rightarrow$ panic
Output
Average post-burn-in draws: $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, $\bar{\mathbf{P}}$
Apply forward filter with posterior means $\rightarrow \pi_t^{\mathrm{filter}}$

Step 1: Sampling the hidden state path via FFBS The full latent regime sequence $(s_1, \ldots, s_T)$ is sampled using Forward-Filtering Backward-Sampling (FFBS; see Frühwirth-Schnatter, 2006, ch. 11), holding the current emission parameters and transition matrix fixed.

The forward pass computes the filtered probability $\alpha_t(k) = P(s_t = k \mid \mathbf{z}_{1:t})$, the probability of being in state k at time t given all observations so far. This quantity is built in two sub-steps. First, a *prediction step* asks: before seeing this month's data, how likely is each state? Since the previous state is unknown, the algorithm sums over both possibilities, weighting each by its filtered probability from the previous month and the transition probability:

$$\alpha_{t|t-1}(k) = \alpha_{t-1}(0)\, P_{0k} + \alpha_{t-1}(1)\, P_{1k}.$$

Second, an *update step* revises this prediction once the current observation $\mathbf{z}_t$ arrives, using Bayes' rule: if the observed features look more like a panic month than a calm month, the probability of panic increases. Combining the two steps yields the recursion

$$\alpha_t(k) = \frac{f(\mathbf{z}_t \mid s_t = k)\; \alpha_{t|t-1}(k)}{f(\mathbf{z}_t \mid s_t = 0)\; \alpha_{t|t-1}(0) + f(\mathbf{z}_t \mid s_t = 1)\; \alpha_{t|t-1}(1)}, \tag{7}$$

where $f(\mathbf{z}_t \mid s_t = k) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the Gaussian emission density.[3] The recursion is initialized with $\alpha_0(k) = 1/2$, reflecting no prior preference for either regime.

The forward pass tells us how likely each regime is at every month, but if we simply flipped a weighted coin independently at each t we could produce implausible paths – for example, rapid calm-panic-calm-panic switching even though the transition matrix says regimes persist for roughly ten months. The backward pass prevents this by sampling states in reverse: once a future state has been drawn, the current state is sampled conditional on it, so the entire path respects the Markov transition dynamics.

The backward pass proceeds in two steps:

Terminal state At $t = T$, the filtered probability already conditions on all observations, so the terminal state is drawn directly:

$$s_T \sim \mathrm{Cat}(\alpha_T(0),\, \alpha_T(1)),$$

[3] In practice, these densities can be extremely small (or large) for four-dimensional observations, leading to numerical underflow. The implementation avoids this by working in log-space: log-likelihoods $\log f(\mathbf{z}_t \mid s_t = k)$ are computed directly, and the log-sum-exp identity $\log(e^a + e^b) = \max(a, b) + \log(1 + e^{-|a-b|})$ is used whenever probabilities must be normalised.

that is, the algorithm draws $s_T = 0$ (calm) with probability $\alpha_T(0)$ and $s_T = 1$ (panic) with probability $\alpha_T(1)$.[4]

Backward recursion For each earlier time step $t = T-1, T-2, \ldots, 1$, the algorithm chooses a state consistent with two pieces of information: (i) the filtered belief $\alpha_t(k)$ from the forward pass, and (ii) the already-sampled future state s_{t+1}, which constrains the current state via the transition probability $P_{k,s_{t+1}}$. Combining these two terms and normalising gives:

$$P(s_t = k \mid s_{t+1}, \mathbf{z}_{1:t}) = \frac{\alpha_t(k)\, P_{k,s_{t+1}}}{\alpha_t(0)\, P_{0,s_{t+1}} + \alpha_t(1)\, P_{1,s_{t+1}}}. \tag{8}$$

For example, if the forward filter assigns high probability to calm at time t ($\alpha_t(0)$ is large) and the already-sampled s_{t+1} is also calm (so $P_{0,0}$ is large), then the numerator for $k = 0$ dominates and the algorithm will very likely sample $s_t = 0$.

The output of the full FFBS procedure is one draw from the posterior over state paths. Across Gibbs iterations, different paths are sampled, exploring the full posterior uncertainty.

Importantly, the backward pass is used only during training to improve posterior parameter estimates. The trading signal π_t^{filter} relies solely on the causal forward filter.

Step 2: Updating the emission parameters Given the sampled state path, each regime k has n_k observations assigned to it. From these, two summary statistics are computed: the sample mean $\bar{\mathbf{x}}_k = \frac{1}{n_k}\sum_{t:\, s_t=k} \mathbf{z}_t$ and the scatter matrix $\mathbf{S}_k = \sum_{t:\, s_t=k}(\mathbf{z}_t - \bar{\mathbf{x}}_k)(\mathbf{z}_t - \bar{\mathbf{x}}_k)^\top$. The posterior hyperparameters are:

$$\begin{aligned} \kappa_n &= \kappa_0 + n_k, \qquad \mathbf{m}_n = \frac{\kappa_0 \mathbf{m}_0 + n_k \bar{\mathbf{x}}_k}{\kappa_n}, \qquad \nu_n = \nu_0 + n_k, \\ \boldsymbol{\Psi}_n &= \boldsymbol{\Psi}_0 + \mathbf{S}_k + \frac{\kappa_0 n_k}{\kappa_n}(\bar{\mathbf{x}}_k - \mathbf{m}_0)(\bar{\mathbf{x}}_k - \mathbf{m}_0)^\top. \end{aligned} \tag{9}$$

Since $\kappa_0 = 0.01 \ll n_k$, the posterior is driven almost entirely by the data.

New regime parameters are then drawn from the updated posterior: $\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_n, \boldsymbol{\Psi}_n)$ and $\boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}(\mathbf{m}_n, \boldsymbol{\Sigma}_k/\kappa_n)$.

Step 3: Updating the transition matrix The sampled path is scanned to count transitions: $n_{ij} = \sum_{t=2}^{T} \mathbb{1}(s_{t-1} = i,\ s_t = j)$ records how many times the path moved from state i to state j. For example, if the sampled path contains 140 calm-to-calm transitions and 10 calm-to-panic transitions, then $n_{00} = 140$ and $n_{01} = 10$. These observed counts

[4] Cat denotes the categorical distribution, the standard notation for a single draw from a discrete probability vector. With two states it reduces to a weighted coin flip.

are added to the Dirichlet prior pseudo-counts to obtain the posterior:

$$\mathbf{P}_{i,\cdot} \mid \{s_t\} \sim \mathrm{Dir}\big(\alpha_{i0} + n_{i0},\ \alpha_{i1} + n_{i1}\big). \tag{10}$$

For row 0 (calm), the prior pseudo-counts are $(\alpha_{00}, \alpha_{01}) = (9, 1)$ from Equation (5), so the posterior becomes $\mathrm{Dir}(9 + 140,\ 1 + 10) = \mathrm{Dir}(149,\ 11)$. The expected persistence probability under this posterior is $149/160 = 0.93$, reflecting that the data reinforced the prior belief in regime persistence. A new transition matrix row is then drawn from this Dirichlet distribution, and the same procedure is applied independently to row 1 (panic).

With all three blocks updated, the sampler returns to Step 1. After sufficient iterations the draws converge to samples from the joint posterior.

3.1.5 Implementation and Convergence

The Gibbs sampler is estimated on the training sample only. To reduce sensitivity to MCMC initialization, the full sampler is run independently with $N_s = 200$ different random seeds. Each chain is run for $M = 2{,}000$ iterations with the initialization described above. The first $B = 500$ iterations are discarded as burn-in, leaving 1,500 posterior draws per seed.

For each seed, posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ are computed by averaging over the post-burn-in draws. A causal forward filter is then applied to the full sample using these parameters to obtain a seed-specific filtered probability $\pi_t^{\text{filter},(s)}$. The final trading signal is the average across seeds:

$$\pi_t^{\text{filter}} = \frac{1}{N_s} \sum_{s=1}^{N_s} \pi_t^{\text{filter},(s)}. \tag{11}$$

This Bayesian model averaging is a methodological requirement, not a performance optimisation: with a single Gibbs chain the out-of-sample Sharpe has mean 1.04 and interquartile range 0.10 across random HMM seed draws, while averaging 50 chains raises the mean to 1.09 and collapses the IQR to 0.02 (Appendix J.2). The 1.11 figure reported in Section 5 therefore reflects the production 200-seed configuration and should be interpreted as a posterior-averaged quantity rather than a single-chain point estimate.

Convergence Diagnostics Convergence is assessed in two ways. First, trace plots of the regime means and persistence probabilities are visually inspected to verify that the chain mixes around a stable posterior region after burn-in. Second, effective sample sizes (ESS) are computed for key parameters. Because successive MCMC draws are autocorrelated, n draws from the chain contain less information than n independent samples. The ESS quantifies the number of truly independent draws the chain is equivalent to. For a post-burn-in chain of length n,

$$ESS = \frac{n}{1 + 2\sum_{\ell=1}^{L^*} \hat{\rho}(\ell)}, \tag{12}$$

where $\hat{\rho}(\ell)$ is the estimated autocorrelation at lag ℓ,[5] and L^* is the first lag at which the autocorrelation becomes negative. High ESS indicates that the chain provides many effectively independent posterior draws.[6]

While ESS measures efficiency within a single chain, it cannot detect whether different chains have converged to different modes. The Gelman–Rubin $\hat{R}$ statistic (Gelman et al., 2013) addresses this by comparing within-chain and between-chain variance across M independent chains, each of length n. Let s_m^2 denote the variance of chain m and $\bar{\theta}_m$ its mean. The within-chain variance is:

$$W = \frac{1}{M}\sum_{m=1}^{M} s_m^2, \tag{13}$$

and the between-chain variance is:

$$B = \frac{n}{M-1}\sum_{m=1}^{M}(\bar{\theta}_m - \bar{\theta})^2, \tag{14}$$

where $\bar{\theta} = \frac{1}{M}\sum_m \bar{\theta}_m$. If all chains have converged to the same posterior, B should be small relative to W. The diagnostic is:

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}, \qquad \hat{V} = \frac{n-1}{n}W + \frac{B}{n}. \tag{15}$$

Values of $\hat{R}$ near 1.0 indicate convergence; the standard threshold is $\hat{R} < 1.1$. To evaluate convergence, five of the 200 chains are selected at random. For the selected feature set (DD+DISP+REL_N+CS), all parameters have $\hat{R} < 1.01$ (Appendix J.1).

Trace plots of the regime means and persistence probabilities (Figure 11, Appendix J) provide visual confirmation that all chains converge to stable posterior values after the burn-in period.

3.1.6 Regime Identification and Trading Signal

Label Switching and Regime Naming In a two-state HMM, the labels 0 and 1 are intrinsically arbitrary: relabeling the states leaves the likelihood unchanged. To identify

[5] $\hat{\rho}(\ell) = \frac{1}{(n-\ell)\,\hat{\sigma}^2}\sum_{i=1}^{n-\ell}(\theta_i - \bar{\theta})(\theta_{i+\ell} - \bar{\theta})$, where θ_i is the i-th post-burn-in draw, $\bar{\theta}$ its sample mean, and $\hat{\sigma}^2$ its sample variance.

[6] An ESS of at least 100 per parameter is a common minimum for reliable posterior inference (Gelman et al., 2013).

which state corresponds to panic, the known crisis-period behavior of each feature is used.[7] Two features rise during crises (DISP: +1, CS: +1) and two fall (DD: −1, REL_N: −1). The regime whose posterior mean best aligns with these crisis directions is designated as "panic"; if the Gibbs sampler assigns the panic characteristics to state 0 rather than state 1, the labels are swapped so that π_t^{filter} always represents the probability of the panic state.

Two caveats follow from this naming convention. The HMM identifies two regimes by clustering on the emission features; the label "panic" is then assigned to the cluster whose posterior mean aligns with crisis directions. The label therefore reflects a statistical resemblance to historical crisis periods rather than a directly observed market state. Relatedly, the economic interpretations used later in the thesis (liquidity pressure, constrained arbitrage) are readings of what this cluster *likely* corresponds to given the features it loads on, not direct identification claims about the underlying mechanism.

Filtered Regime Probability Using the posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ from Section 3.1.5, the forward recursion (Equation 7) is applied to the full sample to obtain:

$$\pi_t^{\text{filter}} = P(s_t = \text{panic} \mid \mathbf{z}_{1:t};\ \bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}}, \bar{\mathbf{P}}). \tag{16}$$

A value near 0 indicates high confidence in calm, near 1 indicates panic, and intermediate values reflect genuine ambiguity. This filtered probability serves as the regime signal passed to the portfolio construction methods in Section 3.2.1. Figure 4 and Table 4 in Section 4 visualise the resulting signal and confirm that the two regimes are meaningfully distinct.

3.2 Stage 2: Regime-Conditioned Portfolio Construction

The filtered regime probability π_t^{filter} from Stage 1 provides a monthly measure of market stress, but on its own it does not select stocks. This stage translates the regime signal into cross-sectional portfolios. Three cross-sectional stock selection methods of increasing complexity are developed: a deterministic regime-adaptive momentum formula (Method 0), a logistic regression classifier (Method 1), and an XGBoost gradient-boosted tree regressor (Method 2). Each method combines π_t^{filter} with momentum signals at multiple horizons to produce a monthly score for every stock, which is then used to rank stocks into a long-short portfolio: the top decile (using NYSE breakpoints) forms the long leg, the bottom decile forms the short leg, and both legs are value-weighted. The stock-level inputs (momentum horizons and the regime signal) are described in Section 4.

A key distinction from standard momentum strategies is worth emphasising. In the

[7] The sign directions are determined by a data-driven procedure comparing crisis and non-crisis 95th percentiles; see Appendix I for details.

traditional approach (Jegadeesh and Titman, 1993), stocks are ranked directly by their raw trailing return at a fixed lookback, so the long leg always holds the stocks with the *highest* momentum and the short leg holds the *lowest*. Here, by contrast, stocks are ranked by their predicted forward return – a learned function of multiple momentum horizons and the regime signal. Because the model learns which momentum *level* is associated with future outperformance in a given market state, the long leg need not contain high-momentum stocks at all: in deep crisis, the model may assign the highest predicted returns to stocks whose momentum is below the cross-sectional average at every horizon (Section 5.2.1). The portfolio construction remains identical (top and bottom deciles), but the criterion for entering each decile is regime-dependent rather than fixed.

The deterministic formula serves as a transparent baseline that tests whether simply varying the momentum lookback horizon with market conditions adds value. The two machine learning methods go further by jointly learning how momentum at multiple horizons and the regime signal interact to predict forward returns. We evaluate every method in long-short construction rather than long-only, for three reasons. First, a long-short portfolio nets out market beta on the two sides, so the remaining return reflects the cross-sectional ranking itself rather than a levered market bet. Second, the regime mechanism this thesis studies is two-sided: in panic the model predicts reversal, meaning it identifies stocks expected to fall as well as stocks expected to rise, and the short leg directly tests whether those downward predictions are informative. Long-only construction would discard that test. Third, long-short is the standard construction in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015), which makes our Sharpe ratios and factor alphas directly comparable to existing results. The results that follow (Section 5) not only compare the out-of-sample performance of the three portfolios but also use SHAP values and coefficient analysis to examine *which* features drive the predictions and *how* their importance shifts across calm and panic regimes. This feature-level analysis provides economic insight into the signals that distinguish regime-conditioned stock selection from standard momentum.

3.2.1 Cross-Sectional Stock Selection Methods

Method 0: Deterministic Regime-Adaptive Formula The simplest approach exploits the economic intuition that the optimal momentum lookback horizon varies with market conditions. The lookback horizon and stock score are:

$$\ell_t = \text{clip}\big(\text{round}(12 - 11 \cdot \pi_t^{\text{filter}}),\ 1,\ 12\big)\,, \tag{17}$$

$$\text{score}_t^s = \prod_{j=1}^{\ell_t}(1 + r_{t-j}^s) - 1. \tag{18}$$

Stocks are then ranked by this score; the top decile forms the long leg and the bottom decile the short leg, as described in Appendix B.1.

When $\pi_t^{\text{filter}} \approx 0$ (calm), $\ell_t = 12$, recovering the standard 12-month momentum signal that captures continuation in trending markets (Jegadeesh and Titman, 1993). When $\pi_t^{\text{filter}} \approx 1$ (panic), $\ell_t = 1$, switching to 1-month reversal to exploit the tendency of oversold stocks to recover after panic selling (Goulding et al., 2023). At intermediate values, the lookback smoothly interpolates between these extremes.

Method 1: Logistic Regression A logistic regression classifier is trained to predict whether a stock's forward return exceeds the cross-sectional median in each month. Classification is chosen over direct return regression because monthly stock returns are extremely noisy, and predicting the binary outcome "above or below median" is a more stable learning target that still produces a useful ranking of stocks. Classification targets have been explored as an alternative to regression in cross-sectional asset pricing, though regression targets generally produce better portfolio performance (Gu et al., 2020); the Ridge regression baseline (Section 3.2.1) confirms that the choice of target does not drive the linear model's failure.[8]

$$P(y_t^s = 1 \mid \mathbf{x}_t^s) = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_t^s), \tag{19}$$

where $y_t^s = \mathbb{1}(r_{t+1}^s > \text{median}_s\{r_{t+1}^s\})$ is a binary label indicating whether stock s outperforms the cross-sectional median in the following month. The feature vector $\tilde{\mathbf{x}}_t^s = (mom_{1,t}^s, \ldots, mom_{12,t}^s, \pi_t^{\text{filter}})$ contains the 12 momentum lookbacks and the regime signal, standardized using training-set means and standard deviations. The model is estimated by maximizing the ℓ_2-regularized log-likelihood:

$$\hat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta}} \left[\sum_i \Big(y_i \log \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i) + (1 - y_i) \log(1 - \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)) \Big) - \frac{1}{2C} \|\boldsymbol{\beta}\|_2^2 \right], \tag{20}$$

with regularization strength $C = 1.0$. The ℓ_2 penalty shrinks coefficients toward zero to guard against overfitting given the large number of correlated momentum features. Once $\hat{\boldsymbol{\beta}}$ is estimated on the training sample, the stock score for any observation is obtained by plugging $\hat{\boldsymbol{\beta}}$ back into the prediction equation:

$$score_t^s = \sigma(\hat{\boldsymbol{\beta}}^\top \tilde{\mathbf{x}}_t^s) = P(y_t^s = 1 \mid \mathbf{x}_t^s; \hat{\boldsymbol{\beta}}). \tag{21}$$

Stocks with higher predicted probability of beating the cross-sectional median receive higher scores and are more likely to enter the long leg; those with the lowest scores enter the short leg.

[8] The logistic sigmoid function $\sigma(z) = 1/(1 + e^{-z})$ maps the linear combination $\boldsymbol{\beta}^\top \tilde{\mathbf{x}}$ from the real line to the interval $(0, 1)$, so the output can be interpreted as a probability.

Method 2: XGBoost Regressor An XGBoost gradient-boosted tree regressor (Chen and Guestrin, 2016) is trained to predict the forward monthly return directly. Unlike Method 1, which classifies stocks as above or below the median, XGBoost predicts the forward return directly. This preserves magnitude information: a stock expected to gain 10% ranks higher than one expected to gain 2%, a distinction that binary classification discards. The feature vector for each stock-month observation comprises the 13 features described in Section 4.4: the 12 momentum lookbacks ($mom_1, \ldots, mom_{12}$) and the regime signal (π_t^{filter}). By including all horizons jointly, XGBoost can learn how the full momentum term structure interacts with the regime signal – for example, conditioning its joint use of all 12 horizons on the level of π_t^{filter} in ways that cannot be decomposed into per-horizon slope changes.

XGBoost builds an additive ensemble of M regression trees, where each successive tree fits the residuals of the current ensemble:

$$F(\mathbf{x}) = \sum_{m=1}^{M} \eta \cdot h_m(\mathbf{x}), \tag{22}$$

where h_m is the m-th decision tree and η is the learning rate that shrinks each tree's contribution to prevent overfitting. Each tree h_m is grown to minimize a regularized objective combining the squared-error loss on the current residuals with penalty terms on tree complexity (number of leaves and leaf weights). The predicted return $\hat{r}^s_{t+1} = F(\mathbf{x}^s_t)$ serves as the stock score. The model is trained with $M = 500$ trees, a learning rate of $\eta = 0.05$, maximum tree depth of 4, and row and column subsampling rates of 0.8. To reduce sensitivity to random initialization, an ensemble of 50 independent XGBoost models (each with a different random seed) is trained; their predictions are averaged before ranking stocks into portfolio deciles.

A maximum depth of 4 allows the model to capture up to four-way feature interactions. The choice of depth is consequential: depth 1 permits no interactions between features, depth 2 permits only pairwise interactions (e.g., one split on π_t^{filter} followed by one split on a single momentum horizon), and depth ≥ 3 permits higher-order conditioning across multiple horizons simultaneously. Table 23 and Figure 9 show that performance increases monotonically from depth 1 (Sharpe 0.53) through depth 4 (Sharpe 1.11), confirming that the regime-momentum interaction requires multi-dimensional conditioning that cannot be captured by pairwise splits alone. The value of the regime signal, however, is architecture-dependent, not just nonlinearity-dependent: a depth-matched Random Forest on the same 13 features achieves only Sharpe 0.73 and demotes π_t^{filter} to 18% of total SHAP (versus 45% for XGBoost; Appendix F.3). XGBoost is chosen because its sequential residual fitting concentrates splits on whichever feature most reduces error, turning the regime signal into the model's primary gating variable rather than one feature among thirteen.

Isolating Linearity from the Training Target Methods 1 and 2 differ not only in functional form (linear vs. nonlinear) but also in their training targets: M1 predicts a binary classification label, while M2 predicts the continuous forward return. To ensure that any performance difference reflects linearity rather than the choice of target, a Ridge regression baseline is included. Ridge regression is a linear model (identical functional form to logistic regression) that predicts continuous forward returns (identical target to XGBoost), using the same 13 standardised features:

$$\hat{r}^s_{t+1} = \boldsymbol{\beta}^\top \tilde{\mathbf{x}}^s_t, \qquad \hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left[\sum_i (r_{i,t+1} - \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)^2 + \alpha \|\boldsymbol{\beta}\|_2^2 \right], \tag{23}$$

with $\alpha = 1.0$. Results are robust across regularisation strengths from $\alpha = 0.01$ to $\alpha =$ 1,000 (Appendix G.8). If Ridge regression succeeds where logistic regression fails, the classification target is the bottleneck; if both linear models fail, linearity itself is the constraint.

The Role of $\pi_t^{\textbf{filter}}$ A key distinction between the linear models (M1 and Ridge) and M2 lies in how each model uses the regime signal. A linear model assigns π_t^{filter} a single coefficient, which is near zero because the regime signal is not a direct return predictor. XGBoost, by contrast, uses π_t^{filter} as a context variable through tree splits, conditioning its joint use of all momentum horizons on the current market state. This conditioning operates through multi-dimensional feature interactions (Gu et al., 2020) and cannot be represented by a single linear coefficient or decomposed into per-feature slope changes. The empirical consequences of this distinction are analysed in Section 5.2.

With the two-stage framework in place, the next section describes the data, sample construction, and the feature selection procedure used to choose the HMM's input features.

4 Data

This section describes the data and features that implement the two-stage framework of Section 3. The HMM features are chosen so that the resulting regime signal can serve as a context variable for momentum's term structure; the cross-sectional features provide the full set of momentum horizons whose regime-dependent predictability the models must learn. The stock-level sample spans January 1990 to November 2025 and includes all ordinary common shares (CRSP share codes 10 and 11) listed on the NYSE, AMEX, or NASDAQ with a price above \$1. The \$1 price floor admits small and illiquid stocks into the universe, but NYSE breakpoints for decile classification tilt portfolio composition toward stocks large enough to be actively traded, and value-weighting within each leg further reduces the influence of the smallest names. Following Shumway (2001), delisting returns are incorporated to prevent survivorship bias: actual delisting returns are used when available, and −30% is imputed for performance-related delistings with missing returns.[9] The sample is split into a training period (1990–2010) used to estimate all models and a test period (2011–2025, 167 months) used exclusively for out-of-sample evaluation; no model parameters are re-estimated during the test period. Table 1 provides an overview.

	Train (1990–2010)	Test (2011–2025)	Full
Unique stocks	13,363	7,037	16,657
Stock-month observations	1,161,014	557,087	1,718,101
Avg. stocks per month	4,838	3,336	4,221
Market-level months	240	167	407

Table 1: Sample summary. The sample includes all ordinary common shares listed on NYSE, AMEX, or NASDAQ with price above $1.

4.1 HMM Input Features

The HMM classifies market regimes using four monthly macrofinancial stress indicators. Each captures a distinct dimension of market stress:

1. **DD (Drawdown):** The percentage decline of the cumulative market price index from its rolling 12-month peak:

$$DD_t = \frac{P_t - \max(P_{t-11}, \ldots, P_t)}{\max(P_{t-11}, \ldots, P_t)}. \tag{24}$$

   DD directly measures the depth of ongoing market declines. Daniel and Moskowitz (2016) show that momentum crashes are predictable from past market declines, making DD the natural anchor for regime identification.

2. **DISP (Return Dispersion):** The log cross-sectional standard deviation of individual stock returns:

$$DISP_t = \log(\mathrm{std}_s(r_{s,t}))\,. \tag{25}$$

   DISP captures cross-sectional *disagreement* among stocks. During stress, individual stocks move in different directions rather than together, causing dispersion to spike.

3. **REL_N (Relative Market Participation):** The log ratio of the number of actively traded stocks to its trailing 12-month average:

$$REL\_N_t = \log\left(\frac{N_t}{\frac{1}{12}\sum_{\ell=1}^{12} N_{t-\ell}}\right). \tag{26}$$

[9] Full data sources and detailed variable definitions are provided in Appendix A.

REL_N turns negative during crises when stocks delist, halt trading, or are acquired at elevated rates. It captures structural market damage that the other features do not measure.

4. **CS (Credit Spread):** The difference between Moody's BAA and AAA corporate bond yields:
$$CS_t = Y_t^{\text{BAA}} - Y_t^{\text{AAA}}. \tag{27}$$
The credit spread provides an independent signal from the corporate bond market, helping the HMM distinguish genuine stress episodes from equity-specific turbulence.

All four features are winsorised at the 1st and 99th percentiles (i.e., values below the 1st or above the 99th percentile are capped at those bounds to limit the influence of outliers) using training-sample statistics, and standardised to zero mean and unit variance.

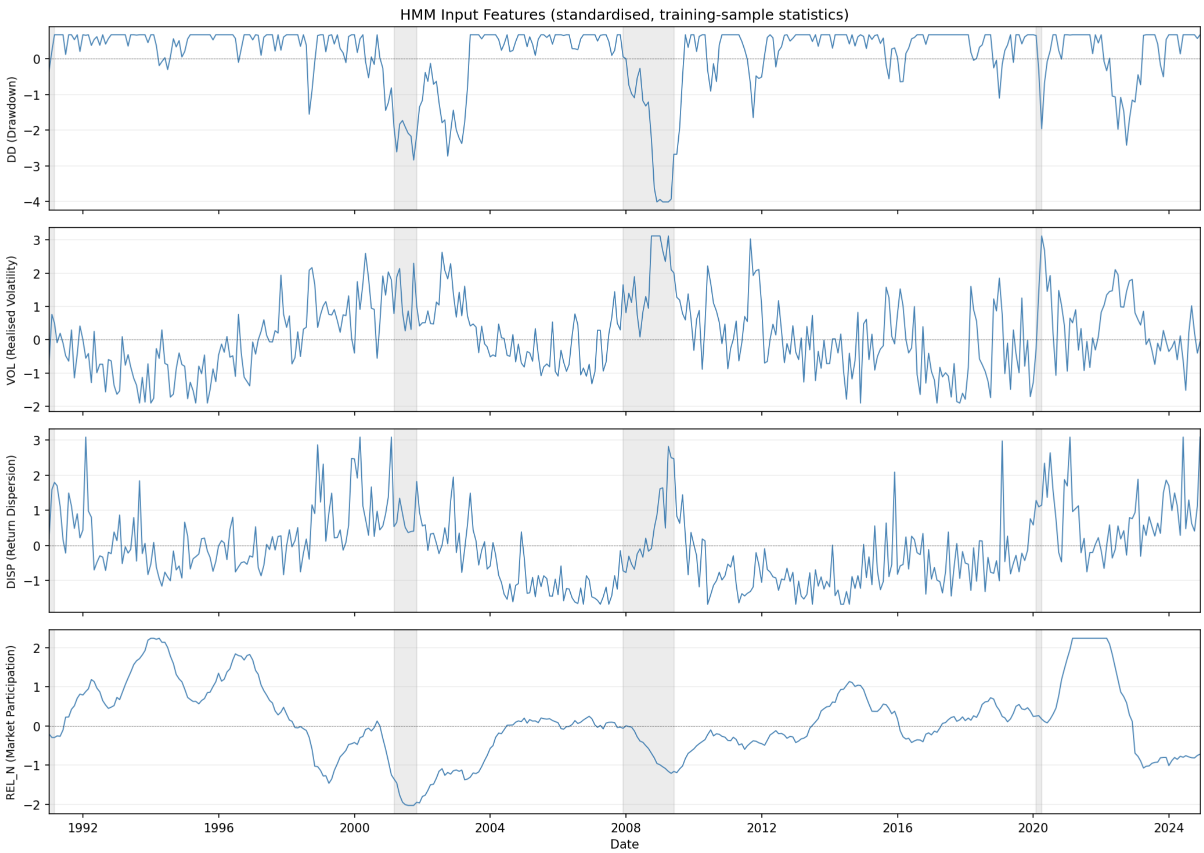


Figure 3: Time series of the four HMM input features (DD, DISP, REL_N, CS), standardised using training-sample statistics. Shaded regions indicate NBER recession periods.

4.2 Feature Selection

The regime signal's ability to capture the shift in momentum's cross-sectional structure depends critically on which stress indicators the HMM observes. The selection problem has two constraints. First, market drawdown (DD) must be included: it is the most economically direct measure of the mechanism identified by Daniel and Moskowitz (2016) – momentum crashes follow market declines – and consistently emerges as the dominant regime-identifying feature. Second, the number of features is capped at four. With 241 training months split roughly 150/91 between calm and panic regimes, each regime has sufficient observations to estimate a 4-dimensional mean vector and covariance matrix (10 free covariance parameters per state). Increasing beyond four features risks estimation instability.

Candidate pool Nine monthly macrofinancial indicators, all available from 1990 onward, are considered:

- *Equity market:* market drawdown (DD), realised volatility (VOL), cross-sectional return dispersion (DISP), relative market participation (REL_N), advance-decline ratio (ADR), and cross-sectional return skewness (SKEW).
- *Credit and macro:* BAA–AAA credit spread (CS), log VIX (LVIX), and the 10-year minus 2-year Treasury term spread (TERM).

All candidates are observable in real time at the end of each month and standardised to z-scores using training-period statistics.

Selection procedure The selection proceeds through three passes of increasing rigour. In the first pass (*quality screen*), all DD-inclusive combinations of up to four features are considered. With 8 remaining candidates, this gives $\binom{8}{0} + \binom{8}{1} + \binom{8}{2} + \binom{8}{3} = 1 + 8 + 28 + 56 = 93$ combinations (DD alone, DD paired with one other, DD with two others, DD with three others). Each is estimated with a single Gibbs sampler seed, and combinations are eliminated if the HMM fails basic sanity checks: the two states must separate meaningfully, and the model must correctly classify known crises (the dot-com bust, the GFC, and COVID) as panic while keeping non-crisis periods calm.[10]

In the second pass (*portfolio ranking*), surviving combinations are evaluated through the full cross-sectional pipeline: HMM to XGBoost to long-short portfolio. This answers the question that matters most – does a better regime signal translate into better stock selection? Combinations are ranked by out-of-sample Sharpe ratio. The top 10 advance.

In the third pass (*stability validation*), the finalists are tested for robustness across random seed partitions. A combination that achieves a high Sharpe on one lucky seed combination but collapses on another is unreliable. We run 20 independent experiments with different seed draws and classify combinations as stable ($\sigma < 0.10$), moderate ($0.10 \leq \sigma < 0.15$), or unstable ($\sigma \geq 0.15$).

[10] Specific thresholds: ESS > 50, regime separation significant for at least $\min(2, D)$ features, average $\pi_t^{\text{filter}} > 0.5$ during GFC and COVID, < 0.4 during 2003–2006 and 2013–2019, > 0.3 during the dot-com crash.

Selected features The four-feature combination DD+DISP+REL_N+CS ranks first on both ensemble Sharpe and stability mean (Table 2). The Sharpe ratios in the table are based on the selection pipeline's seed budget (10 HMM seeds, 20 XGBoost seeds); the production configuration (200 HMM seeds, 50 XGBoost seeds) raises the Sharpe to 1.11 ($\sigma = 0.08$; Section 5) because additional seed averaging reduces MCMC and model noise.

D	Combination	Sharpe	Mean	σ
4	DD+DISP+REL_N+CS	0.91	0.84	0.11
4	DD+DISP+CS+TERM	0.82	0.55	0.15
4	DD+DISP+REL_N+SKEW	0.80	0.69	0.08
4	DD+VOL+DISP+REL_N	0.80	0.77	0.09
3	DD+DISP+REL_N	0.77	0.77	0.10
4	DD+VOL+DISP+TERM	0.71	0.67	0.16
3	DD+VOL+CS	0.69	0.43	0.13
1	DD	0.68	0.67	0.14
4	DD+DISP+REL_N+TERM	0.62	0.45	0.18
3	DD+ADR+TERM	0.57	0.69	0.07

Table 2: Top 10 feature combinations from the three-pass selection pipeline (long-short, 10 bps costs). D = number of HMM features. Sharpe = ensemble Sharpe ratio (10 HMM seeds, 20 XGBoost seeds). Mean and σ are the mean and standard deviation of the Sharpe ratio across 5 independent experiments with different seed partitions, measuring robustness to random initialisation. DD+DISP+REL_N+CS ranks first on both ensemble Sharpe and stability mean.

Table 3 confirms the individual contribution of each feature. DD alone achieves the highest standalone Sharpe, but removing DD or REL_N from the full model causes the largest drops (-0.46 and -0.54). DISP and CS contribute marginally in isolation but reduce the Sharpe ratio's standard deviation across seed partitions from 0.12 (DD alone) to 0.08 (full model), providing independent information from cross-sectional disagreement and corporate credit conditions that stabilises the regime classification.

HMM Input Features	M2 (XGBoost) Sharpe	Δ vs Full
Full 4F baseline (DD+DISP+REL_N+CS)	0.97	—
Include-one (single-feature HMM)		
DD only (market drawdown)	0.66	−0.31
Leave-one-out (three-feature HMM)		
Drop DD	0.51	−0.46
Drop DISP	0.96	−0.01
Drop REL_N	0.43	−0.54
Drop CS	0.96	−0.01

Table 3: HMM input feature ablation (long-short, feature-selection seed counts). The absolute Sharpe levels differ from the production model (Table 5) due to fewer seeds; the relative comparisons (Δ) are the relevant quantities. DD alone nearly matches the full model. Removing DD or REL_N causes the largest drops (−0.46 and −0.54), while DISP and CS contribute marginally but improve seed stability (Section 4.2).

A note on feature selection and out-of-sample discipline The second pass of the feature selection procedure evaluates candidate feature sets using out-of-sample portfolio performance, which means the selected combination (DD+DISP+REL_N+CS) partly reflects its test-period Sharpe ratio. Although no model parameters are re-estimated on the test set, the model *specification* – which features to include – is informed by test-period outcomes. This introduces a mild form of specification search. Three factors mitigate this concern. First, the quality screen in Pass 1 eliminates most combinations on purely in-sample criteria (regime separation, crisis classification) before any test-period evaluation. Second, the final selection in Pass 3 is based on cross-seed stability, not peak Sharpe: DD+DISP+REL_N+CS is chosen because it has the lowest standard deviation across seed partitions, not because it has the highest mean Sharpe. Third, and most importantly, the result is not unique to the selected combination. Table 2 shows that all feature counts from $D = 1$ to $D = 4$ produce positive long-short Sharpe ratios (0.57 to 0.91), and Table 3 confirms that even single-feature specifications (DD alone, Sharpe 0.68) and leave-one-out variants produce qualitatively similar results. The alternative train/test splits in Appendix G.1 further demonstrate that the regime-momentum interaction holds across different sample periods. Nevertheless, the reported Sharpe ratio of 1.11 should be interpreted with the caveat that the HMM feature set was selected with reference to test-period performance.

4.3 Regime Estimation Results

The convergence diagnostics described in Section 3.1.5 confirm that the Gibbs sampler with the selected features has converged. All posterior parameters exceed an ESS of 100, meaning the post-burn-in draws contain at least 100 effectively independent samples per parameter. The Gelman–Rubin $\hat{R}$ statistic is below 1.01 for every parameter across five independent chains (Appendix J.1), indicating that chains started from different initializations have converged to the same posterior distribution.

Table 4 and Figure 4 summarize the HMM output. All four features exhibit statistically significant separation between calm and panic regimes (95% credible intervals excluding zero), and the resulting signal correctly identifies all major stress episodes while remaining near zero during calm expansions.

	$\hat{\mu}_{\text{calm}}$	$\hat{\mu}_{\text{panic}}$	Δ	95% CI
DD_z	0.531	-0.976	-1.507	$[-1.794, -1.215]$
DISP_z	-0.423	$+0.689$	$+1.113$	$[+0.869, +1.353]$
REL_N_z	0.525	-0.895	-1.419	$[-1.591, -1.231]$
CS_z	-0.697	$+0.203$	$+0.900$	$[+0.626, +1.159]$

Table 4: HMM posterior regime-mean separation.

Notes: This table reports posterior mean estimates of regime-conditional feature means from the Bayesian two-state HMM, estimated via Gibbs sampling on the 1990–2010 training period. All features are standardised to zero mean and unit variance. $\Delta = \hat{\mu}_{\text{panic}} - \hat{\mu}_{\text{calm}}$ is the separation between regimes. The 95% CI is the Bayesian posterior credible interval for Δ; zero lies outside all intervals, confirming significant regime separation. DD = market drawdown; DISP = log cross-sectional return dispersion; REL_N = relative market participation; CS = credit spread (BAA−AAA).

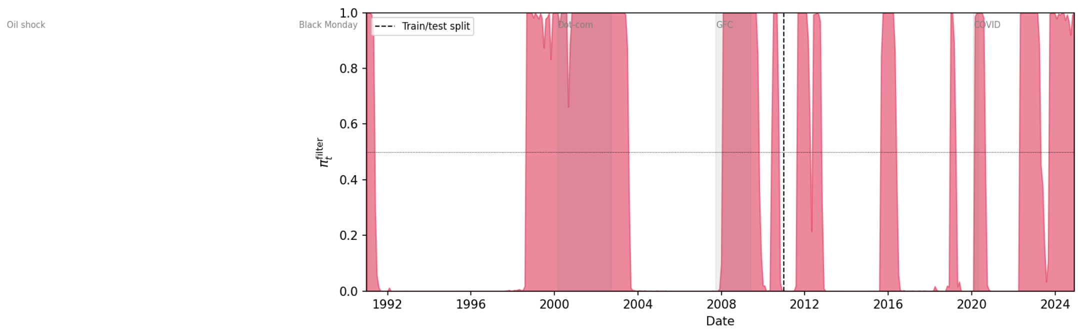


Figure 4: Filtered regime probability π_t^{filter} (1990–2025). The dashed vertical line marks the train/test split (January 2011). Shaded regions indicate months where $\pi_t^{\text{filter}} \geq 0.5$ (panic classification).

4.4 Cross-Sectional Features

The cross-sectional models rank stocks each month using momentum signals and the filtered panic probability π_t^{filter}.

Momentum lookbacks Twelve trailing momentum signals are computed for each stock, corresponding to lookback horizons of 1 to 12 months:

$$\text{mom}^s_{k,t} = \prod_{\ell=1}^{k}(1 + r^s_{t-\ell}) - 1, \qquad k = 1, \ldots, 12, \tag{28}$$

where $r^s_{t-\ell}$ is the adjusted return of stock s in month $t - \ell$. The current month's return is excluded from all lookbacks to avoid mechanical correlation with the forward return being predicted. Including all twelve horizons rather than a single lookback allows the models to learn how the full momentum term structure interacts with the regime signal.

Regime signal The filtered panic probability π^{filter}_t from the HMM enters as a single feature. Because it varies only at the monthly market level (every stock in the same month receives the same value), it functions as a conditioning variable rather than a stock-level predictor.

With the data and features defined, the next section presents the out-of-sample results of the three portfolio construction methods.

5 Results

All strategies are evaluated out-of-sample on 2011–2025 (167 months), using the long-short construction described in Section 3.2. The central question is whether momentum's cross-sectional structure is regime-dependent, and whether exploiting this dependence requires nonlinear modelling.

5.1 Portfolio Performance

5.1.1 Strategy Comparison

Table 5 reports the out-of-sample performance of all strategies.[11]

Momentum's cross-sectional predictability is regime-dependent, and exploiting this dependence requires both a probabilistic regime signal and a nonlinear model. XGBoost combined with the HMM filtered probability (M2, mom+π) achieves an annualised Sharpe of 1.11 with a six-factor alpha of 24.1% ($t = 4.81$; Appendix G.4). Block bootstrap paired tests reject equality of Sharpe ratios between M2 and each benchmark in Table 5 at

[11] All Sharpe ratios are computed using raw returns rather than excess returns above the risk-free rate. Since the same convention is applied to every strategy, relative comparisons are unaffected. The risk-free rate averaged roughly 2% annualised over the 2011–2025 window, so absolute Sharpe ratios are overstated by approximately 0.05–0.10. Sharpe is computed from the arithmetic monthly mean annualised by $\sqrt{12}$, while annualised returns reported elsewhere are compound (geometric); under high volatility the two definitions of average return differ, so Sharpe should not be inferred from a table by dividing annualised return by annualised volatility (e.g. at depth 1, the reported Sharpe is 0.53 but 8.7%/19.4% = 0.45).

	Ann. Ret	Ann. Vol	Sharpe	Max DD	β	NW t	Final $
Market	11.9%	14.6%	0.84	-24.7%	1.00	–	4.8
Fixed 12-mo mom	-1.9%	26.1%	0.06	-72.2%	-0.66	0.24	0.8
Fixed 1-mo mom	3.1%	17.6%	0.26	-30.2%	-0.34	1.08	1.5
M0: Formula	-0.2%	23.6%	0.11	-63.8%	-0.59	0.43	1.0
M1: LR	-1.3%	15.2%	-0.01	-42.4%	0.05	-0.04	0.8
M2: XGB (mom+π)	21.7%	19.5%	1.11	-22.8%	0.45	4.37***	15.3

Table 5: Out-of-sample portfolio performance.

Notes: Long-short (top/bottom decile, NYSE breakpoints), value-weighted, net of 10 bps one-way costs. β is the market beta. NW t uses Newey–West standard errors (6 lags). Test period: January 2011 to November 2025 (167 months). $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

$p < 0.025$; M2's own 95% Sharpe confidence interval is $[0.67, 1.53]$, comfortably above zero, while every benchmark's interval straddles zero (Appendix G.5). An expanding-window backtest extending the OOS sample to 30 years (1995–2024) reproduces this Sharpe within sampling noise on the 167-month overlap with the main test period (0.88; see Section 5.4.3, which also covers the dot-com bust and GFC out-of-sample). Every other specification fails to produce economically meaningful long-short performance, and the pattern of failures is informative.

Unconditional 12-month momentum achieves a Sharpe of 0.06 with a maximum drawdown of -72.2%, exposing the crash vulnerability documented by Daniel and Moskowitz (2016) and Barroso and Santa-Clara (2015). A deterministic regime switch (M0) improves this marginally to 0.11. The logistic regression (M1), given identical features to M2, collapses to -0.01; a Ridge regression with the same 13 features but a continuous-return target (matching XGBoost's) fails more decisively, with Sharpe -0.57 and a coefficient on π_t^{filter} of $+0.003$ (Appendix G.8). The failure is therefore linearity itself, not the classification target. Hand-engineering the missing nonlinearity into M1 through π^2 and $\pi \times mom_k$ terms lifts it to 0.32, establishing that the regime-momentum interaction is real while also showing that pre-specified interactions recover less than a third of what XGBoost finds automatically. A fund-augmented variant (M2: mom+π+fund) adds ten firm-level fundamentals to M2's feature set; it reduces the Sharpe to 0.89 while lowering beta from 0.45 to 0.07 and tightening the maximum drawdown from -22.8% to -18.3%. Section 5.1.3 analyses this variant alongside several feature-set ablations, which together provide additional evidence that π_t^{filter}'s role is specifically to condition the momentum term structure.

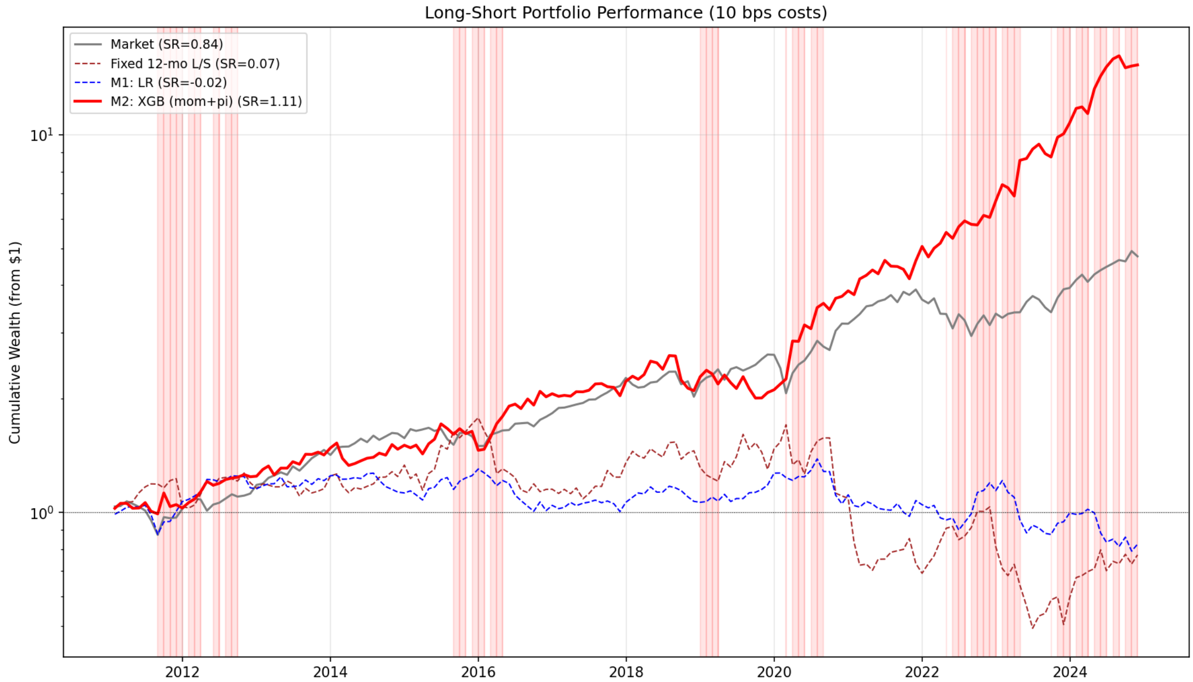


Figure 5: Cumulative wealth paths (log scale) starting from \$1 in January 2011. Shaded bands indicate panic months ($\pi_t^{\text{filter}} \geq 0.5$).

5.1.2 Regime Signal Ablation

The regime signal is essential. Removing π_t^{filter} and retraining on the 12 momentum features alone drops the Sharpe to 0.43 (Table 6). Replacing π_t^{filter} with the four raw stress indicators (DD, DISP, REL_N, CS) performs even worse (Sharpe 0.26): with four additional dimensions the model must reconstruct the regime structure at each split, while the HMM distils those indicators into a single dimension that one tree split can condition on. The HMM's role is not to provide a convenient feature but to solve regime identification *before* stock selection, freeing the cross-sectional model to use the regime signal rather than construct it. If the model cannot extract a meaningful regime signal even when given the four raw indicators directly, it certainly cannot do so from momentum features alone, which are one further step removed from the underlying regime dynamics.

Regime Signal Variant	Ann. Ret	Ann. Vol	Sharpe	Max DD
XGB + π_t^{filter} (HMM)	21.7%	19.5%	1.107	−22.8%
XGB + raw indicators (DD, DISP, REL_N, CS)	3.2%	19.1%	0.257	−45.4%
XGB (no regime signal)	7.0%	20.4%	0.429	−41.0%

Table 6: Regime signal ablation (50-seed ensemble). The HMM signal outperforms both raw stress indicators and no regime signal.

5.1.3 Feature-Set Ablation

The regime signal ablation shows that removing π_t^{filter} reduces Sharpe by more than 60%. A complementary question is whether adding widely used firm-level fundamentals strengthens the strategy, or whether the regime-momentum interaction is load-bearing on its own. This subsection tests both directions by adding, substituting, and removing feature blocks from the production configuration.

Ten firm-level fundamentals are tested: seven standard value-quality-size characteristics (book-to-market, return on equity, leverage, asset growth, gross profitability, earnings growth, and log market equity) and three cash-flow-based earnings-quality measures (operating cash flow over assets, free cash flow over assets, and the accrual component of earnings scaled by assets). All are computed from Compustat quarterly data with a four-month reporting lag to ensure real-time availability; definitions are given in Appendix L. Each variant is trained at production settings (50 XGBoost seeds, depth 4, learning rate 0.05) and evaluated identically to M2.

	Feat.	Sharpe	Calm	Panic	β	NW t (exc.)	Fund SHAP
Baseline (mom + π)	13	1.11	0.84	1.53	0.45	1.83*	0%
+ 10 fundamentals	23	0.89	0.69	1.18	0.07	0.61	44%
Fundamentals only	10	0.59	0.86	0.38	0.19	−0.49	100%
Fundamentals + π	11	0.64	0.59	0.72	0.03	−0.52	60%
Momentum + fund. (no π)	22	0.63	0.68	0.57	0.16	−0.29	54%

Table 7: Feature-set ablation (50-seed XGBoost ensembles, production hyperparameters). Full Sharpe, calm-period Sharpe, and panic-period Sharpe are reported alongside market beta and the Newey-West t-statistic on excess returns over the market (6 lags), matching the convention of Table 5. "Fund SHAP" is the share of total SHAP attention absorbed by the ten fundamental features.

Adding the ten fundamentals to the baseline reduces Sharpe from 1.11 to 0.89 (Table 7). The new features absorb 44% of total SHAP attention, redirecting it away from the momentum term structure (54% to 27%) and the regime signal (46% to 29%) without net improvement in portfolio performance. Beta falls from 0.45 to 0.07 and maximum drawdown tightens from −22.8% to −18.3%; the strategy's excess-return Newey-West t-statistic falls from 4.37^{***} to 3.56^{***}, still highly significant but materially weaker. The fund-augmented variant therefore behaves more like a market-neutral low-beta strategy than a high-alpha regime-conditioned one.

The fund-only variant (ten fundamentals, no momentum, no regime signal) identifies the source of the dilution directly. Its calm Sharpe of 0.86 is essentially equal to the baseline's 0.84, but its panic Sharpe of 0.38 is a fraction of the baseline's 1.53. Firm-level characteristics describe stable attributes (leverage, profitability, valuation) that do not flip direction when the market enters panic. Adding the regime signal to fundamentals (fund +

π, panic Sharpe 0.72) roughly doubles the fund-only panic Sharpe but remains well short of the full model, because without the momentum term structure there is no directional relationship for the regime signal to reorganise. Removing π_t^{filter} from the momentum + fundamentals variant drops Sharpe to 0.63 (panic 0.57), close to fund-only's 0.59. Across both ablations, removing either the momentum term structure or the regime signal causes the same collapse: fundamentals cannot substitute for the regime-momentum interaction, which is the load-bearing mechanism.

These ablations support the interpretation that π_t^{filter} is a context variable for the momentum term structure specifically, rather than a general-purpose conditioning input. Characteristics that describe stable firm quality are useful in calm markets, where the fund-only calm Sharpe matches the baseline's. But those same characteristics carry no information about regime shifts, and adding them to the joint model dilutes the signal that does. A regime-conditional feature-set design, in which the model uses different inputs in calm and panic, is discussed as future work in Section 6.3.

5.2 The Regime-Momentum Interaction

M2 uses momentum differently across regimes (Table 8): Sharpe 1.53 in panic against 0.84 in calm. The regime signal adds the most value precisely when the model switches from continuation to reversal. Where exposure-scaling methods reduce positions during stress to limit losses, M2 leans into panic by reconstructing the portfolio to exploit the reversal (Section 5.4.4). This composition-side mechanism leaves a measurable signature on the legs' market betas: M2's long-leg β rises from 1.40 in calm to 1.65 in panic while the short-leg β falls from 1.22 to 0.98 (Appendix G.6). The cross-leg spread $\beta_L - \beta_S$ widens from 0.18 to 0.67 but never inverts – both legs remain positive-β in panic, in contrast to the leg-beta inversion that drives traditional momentum's crashes (Daniel and Moskowitz, 2016).

	Full	Calm	Panic
Market	0.84	0.64	1.13
Fixed 12-mo mom	0.06	0.08	0.05
Fixed 1-mo mom	0.26	0.52	−0.01
M0: Formula	0.11	0.18	−0.01
M1: LR	−0.01	−0.29	0.39
M2: XGB (mom+π)	1.11	0.84	1.53

Table 8: Regime-conditional Sharpe ratios (108 calm months, 59 panic months).

The regime signal is structurally embedded in the ensemble rather than applied selectively. Across the 25,000 trees of the 50-seed production ensemble, 18,463 (73.9%) contain at least one split on π_t^{filter}, and a typical long-leg stock's path through the ensemble passes through a π-split 55% of the time: regime conditioning is built into roughly three-quarters of the model's decision rules and consulted in the majority of individual stock evaluations. The economically informative statistic is how this reliance is state-dependent. In panic months, 65% of the long leg's total predicted return flows from π-splitting trees; in calm months, only 17% does. The model leans on the regime signal predominantly during stress, and its calm-period predictions come mostly from trees that operate on momentum alone. This pattern is consistent with the Sharpe decomposition in Table 8: the regime signal adds the most value in exactly the months where it carries the most weight.

M2's predictions are decomposed using two complementary tools. SHAP values (Lundberg and Lee, 2017) measure how much each horizon contributes to the model's prediction (definitions in Appendix C.1). Cross-sectional z-scores measure where in the momentum distribution the model's selected stocks fall: for each stock in the long or short leg, the z-score at horizon k is $(mom_k^s - \bar{mom}_k)/\sigma_k$, computed from the cross-sectional mean and standard deviation of all stocks that month. Figure 6 reports these measures averaged across all selected stocks and all months in each regime.

Aggregate feature importance is split almost evenly: π_t^{filter} accounts for 46% of total SHAP, the 12 momentum horizons together for 54% (Table 9). Between regimes, momentum's share rises from 50% in calm to 58% in panic while π_t^{filter}'s share falls from 50% to 42%. The model leans on momentum more heavily during stress, but uses it in a fundamentally different way.

Feature	Overall	Calm	Panic
Momentum (12 horizons)	54%	51%	59%
π^{filter}	46%	49%	41%

Table 9: SHAP feature importance shares for M2. Each column sums to 100%.

Figure 6 decomposes these aggregates by horizon, regime, and portfolio leg. The top panels report cross-sectional z-scores (*what* the model selects); the bottom panels report each horizon's SHAP share (*what drives* the selection). The two measures answer different questions and peak at different horizons: in calm, z-scores peak at month 8 while SHAP peaks at months 11–12.

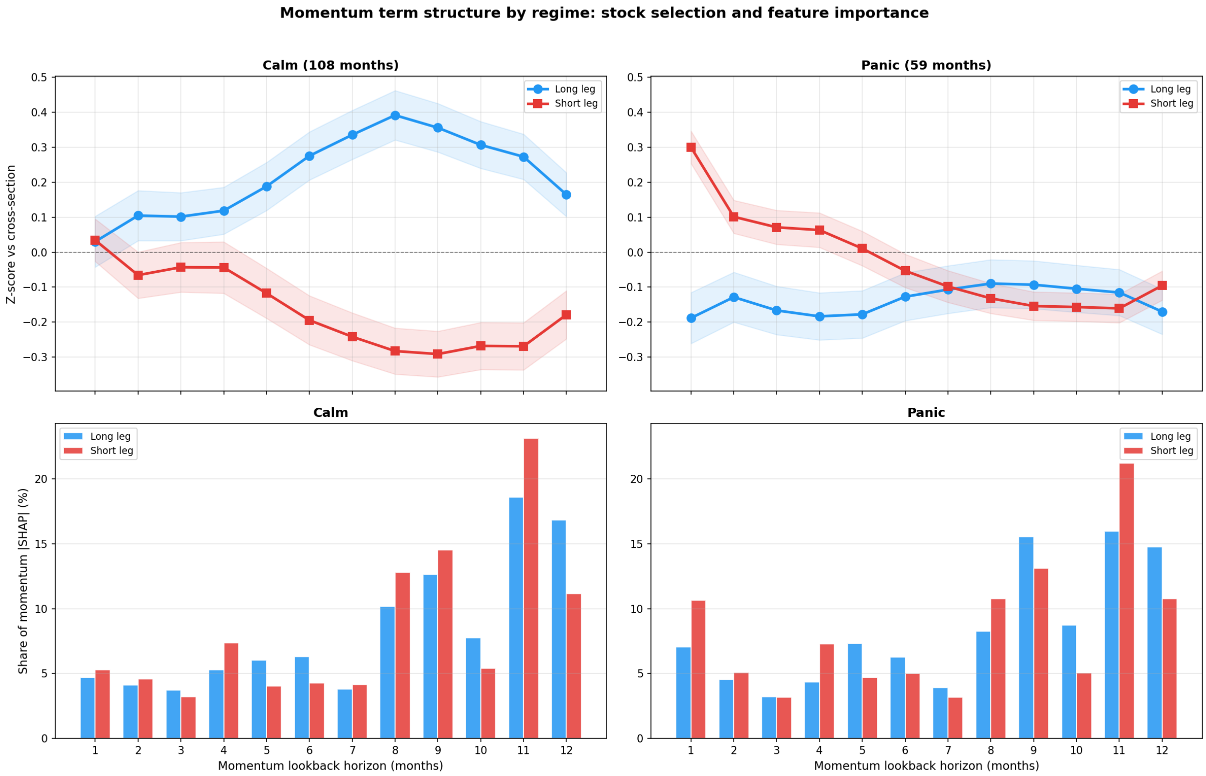


Figure 6: Momentum term structure by regime. Calm (left) and panic (right), split by long and short legs.

5.2.1 Cluster Decomposition of Test-period Picks

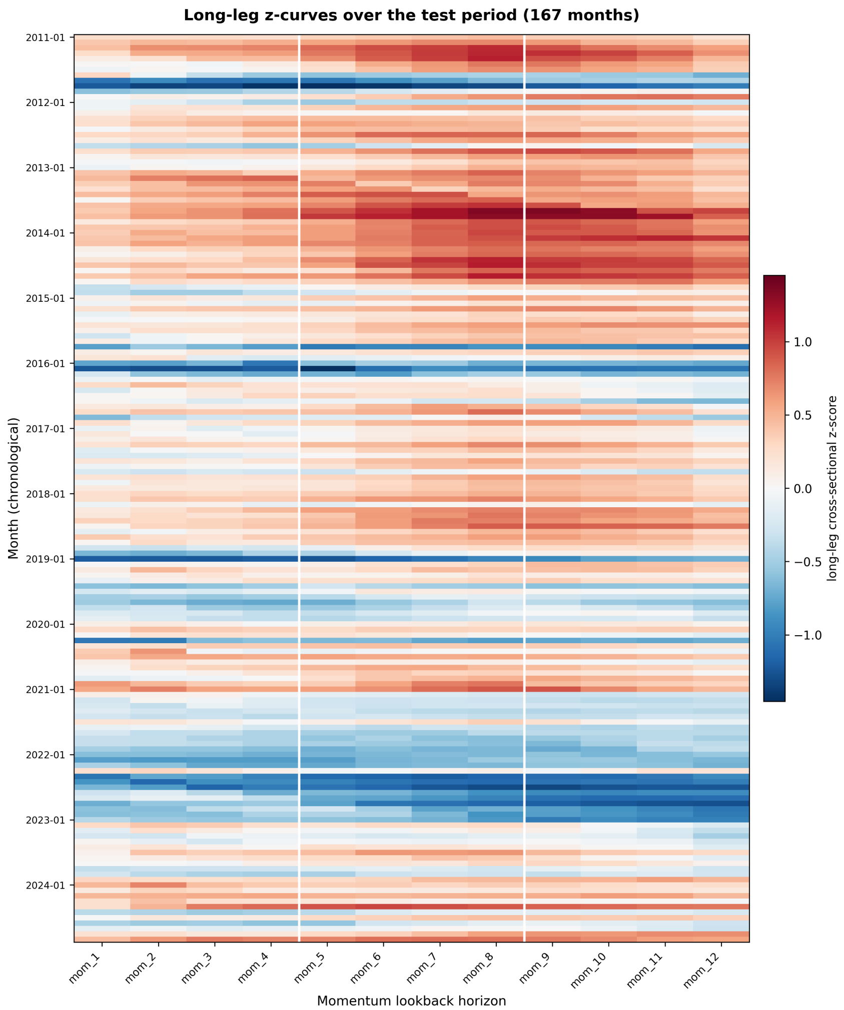


Figure 7: Long-leg cross-sectional z-scores by month and momentum horizon, 2011–2024. Each row is one of 167 test months sorted chronologically; each column is one of the 12 momentum horizons. Colour shows the deviation of the long leg's mean momentum at horizon h from the cross-section average that month. The raw landscape contains structure but is hard to read directly; the K-means decomposition below imposes structure to make the patterns visible.

The most fundamental claim of this section is that the regime signal π_t^{filter} is structurally necessary for M2's selection mechanism. Section 5.2 establishes this at the regime aggregate: π_t^{filter} accounts for 46% of total SHAP, and the leg-beta inversion that drives traditional momentum's crashes (Daniel and Moskowitz, 2016) is absent under M2 but present under unconditional 12-month momentum. The cluster decomposition we develop here sharpens that claim from "π_t^{filter} matters on average" to "π_t^{filter} routes the model between structurally distinct selection modes that no fixed-rank or single-regime model could combine in one ranking". To see this we cluster each test month's long-leg 12-horizon z-profile directly. Figure 7 shows the raw landscape: 167 monthly z-curves stacked vertically, with structure visible to the eye but no clean decomposition. K-means with $k = 4$ on the raw z-vectors (Euclidean distance, no preprocessing or regime pre-split) yields a partition that is highly stable: the same labels emerge across six independent random seeds for 163 of 167 months (97.6%), with mean pairwise Adjusted Rand Index 0.96. The four clusters nest perfectly inside a coarser two-fold partition. Clusters 0 and 1 contain all 115 months in which the long leg's z-curve sits above the cross-section average ("above-average baskets", $\bar{\pi} = 0.28$). Clusters 2 and 3 contain all 52 months in which the long leg sits below ("below-average baskets", $\bar{\pi} = 0.52$). The partition therefore subsumes the calm-vs-panic dichotomy of the regime aggregate while exposing within-regime structure that the binary split obscures, and the per-cluster evidence in Section 5.2.2 shows that the cluster-level differences in selection mode cannot be reproduced without π_t^{filter}.

Headline two-fold finding The above-average state earns a Sharpe of 0.93 (95% block-bootstrap CI $[0.24, 1.45]$) over its 115 months. The below-average state earns 1.46 ($[0.70, 2.38]$) over its 52 months. Mean monthly returns are 1.42% and 2.64% respectively, a ratio of 1.86×. A block-bootstrap permutation test on the Sharpe difference yields $p = 0.17$ (one-sided), suggestive but not significant at conventional levels at this sample size. The point-estimate ordering is consistent with reversal-mode selection being the source of the strategy's edge.

Four sub-clusters Decomposing the two-fold partition into four reveals additional structure within each tier (Figure 8, Table 10). Per-cluster sample sizes, regime composition, Sharpe + 95% CI, hit rate, Sortino, and maximum drawdown are reported in Table 10; the full per-cluster feature signature (cross-section context, raw long-leg pick momentum at each horizon, and long-leg z-curve at each horizon) is in Appendix Table 32. The narrative below emphasises what each cluster *is* (regime context, pick character, when in the test period); the detailed metrics live in the tables.

Cluster	n	$\bar{\pi}$	Sharpe [95% CI]	Hit rate	Sortino	Max DD
0 (calm continuation)	53	0.21	0.92 [0.29, 1.63]	62.3%	1.34	-13.6%
1 (mild continuation)	62	0.34	0.93 [0.05, 1.44]	58.1%	2.29	-14.1%
2 (post-panic recovery)	31	0.32	1.21 [0.18, 2.37]	64.5%	3.40	-15.3%
3 (deep crisis)	21	0.81	1.82 [0.98, 2.68]	61.9%	3.89	-10.7%

Table 10: K=4 cluster descriptors, 2011–2024 test period, $n_{\text{total}} = 167$ months. Sharpes are annualised and computed via block bootstrap with block size 6, 5,000 reps. Hit rate is the fraction of cluster months with positive M2 return. Maximum drawdown is the peak-to-trough drawdown within the cluster's chronological subseries. Sortino uses downside std (negative-return std) in the denominator.

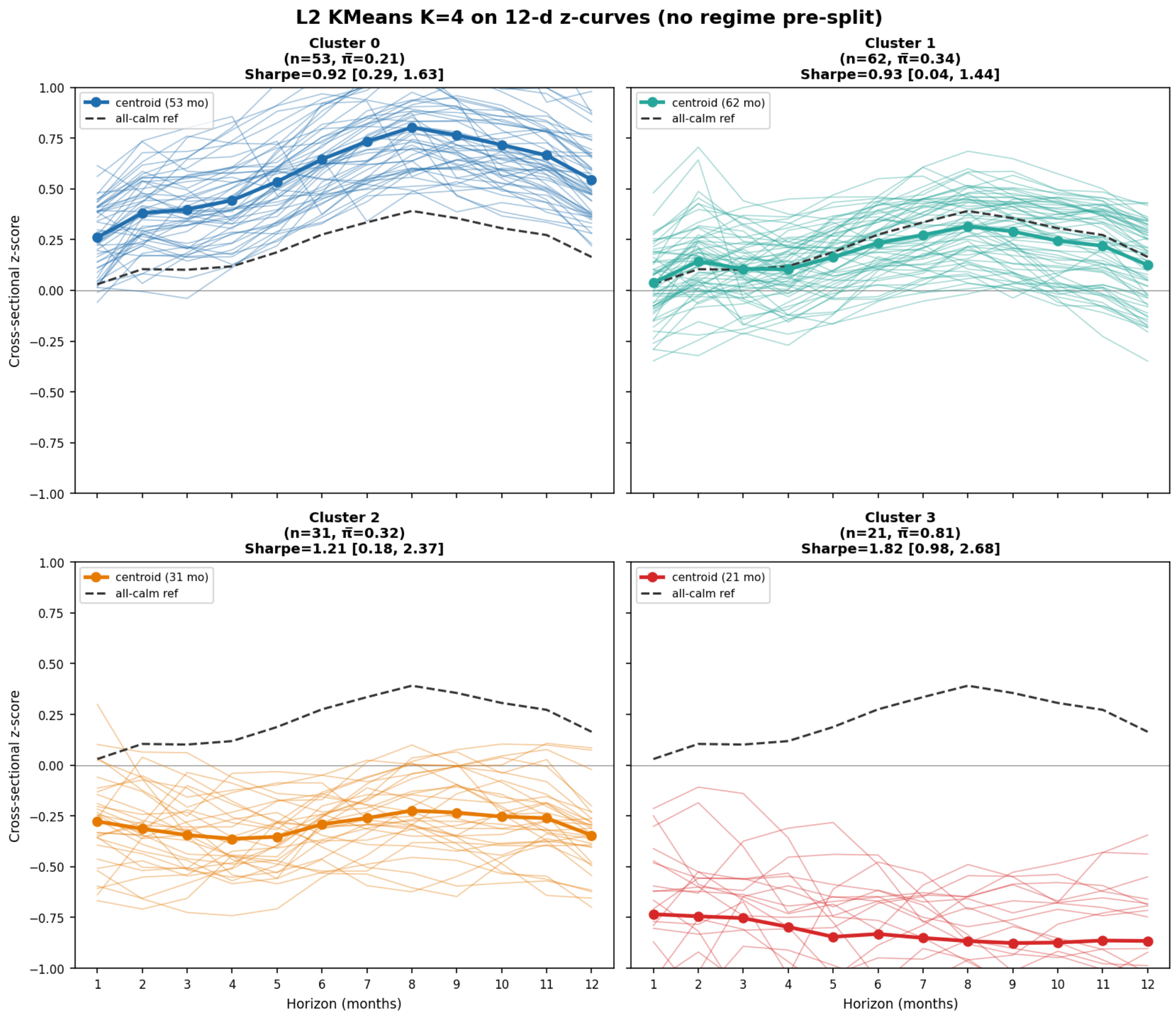


Figure 8: $K = 4$ cluster decomposition of long-leg z-curves over the 167 test months. Each panel shows one cluster: thin lines are member-month z-curves, the bold line is the cluster centroid, and the dashed line is the all-calm reference. Panels are ordered by descending overall z-level (cluster 0 leftmost, cluster 3 rightmost).

Cluster 0 (calm continuation, Sharpe 0.92**)** The strategy's most common state, occurring in long stretches of up to 21 consecutive months with peak frequency in 2013–2014. The long-leg z-curve is humped and uniformly positive, peaking around month 8; raw pick momentum rises monotonically from low single digits at month 1 to roughly +60% at month 9 – the model selects classic momentum winners. The 12-month lagged strategy Sharpe entering these months is 1.22, the highest of any cluster: cluster 0 captures the persistence of an existing momentum environment.

Cluster 1 (mild continuation, Sharpe 0.93**)** The largest cluster, with mixed regime composition and the weakest feature signature. The long leg sits only mildly above the cross-section, and the cluster has the highest long-horizon cross-section skewness (8.1): a few extreme outperformers in the right tail pull the cross-section mean. The best monthly return is March 2020 at +26% – the strategy's largest single-month gain in the test sample – and the cluster's median monthly return sits below its mean, indicating positive return skew. Sharpe is statistically indistinguishable from cluster 0 (point-estimate difference +0.01, two-sided permutation $p = 0.99$).

Cluster 2 (post-panic recovery, Sharpe 1.21**)** Months that follow a panic-heavy year: the 12-month lagged π_t^{filter} frequency is the highest of any cluster. The cross-section is positive but with the highest dispersion of any cluster, and the long leg sits below the cross-section average – raw picks are mild laggards while the broader cross-section is up double digits. This is the model's mild-reversal mode in a healing market: it picks the names that have not yet recovered, anticipating that they will. Heavy in 2021 (9 of the 31 months – the post-COVID rotation) and 2023.

Cluster 3 (deep crisis, Sharpe 1.82**, CI** $[0.98, 2.68]$**)** Active HMM panic with the cross-section uniformly negative across every horizon. The long leg is deeply below the cross-section, and raw pick losses span double digits at short horizons to nearly -50% at long horizons: the model picks the most-beaten-down names in absolute terms and relative to a falling cross-section. Mean monthly return +3.13%. Eleven of 21 months are in 2022 (Fed-hiking cycle), with corroborating instances at April 2020 (COVID trough), October 2015 and January 2016 (China devaluation and oil collapse), and January 2023.

Asymmetry across the two tiers Within the above-average tier (clusters 0 and 1), the two sub-clusters earn essentially identical Sharpes (point-estimate difference 0.01, two-sided permutation $p = 0.99$): the strategy's performance is insensitive to the magnitude or shape of the long leg's positive deviation from the cross-section. Within the below-average tier (clusters 2 and 3), the same comparison yields a point-estimate difference of 0.61 ($p = 0.50$, suggestive but not significant): the depth of the reversal materially affects performance. The model's regime-aware adaptation reveals itself in distinguishing kinds of reversal environments rather than kinds of trend-following environments. This asymmetry is consistent with the leg-beta finding presented in Section 5.2.2: in the above-average tier, both sub-clusters reflect ordinary momentum continuation with similar market-exposure profiles; in the below-average tier, cluster 3 represents the high-β deep-reversal selection that drives the cross-leg spread amplification.

Limitations of the cluster-level Sharpe gradient We report the cluster Sharpes as descriptive of the underlying return patterns and refrain from claiming statistically significant differences between them. None of the six pairwise $K = 4$ Sharpe comparisons reach $p < 0.10$ on a permutation test. Cluster 3's Sharpe rests partly on 2022, which contributes 11 of its 21 months; excluding 2022 leaves a positive point-estimate Sharpe of

2.19 but with a near-uninformative confidence interval. The sample size of 167 monthly observations, with the smallest cluster at 21, is sufficient to identify stable cluster structure (mean pairwise ARI 0.96 across six seeds) but not to formally test cluster-level Sharpe gradients. The structural finding – four feature-distinct, reproducible cluster types tied to regime context, with a clean M2-vs-fixed-momentum leg-beta contrast in every cluster (Section 5.2.2) – is robust. The performance gradient across them is consistent with reversal-mode being the source of the strategy's edge but cannot be claimed as significant at this sample size.

5.2.2 Three Lenses on the K=4 Partition

This subsection cross-validates the $K = 4$ partition through three complementary lenses: the legs' market-beta exposure, π_t^{filter}'s SHAP share, and the short-leg pick characteristics. Each lens decomposes a regime-aggregate finding from Section 5.2 across the four clusters.

Triangulation with leg-betas (parallel to D&M) The leg-beta amplification documented at the regime level in Section 5.2 decomposes across the $K = 4$ partition as a direct contrast against unconditional 12-month momentum (Table 11). M2 retains long-leg $\beta >$ short-leg β in every $K = 4$ cluster, with cluster spreads of $+0.68$, $+0.40$, $+0.49$, and $+0.43$ for clusters 0–3 respectively. The no-leg-beta-inversion claim of Section 5.2 therefore holds at this finer partition, not just on regime average. Unconditional 12-month momentum, by contrast, exhibits leg-beta inversion in every $K = 4$ cluster: spreads of -0.24, -0.87, -0.75, and -0.42. The D&M crash mechanism is latent across the entire test period rather than panic-specific. The contrast is sharpest in cluster 3, where M2's spread $+0.43$ mirrors fixed momentum's spread -0.42: in the cluster where the unconditional ranking would fail most severely, M2 selects high-β names whose recent drawdowns reflect market exposure rather than fundamental deterioration – the cross-sectional realisation of the tilt-amplification mechanism. This sharpens the M2-vs-D&M comparison of Section 5.4.4: where D&M observed leg-beta inversion in fixed momentum during specific stress episodes, the $K = 4$ decomposition shows the inversion pattern across every cluster type, while M2 escapes it everywhere through compositional re-ranking.

		M2 (XGB ensemble)			Fixed 12-mo mom	
Cluster	n	β_L	β_S	Spread	β_L	β_S
0 (calm continuation)	53	1.6485 (0.2199)	0.9688 (0.1318)	0.6797	1.3248 (0.2202)	1.5654 (0.2362)
1 (mild continuation)	62	1.6023 (0.1528)	1.1996 (0.0866)	0.4027	1.0424 (0.0755)	1.9128 (0.1716)
2 (post-panic recovery)	31	1.4606 (0.1554)	0.9723 (0.1121)	0.4883	1.0885 (0.1655)	1.8403 (0.2176)
3 (deep crisis)	21	1.4896 (0.0802)	1.0637 (0.0852)	0.4259	1.0749 (0.0970)	1.4918 (0.1390)

Table 11: Leg-level CAPM betas by K=4 cluster, 2011–2024 test period. M2 (regime-conditional XGB ensemble) versus unconditional 12-month momentum, side-by-side. Standard errors in parentheses. Spread = $\beta_L - \beta_S$. M2 retains positive spread in every cluster (no leg-beta inversion); fixed momentum exhibits inversion in every cluster.

State-dependent role of π_t^{filter} The aggregate π_t^{filter} SHAP share documented at the regime level decomposes across the $K = 4$ clusters (Table 12). Combined long-and-short shares are 0.50 in cluster 0, 0.46 in cluster 1, 0.54 in cluster 2, and 0.24 in cluster 3. The long-leg-only contributions follow the same ordering: 0.34, 0.27, 0.28, 0.11. Weighted across the four clusters, the combined share recovers the overall figure of 0.46 reported in Section 5.2. The pattern matches the routing interpretation introduced in Section 5.3.1: π_t^{filter}'s marginal value is highest in clusters where the cross-section is ambiguous (clusters 0–2, where the picks sit near or moderately above/below the cross-section), and lowest where the cross-section itself is unambiguously crisis-shaped (cluster 3). When the long-leg z-curve is uniformly between −0.7 and −0.9 across all twelve horizons, momentum splits alone suffice to make the selection; the regime signal contributes only marginal additional information. π_t^{filter} is therefore a router whose marginal contribution scales with cross-sectional ambiguity, not a behaviour switch with constant influence.

Cluster	n	Long π share	Short π share	Combined π share
0 (calm continuation)	53	34.0%	50.1%	49.9%
1 (mild continuation)	62	26.9%	46.5%	46.0%
2 (post-panic recovery)	31	27.8%	53.7%	53.6%
3 (deep crisis)	21	11.4%	38.1%	24.4%

Table 12: π^{filter}'s SHAP share by K=4 cluster, 2011–2024 test period. Long and short shares are computed over long-leg and short-leg picks respectively; combined is over all stock-months in the cluster. The weighted average across clusters recovers the overall combined share of 0.46 from Section 5.2.

The short leg is not a mirror of the long leg The $K = 4$ partition was defined on long-leg z-curves; applying the same labels to the short leg (NYSE P10 of $score_{\text{xgb}}$, identical construction to the appendix leg-beta analysis) reveals that the short basket is structurally distinct from a negated long-leg in every cluster (Table 13). For each cluster we measure the correlation between the short-leg cluster centroid z-curve and the negated long-leg cluster centroid z-curve. A perfect "mirror" would yield correlation

+1.00; the observed values are +0.94 in cluster 0, +0.89 in cluster 1, +0.19 in cluster 2, and −0.84 in cluster 3. The above-average tier (clusters 0 and 1) is roughly mirror-shaped at the long-horizon end with attenuation; the below-average tier breaks the mirror entirely. In cluster 2, both legs sit above the cross-section average – short-leg z-curve uniformly between +0.21 and +0.37, long-leg uniformly between −0.22 and −0.36 – consistent with a healing market in which both buy and short candidates have elevated momentum and the model's edge comes from relative selection within an upward-drifting cross-section. In cluster 3, the legs are anti-mirror: short-leg z spikes to +0.57 at month 1 and decays to +0.11 at month 12, while the long leg sits between −0.73 and −0.88 throughout. The month-1 short-leg spike replicates the asymmetric short-side sensitivity documented at the regime aggregate (Nagel (2012) short-term reversal mechanism) and localises it to the deep-crisis cluster: in active panic with broadly negative momentum, the model shorts stocks whose recent (one-month) returns have started to recover, anticipating that those bounces are unsustained. Short-leg basket sizes are 391 / 334 / 382 / 353 stocks per month for clusters 0–3, all smaller than the corresponding long-leg baskets (498 / 530 / 488 / 533).

Cluster	n	z_1	$\bar{z}$	Basket	$\rho(-z^L)$
0 (calm continuation)	53	−0.062	−0.365	391	0.943
1 (mild continuation)	62	0.101	−0.174	334	0.888
2 (post-panic recovery)	31	0.209	0.241	382	0.190
3 (deep crisis)	21	0.568	0.170	353	−0.835

Table 13: Short-leg z-curve summary by K=4 cluster, 2011–2024 test period. z_1 is the cluster-mean cross-sectional z-score of short-leg picks at the 1-month horizon; $\bar{z}$ averages across all 12 horizons. Basket size is mean stocks per month. $\rho(-z^L)$ is the correlation between the short-leg z-curve centroid and the negated long-leg z-curve centroid: values below 0.95 indicate structural asymmetry. Cluster 3 (deep crisis) shows $\rho = -0.84$, driven by short-leg recent momentum elevated at the 1-month horizon while long-leg picks exhibit the opposite pattern – consistent with the panic-period asymmetry documented in Section 5.2.2.

Why the model needs π_t^{filter} The three triangulations together support the framing claim of Section 5.2.1: π_t^{filter} is structurally necessary to M2's selection mechanism, not merely an additional informative feature. First, the leg-beta evidence shows that unconditional 12-month momentum (which sees the same 12 horizons but no regime signal) exhibits leg-beta inversion in every $K = 4$ cluster, while M2 escapes inversion in every cluster: the inversion-vs-no-inversion contrast holds at the same monthly resolution and is therefore attributable to the input difference, π_t^{filter}. Second, π_t^{filter}'s cluster-level SHAP share spans 0.24 to 0.54, with the lowest share in the cluster where momentum splits alone suffice to make the selection (cluster 3) and higher shares in clusters where the momentum cross-section is ambiguous: π_t^{filter} behaves as a context-dependent router rather than a constant additive feature, and removing it would collapse the routing. Third, the long-leg z-curves under M2 swing from peak +0.80 in cluster 0 (continuation) to a uniform −0.73 to −0.88 in cluster 3 (deep reversal): no fixed-rank or single-direction model can produce both shapes simultaneously, and the routing between them is what the regime signal performs. The cluster decomposition therefore upgrades the regime-aggregate evidence of Section 5.2 from "π_t^{filter} matters on average" to "the model would not produce the $K = 4$ cluster structure without π_t^{filter}".

5.2.3 Horizon-level Evidence

Beyond the per-cluster decomposition, the SHAP distribution by momentum horizon at the regime level (calm vs panic, long vs short) characterises which horizons drive the model's predictions independently of how those predictions sort months into clusters. In the calm long leg, months 11 and 12 are the most influential individual horizons (18.6% and 16.8% of long-leg momentum SHAP), with months 8–12 together accounting for 66.0%. The model independently recovers the intermediate-to-long-horizon dominance documented by Novy-Marx (2012). In the panic long leg, months 7–12 dominate even more strongly, accounting for 67.2% of long-leg momentum SHAP – roughly two-thirds of the prediction signal sits at the intermediate-to-long end in both regimes, with the panic distribution mildly shifted toward the shorter end of that band. On the panic short side, month 1 SHAP rises to 10.6% (versus 5.3% in calm), indicating that the model's predictions become more sensitive to recent returns when identifying stocks to short during stress – consistent with Nagel (2012), who shows that returns to short-term reversal spike during crises when liquidity provision is most valuable. Combined with the per-cluster evidence in Section 5.2.2, the picture is that the same horizons drive the model's predictions in every regime, but the direction of selection at those horizons – continuation in the above-average tier, reversal in the below-average tier – reorganises across clusters.

5.2.4 Discussion

Prior work has established that momentum profits depend on market state (Cooper et al., 2004), that momentum crashes are tied to rebounds after stress (Daniel and Moskowitz, 2016), and that blending fast and slow momentum signals across market cycles can improve performance (Goulding et al., 2023). The analysis above contributes three findings that go beyond these results.

The deep-crisis cluster selects a term-structure-wide loser portfolio The $K = 4$ decomposition in Section 5.2.1 shows that the flat, negative z-score profile appears in pure form in the 21-month deep-crisis cluster (cluster 3), where the long leg is uniformly negative across all 12 horizons (z-curve between −0.73 and −0.88). This shape is qualitatively different from short-term reversal (Jegadeesh, 1990; Lehmann, 1990; Nagel, 2012), which targets month 1 only, and from long-term contrarian strategies (De Bondt and Thaler, 1985), which operate at multi-year horizons. In the above-average-basket clusters (0 and 1) the model recovers the intermediate-horizon continuation shape of Novy-Marx (2012), with z-peak at month 8 (+0.80 in cluster 0, +0.32 in cluster 1). The contrast in selection shape across clusters – humped continuation in the above-average tier, flat-negative in deep crisis – has not, to the best of our knowledge, been documented in this form.

The regime signal acts as a router with state-dependent influence The regime does not flip one horizon from continuation to reversal or shift weight between a fast and a slow signal. It alters how the model uses the entire 12-horizon term structure: buying from the upper tail at intermediate horizons in the above-average tier and from the lower tail at all horizons in deep crisis. The SHAP distribution remains stable across regimes (Table 35): the model attends to the same horizons in both states. What changes is the direction of selection at those horizons, and the strength with which π_t^{filter} does the routing. As Section 5.2.2 shows, π_t^{filter}'s long-leg SHAP share is 34% in cluster 0 (calm continuation) where the basket sits clearly above the cross-section, and 11% in cluster 3 (deep crisis) where the cross-section is so extreme that momentum splits alone suffice. The combined long-and-short shares span 0.50 to 0.24 across the four $K = 4$ clusters, matching the routing interpretation: the regime signal functions as a router whose marginal contribution scales with cross-sectional ambiguity, not as a behaviour switch with constant influence.

Selection by momentum level, not momentum rank Traditional momentum strategies rank stocks by trailing return at a fixed lookback (Jegadeesh and Titman, 1993). Here, the portfolio ranks on *predicted forward return* – a learned function of all 12 horizons and the regime signal. The model learns which momentum *level* is associated with future outperformance. In the above-average baskets, the long-leg z-scores peak modestly at +0.32 (cluster 1) to +0.80 (cluster 0), above average but far from the upper tail. In deep crisis (cluster 3 in Section 5.2.1), the distinction becomes decisive: the long leg selects stocks 0.7–0.9 standard deviations *below* the cross-sectional average at every horizon, a selection no fixed-rank strategy would produce.

The role of nonlinearity That nonlinear models outperform linear ones in cross-sectional asset pricing is well established (Gu et al., 2020). What is specific here is *where* the nonlinearity matters: in transforming a macrofinancial regime probability into a conditioning variable for multi-horizon momentum selection. Section 5.3 provides structural evidence that the nonlinearity is not generic feature interaction but specifically the ability of π_t^{filter} to act as a context switch.

5.3 Nonlinearity and Feature Interactions

5.3.1 Tree Path Analysis

The per-horizon decomposition in Section 5.2 treats each horizon independently, leaving open whether the model relies on specific *interactions* between horizons. To test this, 5,000 sampled stocks from each regime×leg group are traced through 50,000 individual trees, recording which momentum horizon groups each decision path splits on (Appendix H). Horizons are grouped into four bands: short (S, months 1–3), medium (M, months 4–6), intermediate (I, months 7–9), and long (L, months 10–12). Table 14 reports the frequency of each of the 15 possible combinations; each column sums to 100%.

Combination	Calm Long	Calm Short	Panic Long	Panic Short
I + L	14.0%	14.7%	12.8%	14.5%
M + I + L	10.6%	10.8%	10.2%	10.7%
S + I + L	8.3%	8.4%	8.5%	8.3%
S + M	7.5%	7.4%	8.0%	7.6%
S + M + L	6.9%	6.7%	7.5%	6.7%
L	6.9%	6.9%	6.6%	6.8%
S + M + I	6.7%	6.5%	6.6%	6.6%
I	6.5%	6.3%	5.4%	5.6%
S	6.0%	5.8%	6.3%	6.0%
M + L	5.7%	5.7%	6.2%	6.0%
M + I	5.5%	5.5%	5.3%	5.6%
S + L	4.7%	4.6%	5.3%	4.8%
S + M + I + L	4.4%	4.3%	4.4%	4.3%
M	3.4%	3.2%	3.5%	3.3%
S + I	3.1%	3.0%	3.5%	3.2%

Table 14: Share of momentum-involving tree paths by horizon group combination and regime×leg. Each column sums to 100%. Horizon groups: S = months 1–3, M = months 4–6, I = months 7–9, L = months 10–12.

The combination frequencies are stable across all four groups: I+L accounts for 13–15% of paths, M+I+L for 10–11%, and S+M for 7–8%. No unexpected interaction emerges. The distribution matches what the per-horizon SHAP decomposition would predict: horizons with higher SHAP weight appear more frequently in tree paths and combine in proportion to their individual importance.

This locates the specific source of nonlinearity that makes XGBoost essential. The model's value does not come from complex momentum-momentum interactions that differ across regimes – the horizon combinations are regime-invariant. It comes from conditioning the *direction* of selection on π_t^{filter}: the same tree structure that buys intermediate-horizon winners in calm buys intermediate-horizon losers in panic, depending on which branch the regime signal routes each stock through. A linear model assigns a single fixed coefficient to each horizon; XGBoost effectively assigns coefficients whose sign flips with the regime. Table 36 in Appendix H quantifies this: the calm-minus-panic z-score difference is positive for all 39 entries in the long leg, with the largest shifts at the intermediate horizon (+0.36 to +0.55).

5.3.2 Tree Depth and Feature Interactions

Tree depth provides direct evidence for why multi-dimensional conditioning is required (Figure 9). Annualised return rises monotonically from depth 1 (8.7%, no feature interactions) through depth 4 (21.7%), with the largest gains at depths 2 and 3 where the model first gains multi-way conditioning. Volatility is nearly constant across depths (≈19.5%): deeper trees improve signal quality rather than amplifying risk-taking. A depth-2 model, which conditions on π_t^{filter} and one momentum horizon per tree, achieves 15.7% – 28% below the full model. Reading the term-structure shape requires comparing multiple horizons simultaneously, which only becomes possible at depth ≥ 3. Performance peaks at depth 4 and declines beyond depth 5.

The $K = 4$ cluster decomposition in Section 5.2.1 provides a more specific account of the depth requirement. The four clusters correspond to distinct multi-horizon shapes: humped continuation in the above-average tier (clusters 0 and 1), mild reversal in cluster 2, and uniform negativity in cluster 3 (deep crisis). Distinguishing these shapes within a π_t^{filter}-conditional tree branch requires reading at least three momentum horizons. Depth 4 provides the depth budget required to encode multi-shape discrimination after a π_t^{filter}-routing split: depth 1 for the regime gate plus depth 3 for within-regime shape reading.

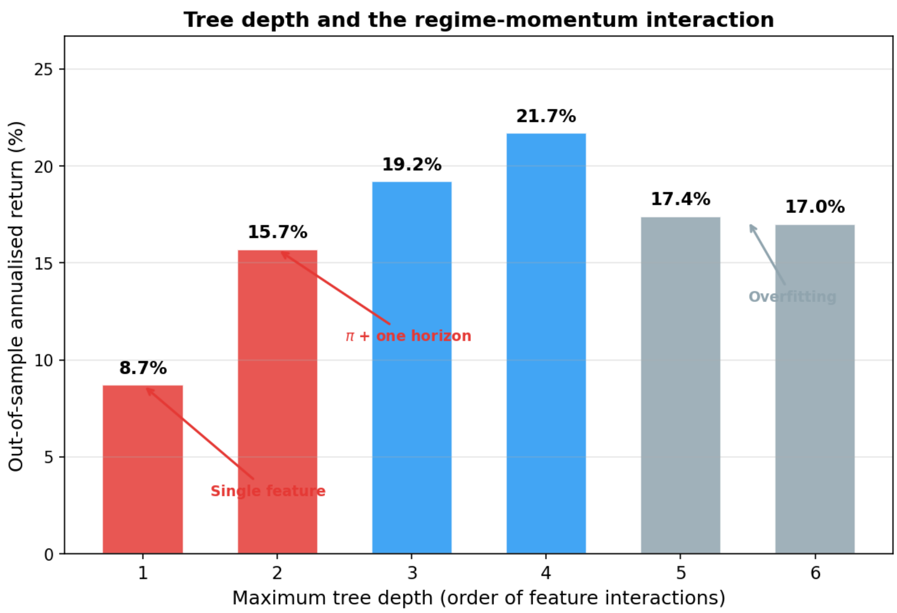


Figure 9: Out-of-sample annualised return as a function of maximum tree depth. Volatility is approximately constant (≈19.5%) across all depths.

5.4 Risk Aversion and Stress Testing

5.4.1 Risk-Aversion Control

An investor may prefer lower portfolio risk at the cost of some expected return. We incorporate a risk-aversion parameter γ into the strategy and trace the out-of-sample trade-off across $\gamma \in [0, 1]$.

The natural starting point is the mean-variance utility $r - \frac{\gamma}{2}\sigma^2$ (Markowitz, 1952), but it is the wrong lens for this problem for two reasons. First, mechanically, applying the penalty as a scoring function in a long-short construction produces a betting-against-beta portfolio: the penalty is subtracted symmetrically from every stock's score, which pushes the long book toward low-volatility names and the short book toward high-volatility names. Realised portfolio volatility then rises rather than falls as γ grows. Appendix D confirms this pattern across several mean-variance variants. Second, economically, mean-variance penalises variance in *absolute* terms: a stock at $\sigma = 30\%$ is penalised nine times more than one at $\sigma = 10\%$, and the penalty's importance scales with portfolio wealth. Real investors behave as if their risk aversion is *relative* rather than absolute, with a stable tolerance for proportional risk that does not change with wealth level. Constant relative risk aversion captures this; mean-variance does not.

We therefore adopt a CRRA utility that penalises $\log \sigma$ instead of σ^2, and evaluate each stock's utility separately under a long direction and a short direction. The volatility penalty enters identically in both: a volatile stock is equally unattractive as a holding or as a short position, so the investor's aversion to risk does not flip sign with trading direction. Longs are the top decile of long-utility across the full cross-section; shorts are the top decile of short-utility. The full specification is given in Appendix E.

Figure 10 shows the resulting Pareto curve.[12] At $\gamma = 0.1$, Sharpe stays at 1.01 while maximum drawdown tightens from -25% to -18%, a near-free reduction in tail risk. At $\gamma = 0.5$, annualised volatility halves from 19.7% to 11.2%, drawdown halves to -12%, and Sharpe settles at 0.78. At $\gamma = 1.0$, volatility falls to 5.2% (roughly a quarter of the market's) and Sharpe remains positive at 0.46. An investor can choose any point on this curve based on their volatility budget or drawdown tolerance. Beyond $\gamma \approx 1.5$, the risk penalty dominates the return signal and the long-short spread collapses into dollar-neutral noise. The framework does not blow up; it simply stops producing a directional bet.

[12]The $\gamma = 0$ baseline reflects the CRRA dual-utility implementation rather than the headline M2 scoring; small numerical differences from Section 5.1 (volatility 19.50%, maximum drawdown -22.8%) are within seed noise.

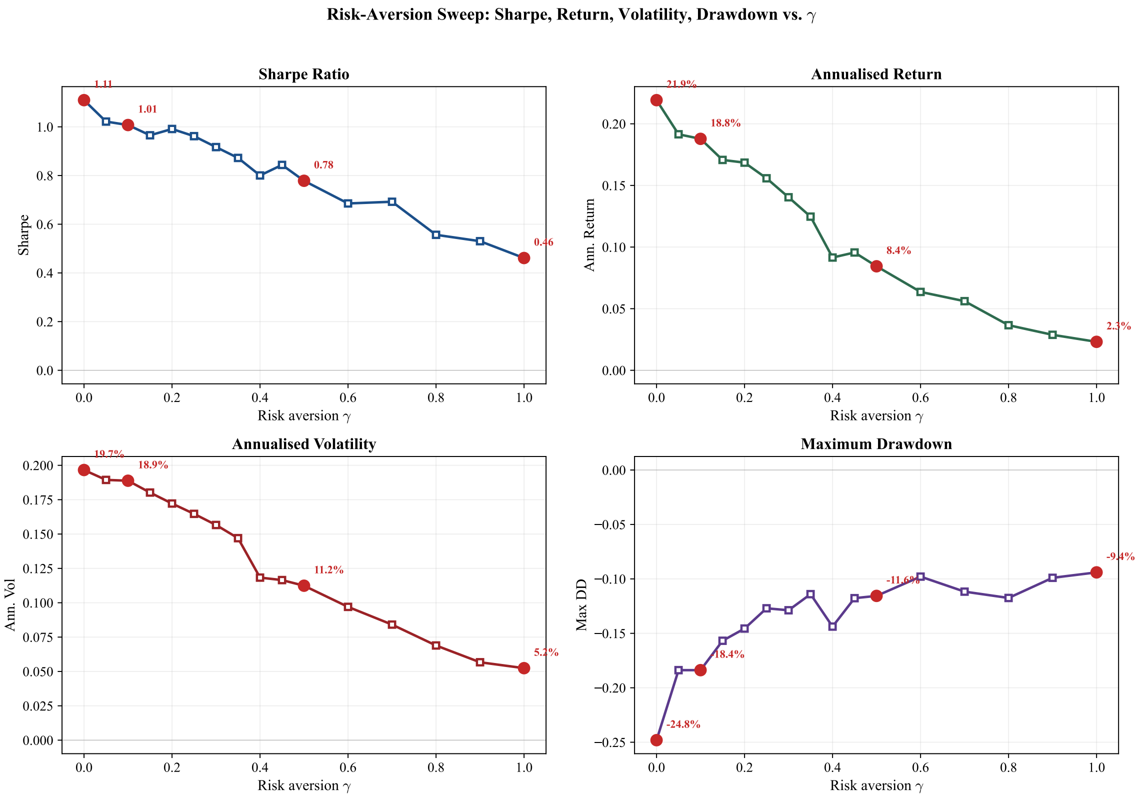


Figure 10: Risk-aversion sweep. Sharpe ratio, annualised return, annualised volatility, and maximum drawdown as a function of γ over $[0, 1]$. Red markers highlight $\gamma \in \{0, 0.1, 0.5, 1.0\}$.

5.4.2 Stress Test: Prolonged Bear Market

The reversal mechanism documented in Section 5.2.1 (cluster 3) rests on a specific premise: that price declines during panic are temporary (driven by liquidity pressure) rather than permanent (driven by economic deterioration). The historical stress periods in the test sample support this: M2 earned +29.9% during the COVID crash (February–April 2020) and +21.5% during the 2022 bear market (January–October 2022). In both cases, the decline was temporary and reversed within months.

But the premise can fail. In a sustained economic deterioration – a depression or structural decline – beaten-down stocks fall because their businesses are failing, not because of temporary mispricing. We call this *reversal failure*: the regime classification is correct, but the momentum structure within that regime has changed. To quantify this vulnerability, we simulate recessions of increasing duration, replacing consecutive months with M2's average losing panic-month return (-3.44% per month) at every possible insertion point.

Recession Duration	Market Loss	M2 Loss	MDD (worst case)
Baseline (no recession)	—	—	−23%
3 months	−10%	−10%	−38%
6 months	−19%	−19%	−42%
12 months	−34%	−34%	−49%
18 months	−47%	−47%	−61%
24 months	−57%	−57%	−69%

Table 15: Prolonged bear market simulation. Each row injects a recession during which M2 loses 3.44% per month (its average losing panic-month return). MDD is the worst-case maximum drawdown across all possible insertion points.

The strategy can absorb substantial downturns. A 12-month recession produces a worst-case drawdown of −49%, after which calm-period continuation resumes. Only a depression-length downturn of 24 months produces a drawdown of −69%. This mirrors the vulnerability identified by Daniel and Moskowitz (2016) for traditional momentum, but on the opposite side: standard momentum crashes when winners stop winning; the regime-conditioned strategy would struggle when losers stop reverting. The hybrid approach combining stock selection with exposure scaling (Section 6.3) is designed for this contingency. Section 5.4.3 provides direct historical evidence on how the strategy actually behaved during the dot-com bust and the global financial crisis, complementing this synthetic stress test.

5.4.3 Historical Stress: Out-of-Sample Evidence

The simulation in Section 5.4.2 estimates worst-case behaviour under synthetic recessions. To complement it with direct historical evidence, the full pipeline (HMM + XGBoost) is re-estimated annually from 1995 onward, with each year's prediction made by a model trained only on data available through the prior year-end. This produces a 30-year out-of-sample monthly return series spanning 1995–2024 that includes the dot-com bust, the global financial crisis, and the 2020 COVID episode. Methodology details are in Appendix G.2; the rest of this subsection focuses on how the strategy actually behaved when stress arrived.

Period	N months	Sharpe	Cumulative	Max DD
Full sample, 1995–2024	359	0.50	1,405.5%	-64.8%
1995–1999	60	0.50	50.8%	-25.5%
2000–2002 dot-com	36	−0.15	-35.9%	-64.8%
2003–2006 bull	48	0.70	37.5%	-12.7%
2007–2009 GFC	36	0.26	11.1%	-36.3%
2010–2019 calm	120	0.67	155.4%	-17.9%
2020–2024 COVID era	59	1.23	299.1%	-22.2%

Table 16: Sub-period decomposition of the 30-year expanding-window out-of-sample back-test. Sub-period rows aggregate by calendar year. The 2011–2024 sub-window of this test (167 months, the same window as the main test in Section 5.1) reproduces the production Sharpe within sampling noise: 0.88 vs production 1.11.

Worst observed stress: the dot-com bust The 2000–2002 dot-com bust produced the strategy's largest historical drawdown – −64.8% peak-to-trough. Restricted to the bear-market proper (March 2000–October 2002), the strategy lost −43.8% cumulative over 32 months (Sharpe −0.39). This is the historical realisation of the reversal-failure vulnerability described in Section 6.2: a prolonged sector-rotating bear in which beaten-down stocks did not promptly recover, so the panic-regime reversal selection bled. The strategy survived the episode and resumed positive returns thereafter (the 2003–2006 sub-sample produces Sharpe 0.70), but the realised drawdown is severe.

Stress at the targeted failure mode: the 2009 momentum crash The single most informative episode is March–December 2009, the period Daniel and Moskowitz (2016) document as a severe momentum crash. Under the same long-short construction used throughout this thesis (NYSE breakpoints, value-weighted, 10 bps cost), unconditional 12-month momentum lost −62.6% cumulative during those ten months. Under expanding-window out-of-sample, the regime-conditional strategy gained +20.5% cumulative (Sharpe 0.73) during the same window. Across the full GFC episode (October 2007 – June 2009) the strategy returned +22.8% cumulative (Sharpe 0.48). This is direct historical evidence that the regime mechanism survives the specific stress event that defeats unconditional momentum.

Synthetic and historical envelopes agree The synthetic stress test in Section 5.4.2 estimates a worst-case 24-month synthetic recession at −70% maximum drawdown. The historical dot-com realisation (−64.8% drawdown over 32 months) sits inside this envelope: the simulation did not understate the risk. Together the two tests cover both scenarios within the 1995–2024 sample (the real bears observed in US history) and beyond it (synthetic prolonged stress), bounding the strategy's stress profile from both sides.

Long-horizon survival Aggregating across all 30 years and all stress events, the strategy produces +1,405% cumulative return with a Sharpe of 0.50. Each historical drawdown is followed by recovery; no stress episode in the sample is unrecoverable. Sub-period samples are small enough that point estimates carry wide confidence intervals, so the qualitative pattern across periods is more informative than the levels.

5.4.4 Comparison with Daniel and Moskowitz (2016)

On the same data and with the same transaction costs, M2 (Sharpe 1.11, CAPM alpha 23.4%, $t = 4.08$) outperforms the D&M managed momentum strategy (Sharpe 0.49, CAPM alpha 10.3%, $t = 1.86$).[13] The two approaches use regime information differently: D&M scale portfolio *exposure* based on predicted crash risk (how much to hold), while M2 changes *which stocks* are held by selecting from the opposite tail of the cross-sectional distribution (what to hold).

A natural extension is to combine the two: use M2's stock selection for portfolio construction and D&M-style exposure scaling as a circuit breaker. Adding D&M-style scaling on top of M2 does not improve performance in the current test period, because the condition it guards against – a sustained reversal failure where beaten-down stocks never recover – did not occur. Although the test period contains panic streaks of up to 15 consecutive months, M2 never loses more than 3 consecutive panic months. The stress test above (Table 15) shows that only a depression-length downturn of 24 months or more would produce drawdowns severe enough to justify exposure scaling. The hybrid approach remains a valuable contingency for deployment in environments where such prolonged deterioration is possible (Section 6.3). The $K = 4$ cluster decomposition in Section 5.2.1 sharpens the complementarity: D&M-style scaling would help most in cluster 3 (deep crisis), where the long leg sits term-structure-wide negative and the reversal bet is most exposed to failure; in the above-average baskets (clusters 0 and 1) it would unnecessarily reduce positions because the cross-section there resembles calm and the strategy continues to earn a calm-like Sharpe.

5.5 International Robustness

To test whether the cross-sectional regime-momentum mechanism extends beyond US data, we replicate the full pipeline on UK and Japanese equities (Compustat Global,

[13]The CAPM alpha is used here rather than the six-factor alpha (24.1%, $t = 4.81$) reported elsewhere so that M2 and D&M are compared under the same one-factor specification used throughout the Daniel and Moskowitz (2016) evaluation. The two alphas differ by 0.9 percentage points for M2 because controlling for the momentum factor (UMD) and the FF5 factors leaves the cross-sectional-regime alpha essentially unchanged.

1990–2025).[14]

Table 17: International cross-sectional model performance, 2011–2025 out-of-sample test period. M2 uses the US specification (12 momentum horizons, 50-seed XGBoost ensemble, decile-spread long-short construction, value-weighted by market equity) with locally-fitted HMM (200 production seeds, DD+VOL+REL_N feature set selected per region). Long and short legs are formed from unconditional decile breakpoints across all listed stocks (no NYSE-equivalent tier exists for either market). Net of 10 bps one-way transaction cost.

	UK			JP		
Strategy	Sharpe	Ann. Ret	Max DD	Sharpe	Ann. Ret	Max DD
Local market	0.62	6.7%	−25.2%	0.86	12.3%	−21.5%
Fixed 12-mo mom	0.45	9.3%	−62.6%	0.07	−0.5%	−66.3%
M2 (XGB)	**0.84**	**16.0%**	**−21.4%**	0.50	6.0%	−19.8%

Both regions support the mechanism. UK M2 attains a Sharpe of 0.84 over 2011–2025, well above unconditional 12-month momentum's 0.45 and the local market's 0.62. JP M2 attains 0.50, substantially above unconditional momentum's 0.07 but below the JP market's 0.86 – consistent with the documented weakness of Japanese momentum (Asness et al., 2013). Maximum drawdown is reduced from −62.6% (fixed momentum) to −21.4% in UK, and from −66.3% to −19.8% in JP.

Taken together, the results show that momentum's cross-sectional predictability shifts between continuation and reversal across market regimes, and that exploiting this shift requires nonlinear conditioning on a regime signal. The signal's value lies not in predicting returns directly but in conditioning how the model jointly uses the full momentum term structure – a multi-dimensional interaction only tree-based models can learn.

[14] Each region's HMM is fit on locally-selected stress features (DD+VOL+REL_N for both, with BANK_REL replacing the US-specific Moody's BAA-AAA spread in the candidate pool); details in Appendix K. The cross-sectional model is unchanged from the US specification, except that long and short legs are formed from unconditional decile breakpoints across all listed stocks rather than NYSE-equivalent breakpoints (no analogous tier exists for either market).

6 Conclusion

The central finding of this thesis is that, in the sample studied, momentum's cross-sectional predictability shifts with market conditions: which lookback horizons carry predictive power, and in which direction, depends on the prevailing market regime. The regime signal captures this shift not by predicting returns directly but by conditioning how the model jointly uses the full momentum term structure – a multi-dimensional interaction that only a nonlinear model can exploit. These findings complement the exposure-scaling approach of Daniel and Moskowitz (2016) and Barroso and Santa-Clara (2015) along two distinct channels. First, the HMM filtered probability provides a continuous, smoothed regime signal – a probabilistic alternative to the binary bear-market indicators used to gate exposure-scaling rules. Second, the cross-sectional model uses that signal to reorganise which stocks fill each leg rather than to scale gross exposure: the regime signal conditions the direction of momentum selection, and the leg-beta inversion that drives traditional momentum's crashes is absent in the resulting portfolio (Appendix G.6).

6.1 Contributions

The three contributions outlined in Section 1 are supported by the empirical evidence. The term-structure reorganisation is visible directly in z-score profiles and does not depend on the modelling framework. The context-variable mechanism is confirmed by SHAP analysis showing that π_t^{filter} conditions the joint use of momentum horizons rather than predicting returns directly. The nonlinearity requirement is established by the failure of both classification-based and regression-based linear models on identical features. Together, these findings suggest that the fragility of momentum documented in prior work may partly reflect unconditional models averaging over a regime-dependent structure, rather than an intrinsic deficiency of the anomaly itself.

6.2 Limitations

Economic vulnerability The strategy's primary vulnerability is a prolonged economic deterioration where beaten-down stocks do not recover because their underlying businesses are failing rather than because of temporary mispricing. The stress test in Section 5.4.2 quantifies this: a 12-month recession produces a worst-case drawdown of -49%, while a 24-month depression pushes the worst-case drawdown to -69% (Table 15). Calm-period continuation provides a floor, but a sustained downturn where reversal consistently fails would be painful.

Stress-tested vulnerabilities The strategy's largest historical drawdown is the dot-com bust (-64.8% peak-to-trough; -43.8% cumulative over the 32-month bear-market proper, March 2000–October 2002, under expanding-window OOS; see Section 5.4.3). This is the empirical realisation of the reversal-failure vulnerability described above and falls within the envelope estimated by the simulation in Section 5.4.2 (−69% worst-case drawdown for a 24-month synthetic recession). Worst-case scenarios beyond the historical envelope – a sustained multi-year decline without any reversal, e.g. a depression-style outcome – remain testable only via simulation.

Test period scope The test period (2011–2025, 167 months) does not contain a prolonged recession. The 30-year expanding-window backtest (Section 5.4.3) confirms that M2 would have survived the 2009 momentum crash, but the dot-com result there is the historical realisation of the reversal-failure vulnerability discussed above. Performance is strongest in 2021–2025 (Sharpe 1.70), a period with frequent regime transitions; in calm-dominated years the model earns modest returns from standard continuation. A permanently calm market would reduce the strategy to ordinary momentum.

Frozen HMM parameters and regime drift The HMM is estimated on 1990–2010 and held fixed during the test period. This assumes that the statistical properties of calm and panic regimes remain stationary. If the market undergoes structural change – a permanent increase in baseline volatility due to algorithmic trading or shifting monetary policy regimes, for example – the trained thresholds may become miscalibrated, with structurally calm periods classified as panic. The sustained elevation of π_t^{filter} in recent years may partly reflect such drift rather than genuine prolonged stress. The 30-year expanding-window backtest in Section 5.4.3 – in which the HMM is re-estimated annually using only data available at the time – reproduces the production result on the 2011–2024 overlap, indicating that the regime structure is reasonably stable; online re-estimation would still provide a stronger guarantee.

Memoryless regime conditioning XGBoost observes only the current month's features and has no memory of the π_t^{filter} trajectory, so a fresh spike and a sustained elevation are treated identically despite their different implications for reversal. Section 6.3 sketches a sequential extension.

Implementation frictions The analysis assumes frictionless short selling and applies a uniform 10 basis point one-way transaction cost. Cost sensitivity (Appendix B.2) confirms M2 remains profitable at 50 bps, but the uniform cost understates the borrow fees and hard-to-borrow constraints that would apply most strongly to the short leg for small or less liquid stocks. Value-weighting and NYSE decile breakpoints tilt the portfolio toward large, liquid names, partially mitigating this concern.

Specification search The HMM feature selection procedure uses out-of-sample portfolio Sharpe in its second pass, meaning the selected combination (DD+DISP+REL_N+CS) is informed by test-period outcomes. Section 4.2 discusses the mitigating factors in full – the quality screen in Pass 1, the cross-seed stability criterion in Pass 3, and the qualitatively similar performance of leave-one-out specifications (Table 3) – but the reported 1.11 Sharpe should be read with this caveat.

Sample scope and causal inference The sample is restricted to US equities. The interpretation that the regime signal conditions momentum's term structure is supported by multiple lines of evidence (SHAP, tree paths, ablation, LR-XGB divergence) but is inferred from observational data and should be read as consistent evidence rather than definitive proof. The Newey–West t-statistic on M2's excess return over the market is 4.37^{***}, well above conventional thresholds, and the factor-model alphas in Appendix G.4 confirm this strength across all five specifications (CAPM, FF3, Carhart, FF5, FF6: every $t > 4$).

6.3 Future Work

Several directions could extend this framework.

Predictive regime signal The current regime signal π_t^{filter} estimates the probability that the market is in panic *now*, conditional on data through time t. Because the HMM defines transition probabilities between states, a natural extension is to predict the regime one step ahead: $P(s_{t+1} = 1 \mid \mathbf{z}_{1:t}) = \sum_k \pi_t^{\text{filter}}(k) \cdot P_{k,1}$. Feeding this predictive probability into the cross-sectional model would allow the portfolio to reposition before a regime shift rather than reacting to it. The expected gain is concentrated in transition months, where the current framework trades on a stale view of the regime in precisely the months when the portfolio is furthest from its regime-conditional optimum. A richer extension applies the same recursion to produce a k-step forecast, turning the regime signal into a short-horizon expected-state distribution that the cross-sectional model can condition on directly.

Alternative nonlinear models XGBoost is one of several nonlinear model classes, and the central mechanism – that π_t^{filter} gates the use of momentum horizons – should in principle be recoverable by any sufficiently expressive architecture. The Random Forest comparison in Appendix F.3 partially tests this: bagging still selects π_t^{filter} as a material feature, but the drop in Sharpe (1.08 to 0.73) and the fall in its SHAP share (45% to 18%) show that the boosted additive structure is what elevates the regime signal from a marginal predictor to a primary gating variable. The more informative test is whether the mechanism survives under genuinely different inductive biases. An attention-based model would learn horizon-by-regime weights directly and produce an interpretable per-horizon attention profile that could be compared against the SHAP decomposition in Table 9; a well-regularised multilayer perceptron with explicit $\pi \times mom_k$ and $mom_k \times mom_j$ interaction inputs would isolate whether the dependence on boosting reflects sequential residual fitting or a shortage of data for richer architectures. Recovering the same mechanism across these alternatives would strengthen the claim that the regime-conditional shift is a property of the data rather than of XGBoost specifically; failure would force a narrower interpretation tied to tree-based models.

Sequential regime conditioning XGBoost observes only the current month's features and treats all months with the same π_t^{filter} identically. Two months both at $\pi_t^{\text{filter}} = 0.75$ carry the same input regardless of whether the signal just spiked from 0.1 – a fresh crash, when mean reversion is most likely – or has been elevated for six consecutive months – prolonged stress, when the reversal mechanism may be breaking down. In practice, the spike is where the long-reversal premium is largest and the sustained elevation is where the strategy is most exposed to failed reversals. A sequential model could distinguish these cases: candidates include a recurrent network such as an LSTM, a transformer over a fixed window of regime features, or a simpler feature-engineering approach that adds lagged π_t^{filter} values, a short-window rate of change, and a duration-since-entry counter to the existing input set. The HMM's forward-backward recursion already integrates the full history into π_t^{filter}, but reducing that history to a single scalar discards exactly the information needed to distinguish a fresh spike from a prolonged one. Exposing the trajectory to the cross-sectional model, rather than only its filtered expectation, should improve decisions in the months where the current specification is weakest.

Regime-conditional feature sets Adding ten firm-level fundamentals to M2 reduces Sharpe from 1.11 to 0.89 (Section 5.1.3) because the new features absorb 44% of SHAP importance and dilute the regime-momentum signal. The feature-set ablation localises the dilution: fund-only attains a calm Sharpe of 0.86 (matching the baseline's 0.84) but a panic Sharpe of only 0.38, because firm-level characteristics describe stable attributes that do not flip direction when the regime shifts. The natural conclusion is not that fundamentals are uninformative but that the optimal feature set may itself be regime-dependent. During panic, when reversal dominates, momentum and the regime signal carry the mechanism, and adding quality or valuation characteristics competes for splits without contributing to the primary gating rule. During calm, when the continuation premium is weaker and more diffuse, fundamentals may provide a useful second dimension. Several designs could test this. The simplest is a layered construction in which M2 produces the initial long-short ranking and fundamentals enter as a post-ranking filter that excludes bottom-quintile-quality names from the long leg and high-leverage names from the short leg. A more ambitious variant trains two separate cross-sectional models, selected by π_t^{filter}, each with its own feature set. Both designs require careful out-of-sample validation because the regime-conditional split reduces effective sample size and is vulnerable to overfitting, but both offer a principled way to recover the information that fundamentals plausibly carry without diluting the mechanism that drives current performance.

Cross-validated feature selection The HMM feature-selection procedure used here (Section 4.2) evaluates candidate feature sets in part by their out-of-sample portfolio Sharpe, a step that draws on test-period outcomes and is acknowledged in Section 6.2. A more disciplined alternative would replace the test-period evaluation with a cross-validation procedure run entirely within the training partition, for instance a five-fold expanding-window cross-validation on the 1990–2010 panel that never touches the 2011–2025 test data, with each fold scoring candidate feature sets on its held-out validation slice. Rolling-window robustness checks at multiple training-end dates would additionally test whether the chosen feature set is stable across regimes or sensitive to the specific window the procedure observes. Adopting cross-validation as the primary selection method, with the in-sample feature ablation reported in Section 4.2 as a complementary stability check, would remove the test-period dependence from the feature choice and produce a more rigorous justification for the four-feature input.

Systematic hyperparameter dissection The XGBoost specification used throughout (depth 4, learning rate 0.05, 500 trees) was chosen via a test-period sweep over a small grid (Appendix F.3) and is therefore subject to the same specification-search caveat as the HMM features. A natural extension is a fine-resolution sweep of the hyperparameter space – tree depth, learning rate, ensemble size, regularisation strength, and column-subsampling rate – with each configuration evaluated not only on out-of-sample Sharpe but also on the cross-sectional z-scores of long- and short-leg selections across regimes. Because the central mechanism is a regime-conditional shift in the momentum term structure rather than headline Sharpe alone, the diagnostic question is whether the cross-sectional z-score reorganisation documented in Section 5.2.1 survives changes in model capacity, or whether the deep-crisis reversal selection pattern is specific to the production specification. A rolling-window cross-validation across multiple training-end dates would further test stability of the optimum across market regimes. The expected outcome is a tight cluster of high-Sharpe configurations with similar z-score profiles, in which case the production specification is robust; a finding that the panic z-score profile flips with model capacity would localise the mechanism more narrowly to the chosen depth and learning rate.

Computational optimisation of stock selection The cross-sectional ranker used throughout this thesis – a 50-seed XGBoost ensemble at depth 4 with 500 trees per seed – is treated as the canonical stock-picker for the analyses in Section 5.2, but it was not chosen by global optimisation. The coarse test-period sweep of Appendix F.3 and the architecture comparison sketched above leave open the likelihood that a more disciplined optimisation – finer hyperparameter grids under Bayesian search, end-to-end training against a portfolio-level objective rather than per-stock predicted return, or domain-specific architectures such as permutation-invariant set encoders that operate directly on the cross-section – would produce a strictly better-performing ranker. The qualitative findings of Section 5.2 are nevertheless robust to this concern: the regime-conditional shift in selection (continuation in the above-average tier, reversal in the below-average tier), the four-cluster heterogeneity in pick selection documented in Section 5.2.1, and the leg-beta amplification mechanism reflect properties of the data and the regime signal rather than artefacts of XGBoost's specific optimum. A more optimal ranker would almost certainly improve absolute Sharpe, but the regime-conditioning story should survive that improvement. Quantifying both the absolute-performance gap left by the current specification and the stability of the qualitative findings across better-tuned rankers would tighten the link between the methodological choice and the strategic claim.

Exposure-scaling overlay The strategy's primary vulnerability, discussed in Section 6.2, is a prolonged economic deterioration where beaten-down stocks do not recover and the reversal mechanism fails. M2's stock-selection channel does not address this directly: its contribution is to choose which stocks to hold, not how much. A natural extension is to combine it with a Daniel and Moskowitz (2016)-style exposure-scaling overlay that reduces gross exposure when a sustained-stress indicator is triggered – for example, when π_t^{filter} has been above a threshold for a minimum number of consecutive months, or when its rolling maximum over a fixed window exceeds a calibrated bound. Preliminary checks on the 2011–2025 sample show no improvement from such an overlay because the sample contains no prolonged recession, so the circuit breaker only cuts profitable panic months; its value would emerge in a sustained downturn outside the current window. The dot-com sub-period of the 30-year expanding-window backtest (Section 5.4.3) is a natural setting to evaluate candidate overlay rules.

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Appendix A: Data Sources and Variable Definitions

Three primary data sources are used. **CRSP** (Center for Research in Security Prices, via WRDS) provides monthly stock-level returns, prices, shares outstanding, exchange codes, and share codes from the Monthly Stock File (`crsp.msf`), as well as daily market returns from the Daily Stock Index file (`crsp.dsi`). **Compustat** (via WRDS) provides quarterly fundamental accounting data from `comp.fundq`, linked to CRSP via the CCM bridge table using primary link types (LU, LC). **FRED** (Federal Reserve Economic Data) provides Moody's BAA and AAA corporate bond yields for computing the credit spread.

Following Shumway (2001), delisting returns are incorporated to prevent survivorship bias. When a stock is delisted, its actual delisting return is used if available; for performance-related delistings (CRSP codes 500–584) with missing returns, −30% is imputed.

A comprehensive overview of all data variables is provided in Table 18.

Variable	Source	Description	Used in
Stock-level variables			
Monthly returns	CRSP `msf`	Adjusted holding-period returns	Sections 3, 5
Price, shares out.	CRSP `msf`	For market equity computation	Section 4
Exchange, share codes	CRSP `msf`	Universe filtering (codes 10, 11)	Section 3.2
Delisting returns	CRSP `msedelist`	Survivorship bias correction	Section 3.2
Book equity (`ceqq`)	Compustat `fundq`	Book-to-market, ROE	Appendix L
Total assets (`atq`)	Compustat `fundq`	Asset growth, gross profit, CF ratios	Appendix L
Income (`ibq`)	Compustat `fundq`	ROE, earnings growth, accruals	Appendix L
Revenue (`revtq`)	Compustat `fundq`	Gross profitability	Appendix L
COGS (`cogsq`)	Compustat `fundq`	Gross profitability	Appendix L
Debt (`dlttq`, `dlcq`)	Compustat `fundq`	Leverage ratio	Appendix L
Operating cash flow (`oancfy`)	Compustat `fundq`	CFO, FCF, accruals	Appendix L
Capital expenditure (`capxy`)	Compustat `fundq`	Free cash flow	Appendix L
Market-level variables (HMM inputs)			
Daily market returns	CRSP `dsi`	Drawdown (DD) computation	Section 4.1
Market price index	CRSP `dsi`	Drawdown (DD)	Section 4.1
Cross-sect. stock returns	CRSP `msf`	Dispersion (DISP), REL_N	Section 4.1
BAA, AAA bond yields	FRED	Credit spread (CS)	Section 4.1
Derived features			
$mom_1, \ldots, mom_{12}$	Computed	12 trailing momentum signals	Section 4.4
π_t^{filter}	HMM output	Filtered panic probability	Section 4.3
10 fundamentals (ablation only)	Computed	bm, roe, gross profit, cash flow, etc.	Appendix L

Table 18: Summary of all data variables used in the analysis.

Notes: All data accessed via WRDS. Compustat linked to CRSP via the CCM bridge table using primary link types (LU, LC). Fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. The sample covers January 1990 to November 2025 (training: 1990–2010; test: 2011–2025).

Appendix B: Portfolio Formation, Transaction Costs, and Performance Metrics

B.1 Portfolio Construction

For each method, a long-short portfolio is constructed each month. Stocks are ranked by their scores, and NYSE-listed stocks are used to determine decile breakpoints. All stocks (including AMEX and NASDAQ) whose score exceeds the 90th percentile breakpoint enter the long leg; all stocks below the 10th percentile breakpoint enter the short leg. Long-short construction is the standard in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015) because it isolates momentum's cross-sectional stock selection from market beta, providing a cleaner test of the strategy's ability to distinguish future winners from losers.

Portfolio weights within each leg are assigned proportionally to market equity (value-weighting):

$$w_t^{s,L} = \frac{ME_t^s}{\sum_{s \in \mathcal{L}_t} ME_t^s}, \qquad w_t^{s,S} = \frac{ME_t^s}{\sum_{s \in \mathcal{S}_t} ME_t^s}, \tag{29}$$

where $\mathcal{L}_t$ and $\mathcal{S}_t$ are the sets of stocks in the long and short legs at month t, respectively. The long-short portfolio return is:

$$r_t^{\text{LS}} = \sum_{s \in \mathcal{L}_t} w_t^{s,L} \cdot r_t^s - \sum_{s \in \mathcal{S}_t} w_t^{s,S} \cdot r_t^s. \tag{30}$$

Value-weighting tilts each leg toward large, liquid stocks, reducing the influence of small-cap return anomalies and improving implementability.

B.2 Transaction Costs

All portfolios are rebalanced monthly.

A one-way transaction cost of $c = 10$ basis points is applied to the turnover in both the long and short legs at each rebalancing date. The net portfolio return at month t is:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot (\tau_t^L + \tau_t^S), \tag{31}$$

where τ_t^L and τ_t^S are the one-way turnover in the long and short legs, respectively:

$$\tau_t^L = \frac{1}{2} \sum_s \left| w_t^{s,L,\text{new}} - w_{t-1}^{s,L,\text{prev}} \right|, \qquad \tau_t^S = \frac{1}{2} \sum_s \left| w_t^{s,S,\text{new}} - w_{t-1}^{s,S,\text{prev}} \right|, \tag{32}$$

and each summation runs over all stocks appearing in either the new or previous portfolio for that leg. The 10 basis point cost reflects the transaction environment for a value-weighted, large-cap-biased strategy where NYSE breakpoints tilt toward liquid securities.

It is important to note that transaction costs are applied *post-hoc* as an accounting adjustment and are not incorporated into the models' training objectives. The cross-sectional models (Methods 0–2) optimise stock-level prediction (classification accuracy or return forecasting) without knowledge of the previous portfolio, the resulting turnover, or the cost of rebalancing. This is standard practice in the empirical asset pricing literature (e.g., Fama and French, 2015; Goulding et al., 2023), where portfolio construction is treated as a fixed downstream step applied uniformly to all strategies. An end-to-end framework that jointly optimises stock selection and portfolio-level objectives (e.g., net-of-cost Sharpe ratio) would require a fundamentally different architecture, such as a reinforcement learning or differentiable portfolio optimisation layer, and remains a potential extension of this work.

B.3 Performance Metrics

Standard Metrics Strategy performance is assessed using four standard risk-adjusted metrics computed on monthly net-of-fee returns.

Annualized return The geometric annualized return over n months is:

$$R_{\text{ann}} = \left[\prod_{t=1}^{n} (1 + r_t) \right]^{12/n} - 1. \tag{33}$$

Annualized volatility The annualized standard deviation of monthly returns:

$$\sigma_{\text{ann}} = \sigma(r) \cdot \sqrt{12}. \tag{34}$$

Sharpe ratio The annualized Sharpe ratio:[15]

$$SR = \frac{\bar{r}}{\sigma(r)} \cdot \sqrt{12}, \tag{35}$$

where $\bar{r}$ is the mean monthly return.

Maximum drawdown Let $C_t = \prod_{\tau=1}^{t} (1 + r_\tau)$ denote the cumulative wealth path. The maximum drawdown is:

$$MDD = \min_{1 \leq t \leq n} \frac{C_t - \max_{1 \leq \tau \leq t} C_\tau}{\max_{1 \leq \tau \leq t} C_\tau}. \tag{36}$$

[15] All Sharpe ratios in this thesis are computed using raw returns rather than excess returns above the risk-free rate. Since the same convention is applied to every strategy, relative comparisons and rankings are unaffected. The risk-free rate was near zero for most of the test period (2011–2021) and averaged roughly 2% annualized over the full 2011–2025 window, so absolute Sharpe ratios are overstated by approximately 0.05–0.10.

Regime-Conditional Evaluation To assess whether the regime signal adds value beyond unconditional stock selection, performance is decomposed by market regime. Months are partitioned into calm ($\pi_t^{\text{filter}} < 0.5$) and panic ($\pi_t^{\text{filter}} \geq 0.5$) periods, and the Sharpe ratio is computed separately within each subset:

$$SR_{\text{calm}} = \frac{\bar{r}_{\text{calm}}}{\sigma(r_{\text{calm}})} \cdot \sqrt{12}, \qquad SR_{\text{panic}} = \frac{\bar{r}_{\text{panic}}}{\sigma(r_{\text{panic}})} \cdot \sqrt{12}. \tag{37}$$

This decomposition helps assess whether the strategy's alpha originates primarily from calm-period stock selection, panic-period risk avoidance, or both.

Appendix C: Feature Importance Methods

C.1 SHAP Values for XGBoost

To interpret the XGBoost model and assess the marginal contribution of each feature, SHAP (SHapley Additive exPlanations; Lundberg and Lee, 2017) values are computed on the test set. The SHAP value of feature j for observation $\mathbf{x}$ measures its average marginal contribution to the prediction across all possible feature orderings:

$$\phi_j(\mathbf{x}) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big[f_{S \cup \{j\}}(\mathbf{x}) - f_S(\mathbf{x})\Big], \tag{38}$$

where N is the full feature set, $p = |N|$, and $f_S(\mathbf{x})$ is the expected model output when only features in S are observed. For tree-based models, the TreeSHAP algorithm (Lundberg et al., 2020) computes these values exactly in polynomial time. The overall importance of each feature is summarised as:

$$I_j = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |\phi_j(\mathbf{x}_i)|. \tag{39}$$

Two additional analyses are performed. First, the mean absolute SHAP value of π_t^{filter} is computed separately for calm and panic months to assess whether the regime signal's influence varies across market states. Second, regime-specific XGBoost models are trained on calm-only and panic-only subsets of the training data, and their SHAP feature importances are compared to identify which features the model relies on differentially across regimes.

Appendix D: Alternative Training Targets for XGBoost

Section 5.4 argues that embedding a risk penalty directly in the XGBoost training target degrades performance, motivating the CRRA specification used in the main text. Table 19 shows this degradation is not an artefact of the mean-variance functional form: the same pattern appears under three families of training targets with varying risk tolerance.

Target family	Parameter	Sharpe	Ann. Ret	Ann. Vol	MDD
Baseline (r)	–	1.107	30.0%	27.1%	−28.3%
Mean-variance: $r - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.061	22.9%	21.6%	−22.3%
	$\gamma = 1$	1.024	16.5%	16.2%	−22.8%
	$\gamma = 2$	1.135	15.5%	13.6%	−22.0%
	$\gamma = 5$	1.062	13.5%	12.7%	−22.5%
	$\gamma = 10$	1.082	13.1%	12.1%	−18.9%
Log mean-variance: $\log(1+r) - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.051	14.6%	13.9%	−17.9%
	$\gamma = 1$	1.039	13.8%	13.4%	−23.8%
	$\gamma = 2$	1.080	14.0%	12.9%	−20.9%
	$\gamma = 5$	1.127	14.1%	12.4%	−18.7%
	$\gamma = 10$	0.995	12.1%	12.3%	−20.2%
Sharpe-like: $r\,/\,\sigma^a$	$a = 0.5$	1.148	24.8%	21.4%	−22.4%
	$a = 1.0$	0.984	14.6%	15.0%	−29.1%
	$a = 1.5$	0.936	12.4%	13.5%	−21.0%
	$a = 2.0$	0.759	12.8%	18.1%	−24.5%

Table 19: Out-of-sample performance of XGBoost (M2) under alternative training targets (long-only construction). Every risk-adjusted target reduces returns faster than volatility.

The pattern is consistent: returns fall faster than volatility under every risk penalty, regardless of functional form. The mildest adjustments (mean-variance with $\gamma = 0.5$, Sharpe-like with $a = 0.5$) come closest to the baseline but still underperform. This confirms that the diagnostic in Section 5.4 is not specific to the mean-variance utility formulation; the single-model training-target approach is structurally unsuited to incorporating risk aversion in a long-short setting, which the CRRA dual-utility framework resolves.

Appendix E: CRRA Dual-Utility Construction

This appendix provides the technical specification behind the risk-aversion sweep in Section 5.4. The strategy uses two forecasts for each stock-month: an expected-return forecast $\hat{r}_t^s$ and a forward-volatility forecast $\hat{\sigma}_t^s$. $\hat{r}_t^s$ is produced by the XGBoost ensemble described in Section 3. $\hat{\sigma}_t^s$ is produced by a second XGBoost ensemble trained on the realised forward volatility

$$\sigma^s_{[t+1,\,t+12]} \;=\; \mathrm{std}\big(r^s_{t+1}, \ldots, r^s_{t+12}\big),$$

using the same feature set augmented with trailing 12-month realised volatility.

For each stock, utility is computed under both trading directions:

$$
\begin{aligned}
u^s_{\text{long}} &= +\,\text{sign}(\hat{r}^s)\,\log\bigl(1+|\hat{r}^s|/\varepsilon\bigr) \;-\; \gamma\,\log(\hat{\sigma}^s),\\
u^s_{\text{short}} &= -\,\text{sign}(\hat{r}^s)\,\log\bigl(1+|\hat{r}^s|/\varepsilon\bigr) \;-\; \gamma\,\log(\hat{\sigma}^s),
\end{aligned}
\tag{40}
$$

with $\varepsilon = 0.01$. The return term is $\pm$ log-compressed expected return under each direction; the $-\gamma\log(\hat{\sigma})$ term enters with the same sign in both, so a low-σ stock is preferred whether considered as a long or as a short.

Each month, longs are the top decile of u_{long} and shorts are the top decile of u_{short}, both using NYSE breakpoints and full-universe membership. Positions are value-weighted within each leg. Stocks appearing in both top deciles (the overlap set, which grows with γ) are retained on both sides; their contributions to the long and short books net in $r_L - r_S$. This is the mechanism by which the long-short spread converges to zero as γ rises beyond the useful range: at high γ, the volatility penalty dominates the return signal, low-$\hat{\sigma}$ stocks occupy both top deciles simultaneously, and the two legs become near-identical value-weighted low-vol portfolios.

The 50-seed ensemble is trained on 1990–2010 and evaluated out-of-sample on 2011–2025. The sweep in Figure 10 uses a 17-point grid over $[0, 1]$ (step 0.05 up to 0.5, step 0.1 thereafter).

Appendix F: Robustness Checks

This section reports six robustness checks that complement the main results in Section 5.

F.1 Sub-Period Stability

To assess whether the main findings are driven by a single favourable sub-period, the test sample is split into three roughly equal windows: 2011–2015, 2016–2020, and 2021–2025.

	2011–2015	2016–2020	2021–2025	Full
Market	0.72	1.04	0.73	0.84
Fixed 12-mo	0.74	-0.20	-0.11	0.06
M1: LR	0.47	-0.02	-0.38	-0.01
M2: XGB	0.59	1.04	1.70	1.11

Table 20: Sub-period Sharpe ratios (long-short). The test period is split into three roughly equal sub-periods. M2 is the only strategy with positive Sharpe ratios in all three sub-periods.

In long-short construction, M2 is the only strategy that produces positive Sharpe ratios in every sub-period (0.59, 1.04, 1.70). M1 collapses in 2016–2020 (Sharpe -0.02) and turns negative in 2021–2025 (-0.38). Fixed 12-month momentum turns negative after

2015, confirming the momentum crash phenomenon documented by Daniel and Moskowitz (2016). M2's strongest performance (1.70) is in 2021–2025, a period characterised by elevated volatility and multiple stress episodes – precisely the conditions where regime-conditional stock selection adds the most value.

F.2 Turnover and Transaction Cost Sensitivity

A strategy's net-of-cost performance depends critically on its turnover. Table 21 reports average monthly one-way turnover for each strategy.

Strategy	Avg. Monthly TO	Annualised TO
Fixed 12-mo	70.1%	842%
Fixed 1-mo	184.3%	2,211%
M1: LR	168.8%	2,026%
M2: XGB	132.3%	1,588%

Table 21: Portfolio turnover by strategy (long-short). Average monthly one-way turnover and annualised turnover (monthly × 12).

In long-short construction, M2's average monthly one-way turnover is 132.3%, reflecting the regime-conditional changes in stock selection that shift the portfolio composition substantially. Table 22 translates these turnover differences into Sharpe ratio sensitivity across a range of transaction cost assumptions.

	0 bps	5 bps	10 bps	20 bps	30 bps	50 bps
Fixed 12-mo	0.10	0.08	0.07	0.03	0.00	-0.06
Fixed 1-mo	0.39	0.32	0.26	0.14	0.01	-0.24
M1: LR	0.12	0.05	-0.02	-0.16	-0.29	-0.57
M2: XGB	1.19	1.15	1.11	1.03	0.95	0.78

Table 22: M2 remains profitable at 50 bps. All other strategies become unprofitable at moderate cost levels.

M2 remains profitable across all cost levels tested: even at 50 bps one-way (five times the baseline), its Sharpe ratio is 0.78. All other strategies become unprofitable at moderate cost levels. The baseline 10 bps assumption is reasonable for a value-weighted strategy using NYSE breakpoints, which tilts toward the most liquid names.

F.3 Model Sensitivity

The XGBoost model uses hyperparameters selected via cross-validation on the training set. To verify that the results are not an artefact of a particular configuration, Table 23 reports out-of-sample performance under alternative tree depths, learning rates, and ensemble sizes. A depth-matched Random Forest baseline is discussed below; full numbers are in `results/thesis/random_forest_results.csv`.

Configuration	Sharpe	Ann. Ret	Ann. Vol
depth=3, lr=0.05, n=500	0.98	19.2%	19.9%
depth=4, lr=0.05, n=500	1.11	21.7%	19.5%
depth=5, lr=0.05, n=500	0.92	17.4%	19.4%
depth=6, lr=0.05, n=500	0.90	17.0%	19.5%
depth=4, lr=0.01, n=500	0.99	18.7%	19.2%
depth=4, lr=0.10, n=500	1.00	19.6%	19.9%
depth=4, lr=0.05, n=200	1.06	20.5%	19.4%
depth=4, lr=0.05, n=1000	0.96	18.7%	19.9%

Table 23: XGBoost hyperparameter sensitivity (long-short). Each row varies one hyperparameter from the baseline. The baseline is not the single best configuration, but all variants outperform unconditional momentum.

Hyperparameter sensitivity. At the baseline depth of 4, the Sharpe ratio ranges from 0.89 to 1.11 across learning rate and ensemble size variations. Performance degrades at depths below 3 (see Section 5.3.2 for the depth analysis) and above 5 due to overfitting. The results confirm that M2's performance is not an artefact of a particular hyperparameter configuration.

Boosting versus bagging. As a complementary check, a Random Forest matched on depth (4), ensemble size (500 trees per seed), and feature set (the same 13 features) was trained over 50 seeds (`results/thesis/random_forest_results.csv`). Sharpe falls from 1.08 to 0.73 and π_t^{filter}'s SHAP share drops from 45% to 18%. The mechanism is the difference between boosting and bagging. XGBoost fits each successive tree to the residuals of the current ensemble, so at every iteration the algorithm preferentially selects whichever feature most reduces the current error; for this problem that feature is π_t^{filter}, and its importance compounds across trees. Random Forest grows each tree independently on a bootstrap sample and subsamples a random subset of features at every split, which by construction prevents any single feature from dominating the ensemble and distributes importance more evenly across the 13 features. Nonlinearity and depth are therefore necessary but not sufficient: the boosted additive structure is what converts the regime signal into a primary gating variable rather than a marginal predictor, which is the basis for the architecture-dependence claim in Section 3.2.1.

F.4 Number of Hidden States

The baseline HMM uses $K = 2$ states (calm and panic). A natural question is whether finer regime granularity – for example, distinguishing mild corrections from severe crises – would improve the downstream portfolio models. Table 24 reports out-of-sample Sharpe ratios for $K = 2, 3, 4, 5$, each averaged over five representative Gibbs sampler seeds (the production regime signal averages 200 seeds, but five suffice for this diagnostic).

	M1: LR		M2: XGB	
K	Mean Sharpe	Std	Mean Sharpe	Std
2	-0.012	0.000	1.107	0.000
3	-0.029	0.004	0.831	0.164
4	-0.029	0.003	0.606	0.291
5	-0.028	0.003	0.724	0.087

Table 24: Out-of-sample Sharpe ratios for HMMs with $K = 2, 3, 4, 5$ states (long-short). Each entry reports the mean and standard deviation across 5 independent Gibbs sampler seeds (2,000 iterations each). M1 is invariant to K. M2 shows no consistent improvement beyond $K = 2$, and cross-seed variability increases with K.

In long-short construction, M2 shows no consistent improvement beyond $K = 2$. Cross-seed variability increases with K, and the intermediate states that emerge for $K \geq 3$ lack the persistence needed to provide a stable conditioning signal. With only 241 training months, the data cannot reliably separate more than two regimes for the purpose of downstream cross-sectional stock selection.

Beyond estimation, $K = 2$ has an interpretability advantage. The two-state model produces a single continuous signal (π_t^{filter}) that XGBoost can condition on in a straightforward way: higher values indicate more stress. With $K \geq 3$, the additional states may capture dimensions of the economy that are orthogonal to the continuation-reversal mechanism, for example a "growth" state characterised by strong fundamentals but moderate stress indicators. Such a state would conflate two economically distinct situations (a genuine expansion vs. a recovery from panic) and could introduce noise into the regime signal, degrading the model's ability to distinguish continuation from reversal. The two-state framework avoids this by restricting the regime classification to a single dimension, the degree of market stress, which is the dimension most relevant to momentum's cross-sectional structure.

F.5 January Exclusion

Momentum is known to reverse in January (Jegadeesh and Titman, 1993): past losers outperform past winners, partially offsetting momentum's annual profitability. To verify that M2's performance is not driven by a January seasonal, Table 25 recomputes all metrics after excluding the 14 January months from the test period. M2's Sharpe ratio declines marginally from 1.11 to 1.06, confirming that the regime-momentum interaction operates throughout the calendar year. Fixed 12-month momentum improves slightly when January is excluded (0.06 to 0.14), consistent with the well-documented January reversal in unconditional momentum.

	Ann. Ret	Ann. Vol	Sharpe	Months
Full sample (baseline)				
M2: XGB	21.7%	19.5%	1.11	167
Fixed 12-mo mom	−1.9%	26.1%	0.06	167
Excluding January				
M2: XGB	20.9%	19.9%	1.06	153
Fixed 12-mo mom	0.1%	25.6%	0.14	153

Table 25: January exclusion test. Performance is recomputed after dropping all 14 January months from the test period.

Appendix G: External Validity

G.1 Alternative Train/Test Splits

Table 26 reports the baseline train/test split performance. The 30-year expanding-window backtest (Section 5.4.3) provides further out-of-sample evidence across alternative training windows and prediction horizons.

Train / Test split	N_{test}	M1 Sharpe	M2 Sharpe
1990–2010 / 2011–2025 (baseline)	167	−0.01	1.11

Table 26: Out-of-sample performance under the baseline train/test split (long-short, mom+π, 50-seed ensemble). M1 collapses while M2 achieves a Sharpe of 1.11 with highly significant factor model alphas.

In the baseline split (1990–2010 training, 2011–2025 test), M2 achieves a Sharpe of 1.11 while M1 collapses (−0.01), confirming the nonlinear regime-momentum interaction documented in Section 5.1.

G.2 30-Year Expanding-Window Out-of-Sample Methodology

The historical evidence reported in Section 5.4.3 comes from a separate analysis from the one above: a 30-year out-of-sample backtest in which the full pipeline (HMM + XGBoost) is re-estimated every January from 1995 onward, with each year's predictions made by a model trained only on data available through the prior December. Each retraining uses the production seed configuration: 200 HMM seeds (Gibbs sampler with 2000 iterations, 500 burn-in), 50 XGBoost seeds (depth 4, learning rate 0.05, 500 trees), and the same long-short construction described in Appendix B.1.

The HMM's panic-state label is fixed by an economically motivated sign convention rather than the data-driven percentile comparison used elsewhere. The convention encodes the known direction of each feature in panic regimes: -1 for drawdown, $+1$ for cross-sectional dispersion, -1 for relative market participation, and $+1$ for the credit spread. The data-driven label-switching procedure used in the main pipeline is unstable on short training windows with few observed crisis months (under 20), so the fixed-sign fallback activates for retrainings before 2002. From 2002 onward the data-driven procedure resumes, by which point the training window contains the dot-com bust as a calibration anchor. The HMM seed loop is parallelised across CPU cores via `multiprocessing.Pool`; the parallel implementation is bit-identical to the serial reference (verified in the test suite by comparing per-seed pi-filter arrays).

Each retraining produces 12 monthly out-of-sample long-short returns, which are stitched into a single 30-year monthly time series spanning January 1995 to November 2024 (359 months). Every prediction is strictly causal: the model used for month t saw no information after month $t-1$. The 167-month sub-window from January 2011 onward overlaps the main test period reported in Section 5.1 and serves as a methodological cross-check; expanding-window OOS produces a Sharpe of 0.88 over that window, within the production result's bootstrap interval.

G.3 Placebo Regime Signals

To assess whether XGBoost benefits specifically from the HMM regime signal rather than from any additional feature, the regime signal is removed entirely and XGBoost is retrained on the 12 momentum features alone.

Regime signal	M2 Sharpe
Real π_t^{filter} (baseline)	1.11
No regime signal	0.43

Table 27: Placebo regime signal test (long-short, mom+π). Removing the regime signal reduces M2's Sharpe from 1.11 to 0.43, confirming that the HMM signal is essential for long-short momentum stock selection.

Removing the regime signal reduces M2's Sharpe from 1.11 to 0.43, a drop of 0.68. The regime signal is essential for long-short momentum stock selection, not merely a marginal improvement.

G.4 Factor Model Regressions

To test whether M2's out-of-sample return is explained by known risk factors, monthly returns are regressed on five standard factor models: CAPM, Fama–French three-factor, Carhart four-factor, Fama–French five-factor, and a six-factor specification that augments FF5 with the momentum factor (UMD). Newey–West standard errors (6 lags) are used throughout. Table 28 reports the intercept (alpha) and its t-statistic under each specification; Table 29 repeats the exercise for the fund-augmented variant. Alphas that survive all five factor models indicate return that cannot be attributed to market, size, value, profitability, investment, or momentum exposure.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	23.4	4.08	-0.14					
FF3	23.2	4.48	-0.14	+0.04	-0.17			
Carhart	24.3	4.75	-0.19	-0.01	-0.22	-0.20		
FF5	22.9	4.62	-0.13	+0.08	-0.23		+0.06	+0.12
FF6	24.1	4.81	-0.18	+0.01	-0.31	-0.22	+0.01	+0.21

Table 28: Factor model regressions for M2 (XGBoost, mom+π). α is annualised. t-statistics use Newey–West standard errors (6 lags).

For completeness, Table 29 reports the same regressions for the fund-augmented variant (M2: mom+π+fund) discussed in Section 5.1.3. Alphas remain statistically significant under all five specifications ($t > 2.9$) but are materially smaller than the baseline's, consistent with the dilution documented in the main text.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	17.6	3.04	-0.16					
FF3	17.1	3.12	-0.14	-0.06	-0.13			
Carhart	18.2	3.22	-0.19	-0.12	-0.18	-0.20		
FF5	16.4	3.02	-0.11	+0.01	-0.30		+0.09	+0.35
FF6	17.7	3.17	-0.16	-0.06	-0.39	-0.23	+0.04	+0.44

Table 29: Factor model regressions for M2 (XGBoost, mom+π+fund). α is annualised. Newey–West t-statistics (6 lags).

G.5 Block Bootstrap Inference

To assess the sampling variability of M2's Sharpe ratio and test its outperformance formally, a block bootstrap is applied to the monthly return series. Returns are resampled with replacement in 12-month blocks to preserve autocorrelation, producing 10,000 synthetic 167-month paths (fixed seed 42). The Sharpe ratio is recomputed on each resample to obtain a 95% confidence interval, and the same block draws are used for M2 and each benchmark to form a paired test of the Sharpe difference. The bottom panel repeats the exercise on M2 returns split by regime.

	Sharpe	95% CI	vs M2 (p)
M2 (XGB)	1.11	$[0.67, 1.53]$	—
M1 (LR)	-0.01	$[-0.41, 0.42]$	0.001^{***}
M0 (Formula)	0.11	$[-0.36, 0.59]$	0.008^{***}
Fixed 12-mo mom	0.06	$[-0.43, 0.59]$	0.005^{***}
Fixed 1-mo mom	0.26	$[-0.21, 0.73]$	0.021^{**}
M2 regime-conditional Sharpe			
M2 Calm (n =108)	0.84	$[0.29, 1.32]$	—
M2 Panic (n =59)	1.54	$[0.85, 2.22]$	—
Panic − Calm	0.70	[-0.16, 1.59]	0.109

Table 30: Block bootstrap confidence intervals for Sharpe ratios and paired tests of M2's outperformance. Uses 12-month blocks and 10,000 resamples with a fixed seed. The "vs M2 (p)" column reports two-sided p-values from paired tests where both strategies are resampled on the same dates. $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$.

M2's Sharpe confidence interval is [0.67, 1.53], comfortably above zero, while every benchmark's interval straddles zero. The paired tests reject equality of Sharpe ratios between M2 and each benchmark at $p < 0.025$. Within M2, the panic-period Sharpe exceeds the calm-period Sharpe in point estimate (1.54 vs 0.84), but with only 59 panic months the paired difference is not statistically significant ($p = 0.109$); the 95% interval $[-0.16, 1.59]$ straddles zero. The point-estimate gap is consistent with the regime-conditional selection mechanism described in Section 5.2, but its statistical confirmation would require a longer panic sample. These intervals describe sampling variability within the 2011–2025 sample; out-of-sample generalisation is addressed separately by the 30-year expanding-window backtest (Section 5.4.3) and the sub-period analysis (Appendix F.1).

G.6 Leg-Level CAPM Betas by Regime

The D&M comparison in Section 5.4.4 characterises M2's contribution as stock selection rather than exposure scaling. A direct test of this framing is to compute leg-level CAPM betas separately by regime: if M2 reorganises which stocks it holds across regimes (cross-sectional re-ranking), the long-leg and short-leg betas should shift between calm and panic in opposite directions; if instead M2 mechanically scales exposure, both legs' betas should move together.

Table 31: Leg-level CAPM betas by regime, 2011–2025 OOS test period. Panic months are those with $\pi_t^{\text{filter}} \geq 0.5$. Newey–West standard errors (3 lags) in parentheses.

Strategy	Leg	Full	Calm	Panic
Fixed 12-mo mom	Long	1.09 (0.06)	1.09 (0.09)	1.07 (0.08)
Fixed 12-mo mom	Short	1.74 (0.11)	1.80 (0.21)	1.69 (0.13)
Method 2: XGB	Long	1.55 (0.08)	1.40 (0.11)	1.65 (0.10)
Method 2: XGB	Short	1.10 (0.06)	1.22 (0.09)	0.98 (0.05)

The pattern in Table 31 is consistent with the cross-sectional re-ranking interpretation. M2's long-leg beta rises from 1.40 in calm to 1.65 in panic (*tilt amplification*: in panic, the long leg holds names with higher market sensitivity), while the short-leg beta falls from 1.22 to 0.98. Fixed 12-month momentum, by contrast, holds both leg-level betas roughly stable across regimes (long: 1.09 calm, 1.07 panic; short: 1.80 calm, 1.69 panic), consistent with the unconditional rank-based construction. The difference is direct evidence that the regime signal acts on *which* stocks are held rather than on *how much*.

G.7 Per-cluster Feature Signature ($K = 4$ Partition)

Section 5.2.1 reports a $K = 4$ partition of the 167 test months and characterises each cluster narratively. Table 32 provides the full per-cluster feature signature for completeness: cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each cluster's months; long-leg pick raw momentum at each horizon, averaged across each cluster's months; and long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 10 in the main text.

Feature	Cluster 0 *calm cont.*	Cluster 1 *mild cont.*	Cluster 2 *post-panic*	Cluster 3 *deep crisis*
Cross-section context				
Cross-section mom (overall mean)	13.8%	7.3%	13.5%	−9.3%
Cross-section mom (short tertile, $h = 1$–4)	6.1%	4.4%	3.0%	−7.1%
Cross-section mom (mid tertile, $h = 5$–8)	14.6%	7.6%	12.4%	−11.1%
Cross-section mom (long tertile, $h = 9$–12)	20.8%	9.7%	25.0%	−9.6%
Cross-section dispersion (short)	0.23	0.27	0.28	0.23
Cross-section dispersion (mid)	0.44	0.46	0.56	0.35
Cross-section dispersion (long)	0.60	0.59	0.85	0.45
Cross-section skewness (short)	4.6	7.2	6.4	3.7
Cross-section skewness (mid)	6.2	7.6	6.2	5.6
Cross-section skewness (long)	6.7	8.1	6.5	5.7
π^{filter} frequency, prior 6 months	0.27	0.37	0.30	0.51
π^{filter} frequency, prior 12 months	0.31	0.32	0.41	0.30
Strategy Sharpe, prior 12 months	1.22	1.04	1.15	1.08
Long-leg pick raw momentum by horizon				
Long-leg pick mom, $h = 1$	7.1%	3.1%	−3.4%	−15.3%
Long-leg pick mom, $h = 2$	13.5%	8.6%	−5.5%	−21.6%
Long-leg pick mom, $h = 3$	18.3%	8.8%	−5.3%	−26.3%
Long-leg pick mom, $h = 4$	23.1%	10.6%	−5.8%	−30.8%
Long-leg pick mom, $h = 5$	30.8%	13.8%	−5.9%	−35.0%
Long-leg pick mom, $h = 6$	40.2%	18.0%	−3.2%	−38.3%
Long-leg pick mom, $h = 7$	49.4%	20.7%	−1.3%	−42.1%
Long-leg pick mom, $h = 8$	59.2%	23.7%	0.0%	−44.9%
Long-leg pick mom, $h = 9$	61.3%	23.1%	1.1%	−46.5%
Long-leg pick mom, $h = 10$	60.7%	21.9%	1.3%	−47.2%
Long-leg pick mom, $h = 11$	61.1%	22.8%	1.2%	−47.3%
Long-leg pick mom, $h = 12$	57.9%	18.5%	−5.3%	−47.5%
Long-leg pick cross-sectional z-score by horizon				
Long-leg z-curve, $h = 1$	0.26	0.04	−0.28	−0.73
Long-leg z-curve, $h = 2$	0.38	0.14	−0.31	−0.74
Long-leg z-curve, $h = 3$	0.40	0.11	−0.34	−0.75
Long-leg z-curve, $h = 4$	0.44	0.10	−0.36	−0.80
Long-leg z-curve, $h = 5$	0.54	0.16	−0.35	−0.85
Long-leg z-curve, $h = 6$	0.65	0.23	−0.29	−0.83
Long-leg z-curve, $h = 7$	0.73	0.27	−0.26	−0.85
Long-leg z-curve, $h = 8$	0.80	0.32	−0.22	−0.87
Long-leg z-curve, $h = 9$	0.76	0.29	−0.23	−0.88
Long-leg z-curve, $h = 10$	0.72	0.24	−0.25	−0.87
Long-leg z-curve, $h = 11$	0.67	0.22	−0.26	−0.86
Long-leg z-curve, $h = 12$	0.54	0.12	−0.35	−0.87

Table 32: Per-cluster feature signature for the $K = 4$ partition, 2011–2024 test period. Features are grouped into three blocks: (1) cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each clusterś months; (2) long-leg pick raw momentum at each horizon, averaged across each clusterś months; (3) long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 10; this table is the full descriptive feature signature.

G.8 Ridge Regression Baseline

To isolate the effect of linearity from the choice of training target, a Ridge regression is estimated on the same 13 standardised features as M1 and M2, predicting continuous forward returns (the same target as XGBoost). Table 33 reports results across a range of regularisation strengths.

Model	α	Sharpe	Ann. Ret	Ann. Vol	MDD	Calm / Panic SR
OLS (no reg.)	—	−0.58	−11.1%	17.4%	−82.0%	−0.61 / −0.56
Ridge	0.1	−0.58	−11.1%	17.4%	−82.0%	−0.61 / −0.56
Ridge	1	−0.58	−11.1%	17.4%	−82.0%	−0.61 / −0.56
Ridge	10	−0.58	−11.1%	17.4%	−82.0%	−0.61 / −0.56
Ridge	100	−0.58	−11.0%	17.4%	−81.9%	−0.60 / −0.56
Ridge	1,000	−0.54	−10.5%	17.5%	−80.3%	−0.60 / −0.47
M1 (LR, classification)	—	−0.01	−1.3%	15.2%	−42.4%	−0.29 / 0.39
M2 (XGBoost)	—	1.11	21.7%	19.5%	−22.8%	0.84 / 1.53

Table 33: Ridge regression baseline predicting continuous forward returns (same target as M2). The Sharpe of −0.57 is robust across all regularisation strengths.

The Ridge regression assigns π_t^{filter} a coefficient of +0.003, which is positive but economically negligible: a one-standard-deviation increase in the panic probability shifts the predicted return by only 0.3 basis points. The linear model cannot use the regime signal to condition how the momentum horizons jointly determine the stock ranking, regardless of the training target.

G.9 Emission Distribution Robustness

To verify that regime identification is robust to the choice of emission distribution, the HMM is re-estimated with multivariate Student-t emissions across a grid of degrees of freedom $\nu \in \{3, 5, 7, 10, 15, 20, 30, 50, 100\}$.

The Student-t model is estimated via data augmentation (Geweke, 1993): each observation receives a latent weight $w_t \sim \text{Gamma}(\nu/2, \nu/2)$, so that $\mathbf{z}_t \mid s_t{=}k,\, w_t \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k/w_t)$. Marginalising over w_t yields a multivariate t with ν degrees of freedom. Crucially, conditional on w_t the emission is Gaussian, so the Normal-Inverse-Wishart conjugate updates carry over with weighted sufficient statistics, preserving the efficiency of the Gibbs sampler.

ν	LL (train)	LL (test)	BIC	Agreement	Corr(π)
3	−882.7	−701.1	1929.9	91.4%	0.872
5	−829.0	−669.8	1822.5	96.8%	0.960
7	−838.3	−698.2	1841.2	98.0%	0.984
10	−815.8	−644.3	1796.1	96.8%	0.960
15	−800.1	−651.0	1764.8	98.3%	0.997
20	−802.9	−621.7	1770.4	99.3%	0.997
30	−802.6	−640.6	1769.8	98.5%	0.993
50	−796.6	−658.1	1757.7	98.8%	0.996
100	−792.3	−643.5	1749.0	98.8%	0.987
Normal	−787.5	−642.5	**1739.5**	—	—

Table 34: Student-t vs. Normal emission HMM comparison. LL = log-likelihood; Agreement = fraction of months with identical binary regime classification as the Normal HMM; Corr(π) = correlation of filtered panic probabilities.

The Normal model achieves the lowest BIC (2257.3 vs. 2257.6 for the best Student-t at $\nu = 100$), indicating no meaningful improvement from heavier tails. Regime classifications agree with the Normal HMM on at least 97.8% of months across all ν values, and the correlation of filtered panic probabilities exceeds 0.98 everywhere. These results confirm that the Gaussian emission specification is robust for this application.

Appendix H: Tree Path Horizon Interaction Analysis

To examine which momentum horizons XGBoost conditions on jointly, we trace stocks through 50,000 individual trees and record which horizon groups each decision path splits on. Momentum lookbacks are grouped into four bands: S (months 1–3), M (months 4–6), I (months 7–9), and L (months 10–12). For each of the four regime×leg groups (calm long, calm short, panic long, panic short), 5,000 stocks are sampled and traced through all trees. A path that splits on at least one feature from both the I and L groups is recorded as the "I+L" combination. Paths that split only on π_t^{filter} (no momentum splits) are excluded; the remaining paths are normalised to sum to 100%.

Tables 36 and 37 report the calm-minus-panic z-score difference for the long and short legs, respectively. The long leg table shows uniformly positive entries: the model selects stocks with higher relative momentum in calm at every horizon. The short leg table shows uniformly negative entries: in panic, the short leg shifts toward shorting stocks with higher relative momentum (closer to the cross-sectional average or above it), rather than the deep losers it targets in calm. Together, the two tables confirm that the regime signal reverses the selection direction on both sides of the portfolio while preserving the same horizon interaction structure.

Table 35 reports the underlying per-horizon z-scores and SHAP shares by regime and leg from which the calm-minus-panic differences below are computed.

Horizon	Z-score				SHAP share (%)			
	Calm		Panic		Calm		Panic	
	Long	Short	Long	Short	Long	Short	Long	Short
1	+0.03	+0.03	-0.19	+0.30	4.7	5.3	7.1	10.6
2	+0.10	-0.07	-0.13	+0.10	4.1	4.6	4.6	5.1
3	+0.10	-0.04	-0.17	+0.07	3.7	3.2	3.2	3.2
4	+0.12	-0.04	-0.18	+0.06	5.3	7.4	4.4	7.3
5	+0.19	-0.12	-0.18	+0.01	6.0	4.0	7.3	4.7
6	+0.27	-0.19	-0.13	-0.05	6.3	4.3	6.3	5.0
7	+0.33	-0.24	-0.11	-0.10	3.8	4.2	3.9	3.2
8	+0.39	-0.28	-0.09	-0.13	10.2	12.8	8.3	10.8
9	+0.36	-0.29	-0.09	-0.15	12.7	14.5	15.5	13.1
10	+0.31	-0.27	-0.10	-0.16	7.7	5.4	8.7	5.0
11	+0.27	-0.27	-0.12	-0.16	18.6	23.1	16.0	21.2
12	+0.16	-0.18	-0.17	-0.10	16.8	11.2	14.8	10.8

Table 35: Z-score and SHAP share by momentum horizon, regime, and portfolio leg. Z-scores measure how far above or below the cross-sectional average the selected stocks are at each lookback. SHAP share measures each horizon's contribution to the model's momentum decision (each leg sums to 100%).

Combination	Pct	S (1–3)	M (4–6)	I (7–9)	L (10–12)	Avg
I	6.0%			+0.28		+0.28
L	6.8%				+0.28	+0.28
M	3.4%		+0.27			+0.27
S	6.1%	+0.09				+0.09
I + L	13.4%			+0.43	+0.36	+0.39
M + I	5.4%		+0.26	+0.40		+0.33
M + L	5.9%		+0.27		+0.35	+0.31
S + I	3.3%	+0.24		+0.51		+0.38
S + L	5.0%	+0.23			+0.44	+0.33
S + M	7.8%	+0.37	+0.51			+0.44
M + I + L	10.4%		+0.27	+0.38	+0.33	+0.33
S + I + L	8.4%	+0.23		+0.52	+0.47	+0.41
S + M + I	6.6%	+0.22	+0.41	+0.56		+0.40
S + M + L	7.2%	+0.16	+0.29		+0.38	+0.28
S + M + I + L	4.4%	+0.19	+0.32	+0.45	+0.39	+0.33

Table 36: Calm-minus-panic z-score difference for the long leg, by horizon group combination. Each entry shows how much higher the average cross-sectional z-score is in calm than in panic. All 39 entries are positive: at every horizon in every combination, the model buys stocks with higher relative momentum in calm and lower relative momentum in panic. The largest shifts occur at the intermediate horizon (I, +0.36 to +0.55), where continuation is strongest. "Pct" is the average share of momentum-involving tree paths using that combination.

Combination	Pct	S (1–3)	M (4–6)	I (7–9)	L (10–12)	Avg
I	5.9%			-0.22		-0.22
L	6.8%				-0.13	-0.13
M	3.3%		-0.22			-0.22
S	5.9%	-0.18				-0.18
I + L	14.6%			-0.30	-0.29	-0.30
M + I	5.6%		-0.25	-0.27		-0.26
M + L	5.9%		-0.19		-0.14	-0.16
S + I	3.1%	-0.18		-0.17		-0.17
S + L	4.7%	-0.20			-0.10	-0.15
S + M	7.5%	-0.21	-0.18			-0.20
M + I + L	10.8%		-0.20	-0.22	-0.18	-0.20
S + I + L	8.4%	-0.21		-0.21	-0.17	-0.20
S + M + I	6.5%	-0.19	-0.18	-0.19		-0.19
S + M + L	6.7%	-0.17	-0.11		-0.06	-0.11
S + M + I + L	4.3%	-0.21	-0.21	-0.23	-0.20	-0.21

Table 37: Calm-minus-panic z-score difference for the short leg. Nearly all entries are negative: in calm, the model shorts stocks deep in the left tail (strongly negative z-scores), while in panic it shifts toward shorting stocks closer to the cross-sectional average or even above it. The short leg's upward z-score shift in panic is the mirror image of the long leg's downward shift (Table 36), confirming that the regime signal reverses the selection direction on both sides of the portfolio.

Appendix I: Label Switching: Sign Correction Procedure

Because the four HMM features respond to stress in different directions, the expected direction of each feature during stress is determined empirically. For each feature j, the 95th percentile is computed separately on training months falling within known crisis windows (the dot-com bust of 2000–2002 and the Global Financial Crisis of 2007–2009) and on all remaining training months:

$$\text{sign}_j = \begin{cases} +1 & \text{if } q_{0.95}^{\text{crisis}}(z_j) \geq q_{0.95}^{\text{normal}}(z_j), \\ -1 & \text{otherwise.} \end{cases} \tag{41}$$

Features that spike upward during crises receive $\text{sign}_j = +1$; features that decline receive $\text{sign}_j = -1$. The panic state is then identified as the regime $k \in \{0, 1\}$ whose posterior mean vector $\hat{\boldsymbol{\mu}}_k$ best aligns with the crisis direction:

$$k^* = \arg\max_{k \in \{0,1\}} \sum_{j=1}^{D} \text{sign}_j \cdot \hat{\mu}_{k,j}. \tag{42}$$

The state labelled k^* is designated as "panic" and the other as "calm."

Appendix J: Convergence Diagnostics

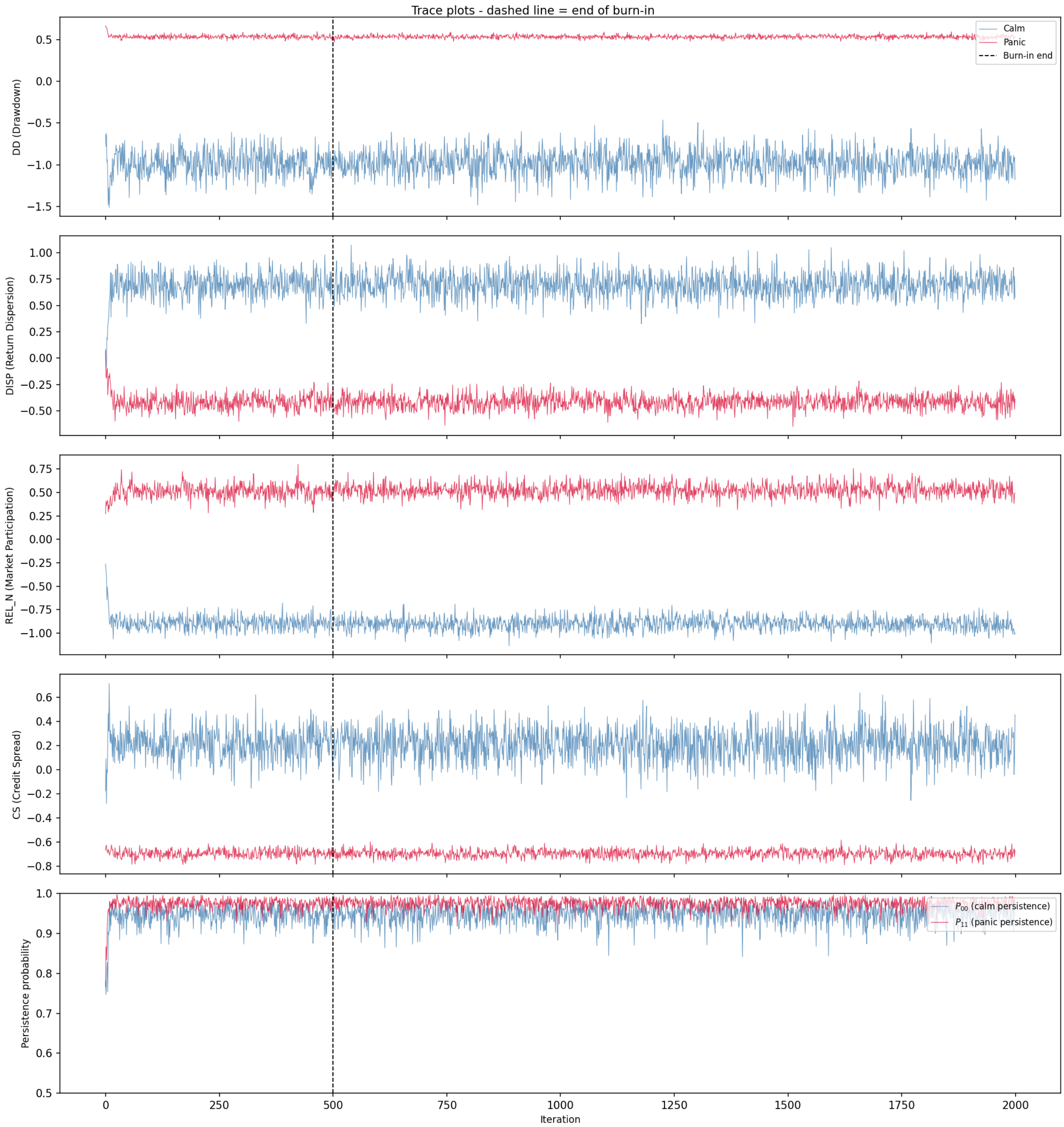

Figure 11: Trace plots of the Gibbs sampler. Each panel shows sampled regime means (calm in blue, panic in red) for one HMM feature. The dashed line marks the end of burn-in.

J.1 Gelman–Rubin Diagnostic

The Gelman–Rubin $\hat{R}$ statistic (Gelman et al., 2013) assesses convergence by comparing within-chain and between-chain variance across the five independent Gibbs sampler chains. For each parameter θ, $\hat{R} = \sqrt{\hat{V}/W}$, where W is the mean within-chain variance and $\hat{V} = \frac{n-1}{n}W + \frac{B}{n}$ combines within-chain variance with the between-chain variance B. Values near 1.0 indicate that the chains have converged to the same posterior distribution; $\hat{R} < 1.1$ is the standard threshold.

Parameter	$\hat{R}$	Status
μ_{panic}(DD)	1.000	OK
μ_{panic}(DISP)	1.000	OK
μ_{panic}(REL_N)	1.000	OK
μ_{panic}(CS)	1.000	OK
μ_{calm}(DD)	1.000	OK
μ_{calm}(DISP)	1.001	OK
μ_{calm}(REL_N)	1.000	OK
μ_{calm}(CS)	1.000	OK
P_{00} (calm persistence)	1.000	OK
P_{01} (calm $\rightarrow$ panic)	1.000	OK
P_{10} (panic $\rightarrow$ calm)	1.000	OK
P_{11} (panic persistence)	1.000	OK

Table 38: Gelman–Rubin $\hat{R}$ statistics for all HMM parameters, computed across 5 independent chains (1,500 post-burn-in draws each). All values are below 1.01, well within the convergence threshold of 1.1.

J.2 Seed Convergence of the Regime Signal

Gelman–Rubin and the trace plots assess whether a single Gibbs chain has converged to its stationary posterior, but they do not quantify how many chains are needed to stabilise the *downstream* Sharpe ratio. The regime signal π_t^{filter} enters the pipeline as the average of N_s seed-specific forward filters (Section 3.1.5), and because XGBoost is a nonlinear function of π_t^{filter}, MCMC noise in any individual chain does not cancel linearly.

To measure this dependence, we subsample $k \in \{1, 2, 3, 5, 10, 15, 20, 30, 50, 75, 100\}$ chains without replacement from the seed pool, average the forward filter over those k chains, retrain the XGBoost ensemble on the resulting signal, and record the out-of-sample Sharpe. For each $k < 100$ we repeat on 30 random subsets; $k = 100$ exhausts the pool.

k	n_{sub}	Mean	Median	Std	P25	P75	Min	Max
1	30	1.040	1.031	0.078	0.985	1.090	0.915	1.205
2	30	1.062	1.081	0.069	1.019	1.103	0.894	1.217
3	30	1.049	1.044	0.043	1.014	1.075	0.973	1.143
5	30	1.042	1.042	0.047	1.002	1.083	0.967	1.154
10	30	1.053	1.050	0.036	1.038	1.066	0.965	1.145
15	30	1.075	1.079	0.029	1.055	1.097	1.008	1.139
20	30	1.078	1.080	0.022	1.065	1.090	1.025	1.122
30	30	1.089	1.082	0.025	1.069	1.109	1.052	1.137
50	30	1.093	1.092	0.015	1.082	1.101	1.067	1.124
75	30	1.098	1.102	0.012	1.091	1.105	1.067	1.118
100	1	1.118	1.118	0.000	1.118	1.118	1.118	1.118

Table 39: XGBoost ensemble seed convergence. For each subset size k, we draw n_{sub} random subsets of k seeds from the full 100-seed pool, average their predictions, and report the resulting out-of-sample L/S Sharpe ratio. Mean Sharpe stabilizes by $k \approx 50$, justifying the production choice; standard deviation across subsets shrinks as $1/\sqrt{k}$.

Two patterns emerge. First, the mean Sharpe rises from 1.04 at $k = 1$ to 1.12 at $k = 100$, so a single-chain point estimate understates the strategy's performance by roughly 0.08 in Sharpe units. The gap closes monotonically, with most of the lift realised by $k = 30$. Second, the interquartile range across random subsets collapses from 0.10 at $k = 1$ to 0.02 at $k = 50$: once $k \geq 50$, any two random subsets of the seed pool produce nearly the same XGBoost portfolio. The production configuration ($N_s = 200$) sits well inside this plateau, which is why the 1.11 Sharpe reported in Section 5 is insensitive to which specific seeds were drawn. The practical implication is methodological rather than cosmetic: a single-chain HMM would not reliably support the main conclusions, so the seed ensemble is a requirement of the design, not a late-stage performance tuning.

Appendix K: International HMM Feature Selection

For the international robustness check (Section 5.5), the HMM feature set is selected per region using the same 4-pass procedure as the US production specification (Section 4.2): a quality screen on MCMC convergence and crisis-window alignment, a portfolio-Sharpe ranking on the surviving combinations, a definitive validation at higher seed counts, and an additional stability check across random seed partitions. The international candidate pool comprises the five locally-available stress features (DD, VOL, DISP, REL_N, BANK_REL); BANK_REL is the value-weighted bank-sector excess return over the broad regional market, sign-flipped so high values indicate stress, and substitutes for the US-specific Moody's BAA-AAA spread. With DD required, the procedure evaluates fifteen combinations of size one to four. Both UK and Japan select *DD+VOL+REL_N* as the regional feature set. The transplanted-from-US template (DD+DISP+REL_N+BANK_REL) fails the Pass-1 quality screen for both regions (minimum effective sample size of 14 and 4 respectively), motivating per-region selection. Pass-by-pass results are reported in `results/thesis/intl_<region>_hmm_feature_selection_pass{1,2,3,4}.csv`; chosen-features manifests in `intl_<region>_hmm_chosen_features.json`.

Appendix L: Fundamental Feature Definitions

Ten firm-level characteristics are computed from Compustat quarterly data (`fundq`), linked to CRSP via the CCM bridge table. All fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. Cash-flow items (`oancfy`, `capxy`) are reported year-to-date and are differenced within each fiscal year to recover the per-quarter contribution before the trailing twelve-month sum is taken.

- **Book-to-market (bm):** Book equity (`ceqq`) divided by market equity.
- **Return on equity (roe):** Income before extraordinary items (`ibq`) divided by book equity.
- **Gross profitability:** Revenue minus cost of goods sold, scaled by total assets: $(\texttt{revtq} - \texttt{cogsq})/\texttt{atq}$.
- **Asset growth:** Year-over-year growth in total assets: $\texttt{atq}_t/\texttt{atq}_{t-4} - 1$.
- **Leverage:** Total debt divided by total assets: $(\texttt{dlttq} + \texttt{dlcq})/\texttt{atq}$.
- **Earnings growth:** Year-over-year growth in earnings, transformed as $\text{sign}(g) \cdot \log(1 + |g|)$.
- **Log market equity:** Natural logarithm of market capitalisation (price $\times$ shares outstanding), lagged one month.
- **Operating cash flow over assets (cfo_a):** Trailing twelve-month operating cash flow (`oancfy`, converted to quarterly and summed) scaled by total assets.
- **Free cash flow over assets (fcf_a):** Trailing twelve-month operating cash flow minus trailing twelve-month capital expenditure (`capxy`), scaled by total assets.
- **Accruals:** The accrual component of earnings, $(\texttt{ibq}_{\text{TTM}} - \texttt{oancf}_{\text{TTM}})/\texttt{atq}$, which captures the non-cash portion of reported earnings.

Fundamentals are winsorised at the 1st and 99th percentiles (5th and 95th for asset growth) using training-sample bounds. The cash-flow features (cfo_a, fcf_a, accruals) are available for roughly 83% of stock-months; the remaining seven characteristics exceed 82% coverage individually, and XGBoost handles any residual missingness natively via its default-branch logic.